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# **Acoustic Analysis of Swiss English Vowel Quality**

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## Abstract

The purpose of this study in second language acquisition (SLA) is threefold. Primarily, it attempts to explore the differences in vowel quality between the English vowels produced by 20 Swiss German (SG) speakers and the vowel quality of both Received Pronunciation (RP) and General American (GA). Language internal as well as language external factors occupy a crucial role in determining a foreign accent; thus, a major component of this study is devoted to their assessment. This study further offers a critical approach towards standardized sociolinguistic and particularly acoustic phonetic procedures. 20 native SG subjects were recorded in school settings. The subjects were prompted to read word lists containing selected English and SG vowels in /hVd/ constructions three times. They further filled out questionnaires that provide information as to their demographic characteristics, language backgrounds, and attitudes towards British English and American English. The recordings were instrumentally analyzed with PRAAT in measuring F1 and F2 values. Measurement was conducted at the formants' most stable phase and where the vowels were least affected by consonantal gestures. A time period of 50 ms was measured at the vowel nucleus which is visualized by the LPC traces on the spectrograms. The obtained Swiss English (SE) formant values were compared to formant values of RP and GA by means of paired-samples t tests in SPSS. Repeated measures ANOVAs were run to test variation among the three attempts of articulating the words. Independent t tests as well as ANOVAs were primarily applied in order to test the SE mean values for external variables gained from the questionnaire. This research design raises several issues, some of which are assessed in the critical approach section. First, these issues are addressed from a general sociolinguistic perspective, followed by the respective implications for the present study. The critical approach section explores the lack of authenticity in language which is recorded under near-laboratory-like conditions, the comparison of formant values to a precarious concept such as Standard English, language variation and its inhibiting implications on sociophonetic second language research, as well as serious methodological difficulties in acoustic phonetic measurement. The implications of these characteristics advocate taking the findings of the present study, as well as the results of empirical studies in sociophonetics and sociolinguistics, with a pinch of salt. The results of the data analyses suggest that the Swiss tend to pronounce English vowels more towards RP than towards GA, particularly in terms of the vowels' F2 values. Equality of means tests showed tendencies that L1 interference occurs principally in the SE vowels /i:/, /ɑ:/, and /ʌ/, as the subjects' SE vowels' F1s and F2s are closer to the F1s and F2s of their SG counterparts than they are to RP or GA. Further statistical tests show potential correlations between the SE mean values and external factors such as age, SG dialect, attitudes towards British or American English, etc. Often, however, these tendencies are statistically insignificant as the resulting p values are not  $\leq .05$ , which is probably due to the small sample of only 20 subjects. Finally, it is suggested that instrumental vowel analyses could be implemented more frequently in foreign language classrooms in Switzerland, because they, as scholarship suggests, supply practical aid in accent reduction, in case a foreign accent is deemed undesirable.



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## Abbreviations Used

|          |                                    |
|----------|------------------------------------|
| $\alpha$ | Level of Significance              |
| ANOVA    | Analysis of Variance               |
| CP       | Critical Period Hypothesis         |
| EFL      | English as a Foreign Language      |
| ESL      | English as a Second Language       |
| EVP      | Electronic Visual Feedback         |
| F1       | First Formant                      |
| F2       | Second Formant                     |
| F3       | Third Formant                      |
| GA       | General American                   |
| Hz       | Hertz                              |
| IL       | Interlanguage                      |
| L1       | First Language                     |
| L2       | Second Language                    |
| LOR      | Length of Residence                |
| LPC      | Linear Predictive Coding           |
| MDH      | Markedness Differential Hypothesis |
| MP       | Mispronunciation                   |
| MV       | Missing Value                      |
| NS       | Native Speaker                     |
| NNS      | Non-Native Speaker                 |
| P        | Significance Value                 |
| RP       | Received Pronunciation             |
| SE       | Swiss English                      |
| SG       | Swiss German                       |
| SLA      | Second Language Acquisition        |
| VOT      | Voice Onset Time                   |

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# 1 Introduction

In a multilingual country like Switzerland, language attitudes, language ideologies, and language practices are often the subject of heated discussions. Not only do they occur at the highest levels of government, but also at many dinner tables. Monica Heller observes the same phenomenon in Montreal, Canada, which offers her prolific grounds for studying sociolinguistics. According to Heller, a principal facet of sociolinguistics is to delve into issues such as how a sociolinguist's work intersects with the major concerns people have about language in their lives" (1999: 263). Deploying this statement as a leitmotif for this paper, it is the author's objective to establish a direct link between concerns Swiss people may have as to their use of English and the work of sociolinguists. To be more precise, it is accented speech that most learners of English in Switzerland have that will be scrutinized in this paper.

The word *accent* is a fuzzy term. The noun has more than one meaning and is often used by the public in peculiar ways. The *Oxford Advanced Learner's Dictionary* (2000) and Rosina Lippi-Green (1997) agree that the noun *accent* designates a manner of pronouncing the words of a language in such a way that the speaker's regional provenance is permeable. Yet *accent* can also refer to distinguish stress in words and as the mark on a letter to point at a particular pronunciation of a word. Both texts discriminate between a further category of accent, that of second language (L2) accent.

L2 accent on the level of phonology refers to the "breakthrough of native language phonology into the target language" (Lippi-Green 1997: 43). Thus, the language articulated with the foreign accent neither wholly pertains to the speaker's first language (L1), nor is it congruent with his target L2.<sup>1</sup> On the basis of this observation, Selinker (1972) coined the term *interlanguage* (IL) in reference to the systematic knowledge of an L2 which is independent of both the target language and the learner's L1. An IL need not inevitably be studied exclusively on the level of its phonology, as is the case in the present study. An IL can be described at various linguistic levels, including its lexicon, morphology, syntax, semantics, and discourse (Major 2001: 1-2). In order to study the accented speech of Swiss English (SE), however, it is this IL phonology that is submitted to careful micro-linguistic observation in this study. Consequently, this paper moves in the direction of sociophonetics.

Sociophonetics, as can be derived from its morphology, can be defined as the work on the borderline between sociolinguistics and phonetics (Foulkes 2005: 1). Thomas (2002: 189) critically asserts that one of the shortcomings of current sociophonetics is that experimental phoneticians do not usually make use

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<sup>1</sup> Although social awareness has changed and the use of the possessive pronoun *his* may seem incongruous, this paper solely applies masculine pronouns as the repetition of *he* and *she*, *him* and *her*, *his* and *hers* etc. at every reference seems clumsy.

of naturalistic data, as the speakers are often recorded in laboratories to improve the quality of the sound signal. Besides, most sociophoneticians only use small samples due to the amount of work involved in their analysis. Although such arguments have some merit, it is worth stressing that sociophonetic research has yielded a great amount of information that has furthered variationist approaches towards language, as is seen in several of Labov's studies (cf. Labov 1966). Hitherto we have come a good way into this paper with the definitions of such topic-relevant terms as *accent*, *L2 accent*, *interlanguage*, and *sociophonetics*, yet the purposes of the present study have not been mentioned. It can be maintained, however, that Thomas' critical voice towards sociophonetics sets the proper tone for the following analyses.

The foci of this study are in fact threefold. The working hypothesis holds that the pronunciation of Swiss German (SG) non-native speakers (NNSs) of English deviates from that of speakers of Received Pronunciation (RP) and General American English (GA). The primary aim is to show how the group under scrutiny differs in its articulation of English vowels. A secondary aim is to shed light on the complex issue of reasons for differences in articulation among the subjects. Finally, a predominantly theory-driven section addresses problematic aspects evoked in empirically studying an IL phonology and sociolinguistics in general. The paper opens with the discussion of these challenging issues, entitled *Section A: Critical Approaches towards Standardized Sociolinguistic Procedures*.

First, this section attempts to highlight methodological concerns with a particular spotlight on the existence of authenticity, or lack thereof, in researched speech. This subchapter is closely interconnected with the consecutive subchapters, which attend to the notions of language variation and standardized language and their implications on the present study. Specific methodological difficulties of pursuing acoustic phonetic research, such as intricacies of recording and spectrogram analyses, are described in a final subchapter.<sup>2</sup> Following Section A is *Section B: Theory underlying the Analysis of 20 EFL Speakers' Vowel Quality*, which explores the theory behind the question as to why Swiss speakers articulate English vowels differently than GA or RP native speakers (NS).

Section B begins by elucidating the criteria according to which subjects were sampled, followed by an in-depth presentation of the word lists and texts the subjects were asked to read. Whereas Section A sets a critical theoretical framework for this research project from a general sociolinguistic viewpoint, Section B further elaborates on how the issues outlined in Section A, inauthenticity in researched language, the inhibiting effects of language

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<sup>2</sup> In this context, it is crucial to emphasize that Section A addresses these issues from a broad sociolinguistic perspective, while Section B presents its repercussions on the present study.

variation on data, as well as difficulties in acoustic phonetic measurements, are tackled in the present study. Sections A and B provide the theoretical background to the understanding of the data analyses in Section C.

The statistical analyses of the data as well as the synthesis of the theory presented in Sections A and B with the results of Section C comprise the core part of this paper: Section C *Data Analyses*. This section starts by outlining the results of the instrumental analysis of the English vowels produced by the Swiss speakers. The resulting values are contrasted to the reference values of RP and GA. The benefit of such a comparative analysis is that it permits conclusions if the Swiss under scrutiny articulate English vowels more like NSs of American or British English. Ultimately, explanations for these findings are sought in running statistical tests with data retrieved from the questionnaires which were filled out by the subjects. The questionnaire largely covers variables that are subsumed in Sections A and B. Final remarks as well as an outlook conclude this paper in second language acquisition (SLA) research. In order to get the reader acquainted with the theoretical framework of this study, key concepts as well as this study's research design are introduced before starting with Section A.

## 2 Key Concepts

The purpose of this introductory section is to familiarize the reader with three fundamental concepts of the present study. First, the object under scrutiny, NNS-produced language, is presented and qualified in brief. Equally important is an issue raised in this qualification, the causes for a foreign accent, which will be addressed secondly. Foreign accents or accents in general, for that matter, only come into existence if an audience inherently compares a speaker's articulation to a norm, which in most cases is one's own language or dialect. Therefore, the last points attended to are the criteria foreign accents are judged by and the bearings on the present study.

### 2.1 Qualifying the Object under Scrutiny

The moment a person speaks in our language in an unusual way, it is likely that one makes judgments about whether or not this person is a native speaker or a non-native speaker.<sup>3</sup> In Major's words (2001: 19): "[T]he overall impression concerning NSs form whether or not and to what degree a person sounds native or non-native is called *global foreign accent*."<sup>4</sup> From what is acoustically perceptible, this *global foreign accent* is manifested in accented speech on different levels: segments, syllables, prosody, stress, length, tone, intonation, and rhythm. The present study confines itself to analyzing the segmental level of IL phonology only.

A segment is a "minimal discrete unit in the sound system of a language, defined physically or auditorily, and generally classified as a vowel or consonant" (Crystal 1992: 345). Segments are the most meticulously studied area in L2 phonology, which is one of the reasons why this paper argues along these lines. The scope of researching the segmental level of an IL phonology in the present study, however, is restricted to vowels only.<sup>5</sup> Monophthongs rather than diphthongs or consonants were chosen for three reasons: One, the amount of time provided for this research project is limited. The second reason regards the characteristics of length of articulation of vowel segments. As opposed to consonants, vowels offer a larger "corroding surface" for individual variation to take place, simply because they are longer. They are also not as categorical as consonants – the continuum of vowels is much smoother than that of consonants. The third motive for choosing monophthongs is based on the disproportionate attention the field has been paying to studying vowels, in particular English vowels. A corollary of that is the backdrop of a large body of scientific literature on which to base this study.

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<sup>3</sup> This is not the place to go into detail about whether this is a conscious or an unconscious process.

<sup>4</sup> Emphasis adopted from Major (2001: 19).

<sup>5</sup> Cf. Chapter 6 *Word Lists* for a detailed analysis of the examined vowels.

## 2.2 Causes for a Foreign Accent

Almost as abundant as the literature on foreign language phonology are academic texts pertinent to the discussion of sociolinguistic variables that are said to cause a foreign accent. Scholarship offers numerous variables that allegedly impinge on foreign language production. Paunovic (2003), who conducted a comparable study with speakers of Serbian, ascertains that the process of acquiring a phonological system is immensely complex, as numerous linguistic variables are involved simultaneously. This is not an isolated view; Piske, Flege, and MacKay (2001) corroborate Paunovic's (2003) statement in providing a tentative list of variables which purportedly determine the degree of foreign accent in an L2, namely age of L2 learning, length of residence in an L2-speaking country (commonly abbreviated as LOR), gender, formal instruction, motivation, language learning aptitude, and amount of L1 use. There are, however, other factors that cause a foreign accent which have received too little attention in the technical literature. These factors can be subsumed as causes related to the personality of the individual.

In his book *Foreign Accent*, Major astutely (2001: 66) observes that motivation, empathy, sense of identity, ego-permeability, self-esteem, risk taking, anxiety, introversion versus extroversion, musicality, and field independence versus field dependence, too, are decisive factors affecting an L2 phonology. In reading this list of variables, one may appreciate why these factors have not been as rigorously analyzed as those previously outlined. The marginality in technical literature is due to the concepts' nebulous, highly complex nature, which poses a serious problem for scientific analysis. Critically accepting that the outlined variables (Paunovic 2003; Piske, Flege, and MacKay 2001; and Major 2001) cause foreign accents, the question remains how and why listeners perceive an accent as foreign at all.

## 2.3 Comparison to Norm

If, in fact, one makes conscious or unconscious judgments of NNSs' speech,<sup>6</sup> what is their speech judged by if not a standard language or standard dialect, i.e. a norm? This obviously raises the issue of the actual existence of a standard language and the ongoing debate about it. This paper, too, deploys the principle of evaluating accented speech against the standard language by comparing SE to RP or GA. As the notion of Standard English is a pivotal concept in the present study, there is need for qualification of the term *standard language*. Standard language need not necessarily be conceived of as a concrete language, as lay people may understand it. Milroy and Milroy (1991: 22-23) once defined Standard English as "an idea in the mind rather than reality – a set of abstract norms to which actual usage may conform to a greater or lesser extent." In

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<sup>6</sup> Cf. 2.1 *Qualifying the Object under Scrutiny*.

comparing Swiss speakers' English vowel quality to that of RP and GA, I dissociate myself from the belief that standardized language exists in reality, but agree with Milroy and Milroy as well as with Lippi-Green's (1997: 44) stance that "[standardized language] certainly does exist as an ideal in the minds of the speakers." In adhering to this understanding of Standard English, I am confident that a comparison between Swiss speakers' accented speech with that of GA and RP will yield plausible explanations.

The fact that the notion of Standard English is a topic of heated debates is one of the many problematic facets of this study. There are several other aspects which are in need of thorough checking. These comprise authenticity of researched language, language variation, and methodological concerns of phonetic-specific data analysis. As will be pointed out in the following section, these issues strongly exacerbate empirical work in sociolinguistics. So as to effectively cope with such multilayered issues, a research design comes in handy.

### 3 Research Design

The general framework for this paper is taken from a similar study I conducted while studying at New York University in 2004. The paper is titled *Tendencies as to how 2 Swiss German Speakers Accentuate 16 English Vowels/Diphthongs Differently than 2 Native Speakers: A Pilot Study*.<sup>7</sup> The research at NYU, although restricted to only four subjects, provided fertile grounds to formulate a pilot study, which at a later stage could be elaborated on and turned into a larger study in the field of SLA research.

Pilot studies fulfill a range of important functions. Primarily, they are the assessment of a full-scale study and they help identify logistical problems. The collection of preliminary data in pilot studies helps identify potential problems or gaps in the data and the data analysis techniques. For the most part, the pilot study conducted in New York identified potential acoustic phonetic measurement problems as well as issues related to subject sampling. It further highlighted phenomena such as the *Observer's Paradox* and the *Hawthorne Effect*.<sup>8</sup> The paper ended on the note that the two Swiss generally articulated high and mid-high vowels higher than the two Americans did. In terms of vowel backness, the paper concluded that mid-low and low back vowels are generally articulated more towards the back by the Swiss. Although this preceding study offered insight as to how vowels could be studied and compared to one another, a research design was not explicitly formulated – a strategy not to be repeated in the present study.

Given the premise of a foregoing pilot study, the present study's research design is based on preconceived notions and expectations about the second language phenomena explored. According to Seliger and Shohamy's (1989: 30) understanding of various types of research designs, the approach used in this paper corresponds to the deductive, hypothesis-driven research design with the aim of producing a theory. We will leave unanswered the extent to which this theory will be produced by qualitative or quantitative methods.

The relationship between qualitative and quantitative methods is a peculiar one. Often, qualitative methods are inseparably linked with quantitative ones and vice versa. Consequently, the dichotomy's boundaries are blurred. Tashakkori and Teddlie (1998: 47) argue in *Mixed Methodology: Combining Qualitative & Quantitative Approaches* that research studies typically begin with "qualitative data collection and analysis on a relatively unexplored topic,

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<sup>7</sup> The paper was submitted to Prof. Iwar Werlen at the General Linguistics Department at the University of Berne in November 2004.

<sup>8</sup> *Hawthorne* was the name of an electricity company in Western Chicago, where a series of studies on productivity was implemented. The term *Hawthorne Effect* is in reference to the initial improvement in a process of production caused by the obtrusive observation of that process.

using the results to design a subsequent quantitative phase of the study” – a strategy this study, too, embarks on. The qualitative aspect of this study resides in what Tashakkori and Teddlie (ibid: 47) name “data collection and analysis on a relatively unexplored topic.” Prior to obtaining numbers and figures that allow the implementation of quantitative analyses, qualitative, often subjective, judgments as to what linguistic phenomena are to be observed need to be made.<sup>9</sup> Once the data are qualitatively collected, quantitative analyses can be put into operation. Given the numerical form of the data, quantitative analyses outweigh qualitative analyses in the present study. Thus, statistical analyses of data, such as correlations, t-tests, analysis of variance (ANOVA), and repeated measures ANOVAs will be deployed. While the running of statistical tests with data may make a great visual impact, it is worth emphasizing that the actual scientific gain from the results remains limited.

Although data are drawn from an empirical knowledge base, this study is not full-scale, nor does it “constitute acceptable evidence about a construct,” as Norris and Ortega (2003: 719) describe studies that make use of a small number of subjects. What this study offers with its findings are tendencies as to how a specific group of Swiss English as a Foreign Language (EFL) speakers, namely 20 speakers from the *Kantonsschule Zofingen* in the canton of Aargau, articulate English vowels. Also, it endeavors to identify trends of how the Swiss’ articulation is different from GA and RP and to offer tentative explanations as to why these differences exist. Having commented on this study’s research design and its position within the framework of quantitative and qualitative research, we will now turn to the first main section of this paper, Section A: *Critical Approaches to Standardized Sociolinguistic Procedures*.

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<sup>9</sup> Cf. 2.1 *Qualification of the Object under Scrutiny*.

## Section A: Critical Approaches to Standardized Sociolinguistic Procedures

### 4 Critical Approaches

#### 4.1 Authenticity of Researched Language

Over the past decades, the issue of whether or not language that is knowingly being researched is still authentic has fueled discussions in various fields of linguistics. In order to understand the issues involved, one has to understand the basic tenets of two conflicting views on language and their relation to society, namely the generative paradigm, which is intimately connected with Noam Chomsky, and the variationist paradigm, which is best represented in the work of William Labov.

Chomsky (1985: 4) postulates that linguistic theory should be concerned with the linguistic competence of an ideal native speaker/listener in a completely homogeneous speech community. This speaker knows the language flawlessly and is not subject to performance errors due to memory limitations, distractions, and shifts of attention. Chomsky asserts that “observed use of language or hypothesized dispositions to respond, habits, and so on, may provide evidence as to the nature of this reality, but surely cannot constitute the actual subject matter of linguistics, if this is to be a serious discipline” (ibid: 4). In other words, it may be interesting to make observations of what Chomsky terms *performance*, but substantial conclusions about language cannot be drawn from those observations. The obverse of this argument is highlighted by Beaugrande, for instance, who maintains that by this substitution of real language with ideal language, generativists fail to consider if they are dealing with language at all (1999: 129). Labov argues along similar lines as Beaugrande by embracing language as a form of social behavior. That is to say, he postulates an inextricable link between language and society, a view different from studying language in isolation.

Labov, for a certain period, was reluctant to use the term *sociolinguistics*. To him, the use of such a term implied the existence of a successful linguistic theory or practice which is not social (1997: 23). As opposed to the generative approach of studying language in isolation, Labov (1972c: 209) maintains that the principal aim of sociolinguistics is to “find out how people talk when they are not being systematically observed.” By definition, sociolinguistics is a “branch of linguistics that studies the ways in which language is integrated with human society” (Crystal 1992: 157), and its main goal is to carry out research by scrutinizing authentic language as it is used in the speech community. The point at issue is, as Labov memorably puts it, that one can only find out how people talk when they are not being systematically observed in systematically observing them (1972c: 209). Labov terms this linguistic-psychological phenomenon the

*Observer's Paradox*. The generative grammarian's approach of studying language in isolation is not directly confronted with this dilemma. Conscious speech monitoring and the resulting biased language, i.e. the *Observer's Paradox*, is part of what generativists call *performance* and is thus only of marginal importance.

Considering the above, the question is whether sociolinguists are at all able to obtain authentic data by recording speech. There are several ways in which one might attempt to tackle this problem. The notion of *authenticity* calls for an explanation and a definition, which, for example, Coupland fundamentally contributes to in "Sociolinguistic authenticities" (2003). Further elaborations on the question of language authenticity are the concepts of authentic speaker and authentic context (cf. Bucholtz 2002 and Eckert 2003), as well as Labov's (1972c) *Attention-to-Speech Continuum* and Bell's (1984) *Audience Design* model, all of which are addressed at a later stage of this chapter. Finally, ways of minimizing and circumventing the *Observer's Paradox* will be outlined.

#### 4.1.1 Sociolinguistic Authenticities

Authenticity of data is not only fundamental to linguistics, but to various other domains. Coupland (2003: 417) maintains that authenticity is more valuable than *inauthenticity*: "[W]e dislike people who dissimulate or are 'above themselves'; [W]e dislike tourist venues that vapidly promise to offer 'the authentic Welsh cultural experience'" (2003: 417). What this case shows is proof of the existence of varied forms of authenticity, of which authenticity in language constitutes only one case in point. With regard to language, according to Coupland (2001b: 415), there are six sociolinguistic authenticities, as seen in Table 4.1.

|                              |                                                 |
|------------------------------|-------------------------------------------------|
| <i>Authentic language 1:</i> | attested and attestable language                |
| <i>Authentic language 2:</i> | naturally occurring language                    |
| <i>Authentic language 3:</i> | language encoding fact and truth                |
| <i>Authentic language 4:</i> | fully owned, unmediated language                |
| <i>Authentic language 5:</i> | language indexing personal authenticity         |
| <i>Authentic language 6:</i> | language indexing authentic cultural membership |

**Table 4.1:** A typology of sociolinguistic authenticities (adapted from Coupland 2001b: 415).

Authentic language 2, i.e. naturally occurring language, is highly relevant to this section and will be turned to following a brief presentation of the various other forms of sociolinguistic authenticities. For decades, authenticities 1 and 2 have been significant empirical criteria in linguistics (Coupland 2003: 421). Authenticity 1 applies to the degree of verification of a linguistic phenomenon in reality. Authenticity in language 3, i.e. language encoding fact and truth, is

reminiscent of Grice's maxim of quality (1975: 45) within his suggested framework of the co-operative principle: "Maxims of quality: Supermaxim: Try to make your contribution one that is true 1. Do not say what you believe to be false 2. Do not say that for which you lack adequate evidence." Put differently, an interlocutor either produces language that encodes fact and truth or he postulates claims that he can endorse with sufficient evidence. Coupland's definition of authentic language 4 assumes that the speakers demonstrate full responsibility for the forms and meanings of their own utterances (2003: 417). 5 and 6, language indexing personal authenticity and language indexing authentic cultural membership, are closely interconnected. 5 alludes to the type of authentic language used to express individual authenticity,<sup>10</sup> i.e. the person must not dissimulate, while 6 corresponds to the community-level correlate of 5 (ibid: 417).

Naturally occurring language, i.e. authentic language 2, is an ideological construct, fundamental to both speakers and linguists (Eckert 2003: 392). Yet to this point it has not been discussed how naturally occurring language is defined, and further how it can be accessed. As for the former, the following sections 4.1.2 *Authentic Speaker* and 4.1.3 *Authentic Context* attempt to address the definition of real language. As for the latter, how to gain access to naturally occurring language, a detailed discussion will follow in 4.1.4 *The Observer's Paradox*. One can, however, anticipate the difficulties of this endeavour, as data from observations of the vernacular have been the most difficult for linguists to acquire (Labov 1972d: 97).

#### 4.1.2 Authentic Speaker

Bucholtz understands real language as "language produced in authentic contexts by authentic speakers" (Bucholtz 2003: 398).<sup>11</sup> Yet what are authentic contexts and authentic speakers? Eckert defines the authentic speaker as a "spontaneous speaker of pure vernacular" (2003: 392), whom Labov, for example, found in a street kid in the inner city (1972b). Authentic speakers can only be authentic if situated in authentic contexts, however. Investigating the speech of a villager who travels to the city, or a working class man who aspires to be middle class, for example, is an endeavor of examining speech produced by authentic speakers who are subject to unnatural social influences, which bears comparison to inauthentic contexts. As will be argued in the next section, there are several approaches towards authentic contexts, two of which will be examined in further detail.

<sup>10</sup> For an example cf. Coupland's article "Dialect stylization in radio talk" (2001a).

<sup>11</sup> Henceforth, *real language* refers to Coupland's (2001b: 415) definition of authentic language 2, i.e. naturally occurring language.

### 4.1.3 Authentic Context

Labov (1972c: 209) contends that systematic observation of a speaker always implies a formal context. As a result, the speaker monitors his speech more than in unobserved situations. Labov assesses this problem by conceptualizing style shifting on a single continuum that ranges from careful to causal speech (1972c: 211); this continuum is frequently referred to as the *Attention-to-Speech Continuum*. A classic illustration often cited is that in a sociolinguistic interview situation questions regarding the topic of speech itself in all probability evoke careful speech, while texts that read well and focus on adolescent themes yield much less formal speech. Labov further adds that lists of isolated words, as deployed by acoustic phoneticians in soundproof booths, are prone to induce speech monitoring (1972c: 211), an assertion highly germane to the present study's methodology.

By refining Erving Goffman's "participant framework" (1981), Alan Bell (1984: 197), in contrast to Labov's line of argument, persuasively stated that "[a]t all levels of language variability, people are responding primarily to other people. Speakers are designing their style for their audience."<sup>12</sup> Bell (1984: 159) argues that people involved in the discourse should be ranked according to whether or not they are known, ratified, or directly addressed by the speaker.<sup>13</sup> He proposes five participant roles: speaker, addressee, auditor, overhearer, and eavesdropper, whereby audience effect decreases with the physical distance from the speaker. Thus, auditor effect, for example, is usually lower than addressee effect due to the larger physical distance between speaker and auditor. The key point to note is that fieldworkers can participate in any of these audience roles, which entails that there is fluctuating bias in observed speech, depending on what audience role the observer occupies.

There are, of course, other approaches towards authentic contexts besides Labov's and Bell's. Nessa Wolfson (1997: 124), for instance, equates natural speech with appropriate speech. Wolfson believes that speech is conventionally perceived as appropriate in a certain context, it can be treated as natural speech in that particular situation regardless of whether or not the context is formal or informal. Unnatural speech, according to Wolfson, only emerges if norms of speaking are uncertain or violated. Consider her understanding of *natural speech*:

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<sup>12</sup> According to Wertheim, Labov's (1972c) *Attention-to-Speech Continuum* has generally fallen into disuse, while Bell's (1984) audience design has been held superior (2003: 511).

<sup>13</sup> The term *ratified*, in Bell's understanding, means "acknowledged but not necessarily addressed" (1984: 159).

Once ‘natural speech’ is recognized to be nothing more than speech appropriate to the occasion, it can cease to occupy the role of an elusive entity continually being sought by investigators in interviews; natural speech is all around us. (Wolfson 1997: 125)

It would be a relief if the complex and complicated search for naturally occurring language could be solved simply by adopting Wolfson’s viewpoint. A fundamental objection to her argument is that observation and research methods, some more than others, influence the speech produced and thus impinge on one of the prime goals of sociolinguistics, namely to explore authentic language in authentic contexts. This stance is extended by Bucholtz who asserts that “if our goal is truly to collect and analyze authentic speech, then we are doomed to failure – or to extremely unethical research practices that no sociolinguist would endorse” (2003: 406). Indeed, it is plausible to argue that without resorting to surreptitious recording, collecting authentic speech is hardly achievable. Yet there are ways of at least minimizing the obstructing effects of the *Observer’s Paradox* and evoking natural language in natural settings, albeit not to the full extent.

Wolfson’s (1997), Labov’s (1972c), and Bell’s (1984) views on authenticity have revealed how authenticity underwrites almost every aspect of sociolinguistics. Although authenticity in researched language as well as methods of gaining access to authentic language are pivotal concepts in this field, they are rarely addressed as topics of investigation in their own right (Bucholtz 2003: 398). For this reason, the following section is dedicated to a brief critical discussion of elementary methodological principles for acquiring authentic linguistic data. Particular attention is paid to the *Observer’s Paradox*’s stern implications for sociolinguistic research, including the results offered in the present study.

#### **4.1.4 The Observer’s Paradox**

As means of gathering and analyzing data, audio and video recordings were introduced in the 1940s and 1950s. Meanwhile, they have come to be widely recognized in various domains of the social sciences (Hutchby and Speer 2003: 315). There are obvious benefits in tape recording speech. Hutchby and Speer (ibid: 316) claim that tape recording of speech is less prone to interpretive filtering than taking field notes, which is thought of as subjective documentation of what had occurred and been said. This line of argument was further developed by Harvey Sacks. He began working with tape-recorded conversations because they can be analyzed and replayed *ad infinitum* (Sacks 1984: 26). Sacks noticed an additional advantage of tape-recorded conversations in being able to offer a play-back of a studied piece of conversation to further interested parties; this would permit the subsequent sharing of opinions.

Despite this apparent optimism, other researchers saw distinct disadvantages in recording language with tape and video recorders (Hutchby and

Speer 2003: 316). Scholars started to comprehend the effects of the researcher and his research instruments on the objects under scrutiny. Accordingly, there was an abrupt increase in the number of terms that denote the same phenomenon as Labov's (1972c: 209) *Observer's Paradox: researcher effect, reactivity, reactive effects, context effects, observer effects, and one-way mirror dilemma*, to name just a few. The understanding of the detrimental effects on the authenticity of the data led, and continues to lead, to strategies to circumvent the *Observer's Paradox*. In linguistics, there are several well-established methods of avoiding, or at least diminishing, the effects of the *Observer's Paradox*. Some of these methods will be considered in further detail below.

#### **4.1.4.1 The Sociolinguistic Interview**

The one-to-one tape-recorded interview is regarded as the classic method for sociolinguistic research. Several advantageous aspects of tape-recording have already been alluded to. To illustrate the advantage of sociolinguistic interviews, Labov (2005) refers to the *Principle of Accountability*, i.e. the principle that "reports of the occurrences of a variant must be accompanied by reports of all non-occurrences" (Labov 2005: 2-3). Successive quantitative analyses elicit whether or not an occurrence is happenstance or systematically patterned (Feagin 2002: 26). The palpable drawback of one-to-one tape-recorded interviews is the formality they create. Accordingly, the naturalness of the language observed is equivocal.

So as to overcome the unnaturalness in language, Labov (1972c: 209) advocates the diversion of an interviewee's attention from speech in order to elicit his true vernacular. In support of his argument, Labov adds that questions of the type "Have you ever been in a situation where you were in serious danger of being killed?" involve the subject in a different context and recreate strong emotions which allow the vernacular to emerge. Undoubtedly, there is a degree of truth to this view. At the same time, it could be argued that this strategy is likely to cause, but is not a guarantee for, a shift in style from careful to less monitored speech.

#### **4.1.4.2 Participant Observation**

Participant observation serves as a complementary method for data collection (Feagin 2002: 33). In *Participant Observation: A Guide for Fieldworkers* (2002), Kathleen and Billie Dewalt give the following definition:

For anthropologists and social scientists, participant observation is a method in which a researcher takes part in the daily activities, rituals, interactions, and events of a group of people as one of the means of learning the explicit and tacit aspects of their life routines and their culture. (Dewalt and Dewalt 2002: 1)

By *explicit*, Dewalt and Dewalt understand what people are able to articulate about themselves, where *tacit* entails those aspects of culture that largely remain outside our awareness (2002: 1). As is perspicuous from this definition, participant observation is not limited to the study of language; rather, it is a methodology “for studies of almost every aspect of human existence” (Jorgensen 1989: 12). In translating this definition to the field of linguistics, however, it essentially means observing and noting down linguistic behavior. One of the prime advantages of participant observation is that the researcher transforms into an insider of the group under study. Consequently, the effects of the *Observer’s Paradox* are greatly lessened as the *Observer* has vanished, and authentic contexts have been “created” in which speakers will produce authentic language.

Such optimistic facets notwithstanding, there is need for qualification to this methodology. In as much as the capacity of humans taking notes is restricted, one can infer that the researcher is not able to note every token of interest during the course of the observation process. Secondly, before writing down the desired tokens, the researcher selects them pursuant to his subjective interest, thus coloring the collected data with researcher bias. This, in turn, exacerbates data quantification, which would allow conclusions as to the relative frequency of a variable (Feagin 2002: 34).

#### **4.1.4.3 Rapid and Anonymous Observations**

This method embeds the desired variables in the answers to a question that is posed to strangers (Feagin 2002: 34). The origins of this method can be traced back to Labov (1966), who first applied this technique in an eminent study in New York City. Labov’s group of researchers confronted employees of numerous department stores with the question “Where can I find women’s shoes?” anticipating the answer “on the fourth floor.” In observing the use of rhotacism in the words *fourth* and *floor* as an indicator for social class, Labov drew vital conclusions: While upper-middle-class New Yorkers articulated the target words with r-ful pronunciations, those workers in less prestigious department stores used an r-less pronunciation, i.e. working-class New York dialect (Feagin 2002: 35). Needless to say this method unmistakably evokes authentic speech in authentic contexts. Yet the most common critique made by authors has been that ethnographic knowledge about the speakers, which is typically needed to generate or confirm research-specific hypotheses, is largely ignored.

#### **4.1.4.4 Telephone Surveys**

Two of the chief advantages of telephone surveys is that they can cover a vast territory and, in sharp contrast to other sociolinguistic methods, linguistic information can be gathered from large samples without much effort. Telephone

surveys consist of an interviewer who follows a prepared script. In case the research group's aim is geared towards the study of a specific speech group's phonological features, such as in Labov's TELSUR project,<sup>14</sup> pre-designed scripts typically include word lists or texts. The collected phonological data are evaluated by acoustic analyses, followed by the plotting of the studied features on maps (Feagin 2002: 36).

With respect to the presence of the *Observer's Paradox*, telephone surveys parallel the observation context outlined in sociolinguistic interviews.<sup>15</sup> When asked to talk to a scientist on the telephone, an artificial, inauthentic context is imposed onto the informants, within which the degree of authenticity of informant-produced language is remote. Depending on what task the informant is to perform on the telephone, authenticity in language may decline even further. Reading off carefully scripted word lists and texts, for example, is an unnatural act of linguistic performance for most informants.

#### 4.1.4.5 Word Lists, Reading Passages, and Minimal Pairs

Whether or not researchers adopt word lists, reading passages or lists of minimal pairs is contingent upon what the research is geared at. As previously mentioned, it makes sense for phonology-oriented research to use carefully scripted texts and word lists because every subject articulates the same words and reads the same texts (Feagin 2002: 31). Yet there is a need not to underestimate the lack of authenticity in speech that is produced when asking an informant to read word lists, a point that Feagin overlooked.

In acoustic phonetics, the informant is often recorded in soundproof booths where he is asked to perform specific reading tasks. In case informants are recorded under such "ideal" conditions, their speech "has the properties of formal, elicited speech we tried to avoid" (Labov 1972c: 190). Nevertheless, Labov (211) recommends two strategies for minimizing speech monitoring in acoustic phonetic investigations: As regards the reading of texts, he suggests if the text reads well, mainly evolves around the vernacular, and focuses on adolescent themes, the informants' speech will probably be less monitored. As for word lists, Labov points at the embedding of minimal pairs within the text.<sup>16</sup> Thereby, it is likely that the speaker will not notice the minimal pair contrasts. It is quite a sensible proposition to embed minimal pairs within the text. Yet in reality, it needs to be emphasized that studies in acoustic phonetics, such as this one, must permit the occurrence of the target words in identical phonological environments, to bypass unfavorable consonant or vowel gestures of neighboring words onto the target features. Apart from the hitherto outlined

<sup>14</sup> Telephone Survey of North America.

<sup>15</sup> Cf. 4.1.4.1 *The Sociolinguistic Interview*.

<sup>16</sup> Labov (1972c: 211) here uses *isolated words* in place of *word lists*.

approaches, there are other methodological means to cope with the obstructing effects of the observer's presence on the observee.

Hutchby and Speer (2003: 315), for example, suggest “turn[ing] participants’ observable orientations to the presence and relevance of recording devices into an analytic topic,” as a means of rerouting the participants’ awareness from the presence of recording devices. There are, however, a number of difficulties with this approach. Both Hammersley (2003: 339) and Labov (1972c: 214) conclude that the researchers nevertheless face the same intractable methodological dilemma. In addition, if this strategy were applied, Labov (1972c: 214) persuasively argues that the subject may deliberately choose not to resort to the vernacular, as it might be equivalent to the standard. This may induce him to “produce stereotyped forms which are simply a collection of the ‘most different’ or ‘worst’ sentence types.”

#### 4.1.5 Preliminary Conclusions and Discussion

Events show that each presented approach towards coping with the *Observer's Paradox* turns out to be somewhat defective.<sup>17</sup> In relation to this given fact, consider how Guy Bailey et al. begin their article “The effects of methods on results in dialectology” (1997). In the first paragraph they state that although dialectology was among the first disciplines to apply surveys, dialectologists have provided little information as to what degree their survey methods have had an effect on their results. To say that this is only the case in dialectology is misleading, as this seems true for most other subfields of sociolinguistics. This raises one crucial question: How can we explain that the effects of methods on results in sociolinguistics are rarely elaborated in the way Bailey et al. (1997) do?

Essentially, this is an issue of the willingness to conduct research. Wanting to do research in the field of sociolinguistics, the researcher has no choice but to adhere to the conventional default methodologies due to ethical constraints. In the case of acoustic phonetics, for instance, either the researcher can opt for a recording in soundproof booths, or he can record the subjects in more natural, authentic contexts. In the former case, he is able to filter out disturbing background noise, in the latter case, weak recording quality, and more importantly, unserviceable data would be the corollary. It is understood that he opts for an optimal high-quality sound signal, sacrificing authenticity of speech in favor of optimal recording conditions. This is not to say, however, that the researcher should not declare this sacrifice – on the contrary. It would be sensible to do so in the research report.

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<sup>17</sup> Trivial as it may seem, this is understood within the boundaries of ethically endorsed procedures. Surreptitious recordings are considered a violation of the subjects’ privacy and, apart from that, repeatedly grant poor quality recordings (Feagin 2002: 33).

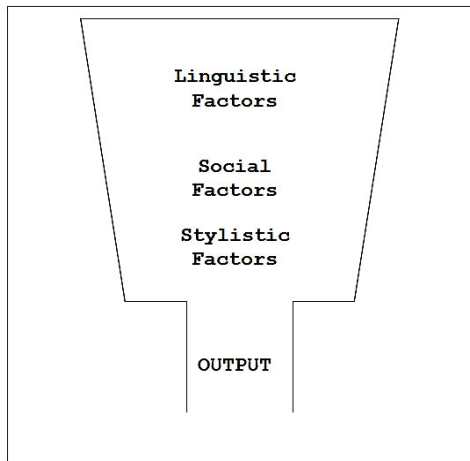
It might well be the case that scholars assume a reader of their text to be in the know about collecting procedures and the supposedly self-evident intricacies of them. However, this does not allow that within the text an author jumps to conclusions without explicitly declaring that the results should be enjoyed with a pinch of salt, as most of them are gathered from inauthentic, observer-biased data. Yet the explanation that a researcher views the effects of the *Observer's Paradox* as self-evident seems too simplistic. It tends to overlook the likeliness of a willful neglect of the *Observer's Paradox's* effects, in order to save face when presenting one's own results. Admitting that one's own research results have to be understood with some precaution entails communicating their disputed credibility. It follows that, as a means of saving face, the scholar only mentions the effects of the *Observer's Paradox* on his own data in the margin, so as to present one coherent, unbiased set of data. This discussion of authenticity, or lack thereof, in researched language, sheds light on only one of the issues that exacerbates work in sociolinguistics.

## 4.2 Language Variation

All too often one tends to overlook that the numbers reported in summary form in a sociolinguistic textbook or journal stand for the product of a long procedure of analysis (Milroy and Gordon 2001: 136). The fact that virtually every data set in sociolinguistic studies contains variation is habitually deemed too trivial to communicate to the audience. Obviously, there must be a reason to avoid commenting on language variation and its effects on sociolinguistic research. Analogous to the issue of authenticity in observed speech, language variation is one of the central problems in linguistics.

Variation in language is largely due to social and stylistic factors. Such variables are frequently referred to as sociolinguistic variables or sociolinguistic variants (Crystal 1992: 408). In the opening lines of "The status of sociological models and categories in explaining language variation" (1996), Suzanne Romaine introduces a prime question on the topic of linguistic variation: "What does it mean to 'explain linguistic variation'?" (1996: 99). There are several approaches towards explaining and coping with language variation. Preston (2001), for instance, has been among the supporters of the view that three distinct factors influence language variation: linguistic factors, social factors, and stylistic factors. This view, represented in Figure 4.1., continues to be influential.

In the "funnel diagram," Preston hierarchically arranges the factors characterizing language variation (2001: 279). According to him, stylistic variation is dependent on the interlocutors' social status, which, in turn, is determined by linguistic factors.



**Figure 4.1:** “Strength” of factors influencing variation (adapted from Preston (2001: 280)).

While Preston’s approach views language as part of social psychology, the generative paradigm, as stated in *4.1 Authenticity in Researched Language*, understands language as a branch of cognitive psychology or biology. Accordingly, the generative approach to language variation is different from the one just outlined. Theoretical linguists are essentially concerned with studying systems of language, which they term *grammar*. Since what Chomsky refers to as *performance*, which would attend to the concept of language variation, is not of primary importance, language variation is random and therefore not worth studying. In attempting to still be able to use variant collected data wisely, generative grammarians impose a hypothetical filter on language data so as to make them invariant. This is done by what is often referred to as the *axiom of categoricity* (Chambers 2002: 8). The following subchapter addresses the generative paradigm’s stance towards language variation in further depth, followed by a brief discussion of the variationist paradigm’s take on language variation. The section ends with potential shortcomings of the two paradigms and preliminary conclusions.

#### 4.2.1 The Generative Paradigm and Language Variation

The study of language *in abstracto* began with Ferdinand de Saussure in the early 20<sup>th</sup> century and gained new impetus from Chomsky in the 1950s. Although Chomsky’s dichotomy of *competence* and *performance* is occasionally equated with Saussure’s postulation of *langue* and *parole*, Chomsky understands Saussure’s conception of *langue* in a somewhat limited way (Labov 1972c: 186). Nonetheless, he acknowledges the similarities between the two binarisms in that “the description of intrinsic competence provided by the grammar is not to be confused with an account of actual performance, as de Saussure emphasized with such lucidity” (Chomsky 1964: 915). As noted in *4.1*

*Authenticity in Researched Language*, Chomsky believes that the study of linguistics means studying an abstract, homogeneous speech community. Typical of this approach is the objective of studying the intuitions a speaker has about language, i.e. which sentences he judges to be grammatical or ungrammatical, rather than studying genuine utterances produced by the individual (Labov 1972c: 186).

The essence of the generativists' argument is fourfold: everyday spoken language is often ungrammatical, there is a limited ability of hearing and recording language, the rarity of syntactic forms, and language variation (Labov 1972c: 189-190). Speech is full of ungrammatical linguistic elements, such as poorly formed sentences, mispronunciations, slips of the tongue, etc., which make speech useless for scientific evidence. Furthermore, recordings are regularly affected by unfavorable background noise that interferes with the sound signal. As emphasized in *4.1.5 Preliminary Conclusions and Discussion*, this can be helped by setting up more ideal recording conditions, which in turn induces speech monitoring on the part of the speakers. For generativists, the existence of this dilemma is an additional motive to study intuitions about language rather than articulated language. They further maintain that an average corpus of linguistic data is insufficient to analyze patterns of language. Most decisively, however, it is language variation that generativists deem an obstacle to the study of articulated speech.

In effect, one could argue that formal theory views variation in language as being outside the scope of the systematic study of language. Dittmar remarks that structuralists "sought to discover the immanent laws of linguistic structures, neglecting their variation" (1976: 112).<sup>18</sup> A linguistic element is analyzed in the context of the total of linguistic tokens it can occur with, largely ignoring the meaning of the linguistic elements. This method, often referred to as *distributional analysis*, serves as one of the prime methods of structuralism. Distributional analysis is most often applied in studying phonological patterns. Structuralism distinguishes between three types of distribution: contrastive and complementary distribution, and what is referred to as free variation (cf. Dittmar 1976: 113).

Contrastive distribution stands for the allocation of linguistic elements, often sounds, in such a way that "the informant responds to this minimal opposition of the two phonemes in the same element," whereas complementary distribution stands for the mutual exclusiveness of linguistic elements. That is to say, one linguistic element occurs in a context where the other does not emerge. Free variation, finally, addresses "linguistic elements that can appear in the same linguistic environment without any particular response from the informant" (ibid: 113). In terms of phonology, for example, free variation refers to the fact that sounds do not contrast or occur in identical phonetic environments

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<sup>18</sup> Emphasis added.

(O'Grady, Dobrovolsky, and Katamba 1996: 714). Whereas generativists largely understand language variation as arbitrary, variationist sociolinguistics rests upon the rudimentary observation that variants occurring in everyday speech do not occur randomly (Bayley 2002: 117).

#### 4.2.2 The Variationist Paradigm and Language Variation

While theoretical linguists draw on idealized versions of homogeneous languages, variationist linguists firmly postulate that language variation and change should be at the center of a valid linguistic theory (Schilling-Estes 2002: 203). One of their principal aims is to “grasp language as it is used in social situations, which is to say variant, continuous and quantitative” (Chambers 2002: 8). Consequently, variationists detect numerous flaws in the structuralists' arguments against studying language as it is used in social contexts.<sup>19</sup>

With regard to ungrammaticality in speech, Labov (1972c: 203) understands this as an unsubstantiated claim. In investigating several empirical studies, he concluded that around 75 percent of the corpora contain well-formed sentences. He accredits the falsehood of the claim that speech is unusable as scientific evidence to two sources. On the one hand, the data which allegedly authenticate the claim are likely to have been retrieved from talks and conferences where academics tried to articulate complex ideas. On the other hand, he finds an even more fundamental objection to the generativists' argument in attributing the falsehood to “the usual tendency to accept ideas that fit into our frame of reference without noticing the data with which we are surrounded” (ibid: 203).

As to the recording of speech in natural settings and the rarity of grammatical features, Labov delineates the following explanations. He categorizes the recording problem as a technical one. If linguists applied state of the art recording devices, which at the time he authored *Sociolinguistic Patterns* (1972c) were battery-operated tape recorders, they are capable of gaining access to excellent recordings in the field. On the issue that an average corpus of linguistic data is supposedly insufficient to thoroughly explore patterns of language, Labov proposes that a linguist could engage in an ordinary conversation with the subject and “elicit the informant's natural use of a given form without using it himself” (1972c: 205). While generativists view linguistic variation as being outside the system being studied, variationists deem heterogeneity in language as the natural product of internal and external variation.

Factors contributing to internal linguistic variability include markedness and language universals (Romaine 2003: 413). A classic illustration often cited is the loss of final consonants that symbolizes a universal phonetic tendency that occurs in a vast variety of languages. As regards second language research, phonetic, morphological, as well as syntactic studies have established that

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<sup>19</sup> Cf. 4.2.1 *The Generative Paradigm and Language Variation*.

“[internal] variability reflects universal principles of markedness rather than the mere influence of L1 and L2” (Romaine 2003: 413), which, in turn, motivates extensive interaction between creolist and SLA researchers. Scholars of various disciplines such as first and second language acquisition as well as of pidgin and creole studies are currently exploring cross-comparatively if there is a tendency to use less complex and less marked forms, i.e. universal forms, more often (Romaine 2003: 418).

Social categories such as class, gender, age, and ethnicity are often referred to as external factors for language variation (Milroy and Gordon 2001: 93). Apologists for this line of argument assume that not only language internal, but also language external factors determine whether, for example, a speaker drops a final consonant or not. It was concluded, for instance, that high-status speakers delete less often than lower-status people (Romaine 2003: 411). In its early days, variationist sociolinguistics was also termed “correlational sociolinguistics” because the supporters of the paradigm attempt to correlate linguistic variation with language external factors (Trudgill 2002: 373). This correlation ultimately helped to solve the myth of alleged random variation.

#### **4.2.3 Shortcomings of the Variationist and Generative Paradigms**

Although the variationists’ as well as the generativists’ approaches towards language variation have some merit, there are a number of qualifications to be made. Labov ascertains several shortcomings in respect to contrastive and complementary distribution, free variation, and the generativists’ general take on how language should be studied.<sup>20</sup> In analyzing speech, so Labov, there are generally many cases that fit into contrastive and complementary distributional categories. In the case of finding proof for free variation, however, one is required to verify that a speaker does not move from one consistent set of co-occurring rules to another. To do so empirically is a hardly possible venture. Labov highlights further general drawbacks in the abstract study of language. Chomsky imagined the intuitive judgments of native speakers as decisive in nature. That is to say, it was hoped that doubtful judgments as to the grammaticality or ungrammaticality of sentences would occur only infrequently, preferably not at all. Labov’s principal criticism has been that the judgments did not turn out the way they were anticipated, as uncertain cases have remained problematic. Labov continues his line of argument in explaining that if a researcher in favor of studying intuitive judgments is challenged in his data at linguistic meetings, he habitually justifies himself by claiming his arguments to be applicable only for one particular dialect. Unquestionably, there is some truth to Labov’s criticisms. However, it is too much to claim that the variationists’

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<sup>20</sup> The generative paradigm as asserted by Fasold (1996: 223), has dampened the quantitative study of observed language performance for an extensive part of the 20<sup>th</sup> century.

approach towards language variation remains flawless (cf. Labov 1972c: 188-192).

For several decades, work within the quantitative paradigm has contributed extensively to the progress of studying language in society. At a high level of sophistication and abstraction, it demonstrated proof for the systematic nature of the variability between linguistic and social variables. Yet work within the quantitative paradigm has paid little attention to the meaning of linguistic variation (Bayley 2002: 135). This, so it is argued, was the reason why several linguists of the quantitative paradigm, with Penelope Eckert as one of the forerunners, began working in studies that

Not only demonstrate the correlations between the use of linguistic variables and social categories, but also show how speakers deploy their linguistic resources, along with other symbolic resources, to construct and reinforce the social categories to which they belong. (Bayley 2002: 135)

Eckert believes that generalizations about gendered language use, for example, can only be made by virtue of a thorough understanding of social practices and social relations of the community members – a fundamental objection to the technique of formulating assumptions about gendered language practices solely on the basis of properties women and men may have (1999: 198). This recently established branch of the variationist paradigm is often referred to as variationist ethnography. Prior to variationist ethnographic studies, longitudinal studies started to gain grounds in the early 1990s.

Sankoff observed that longitudinal studies are ideally suited “to study language acquisition (both L1 and L2) in which the goal is to trace the progress or chart the path of learners as they progress toward the target” (2002: 1). This assumption is extended in Sankoff’s current major research project “Language Change Across the Lifespan,” which, as can be derived from its heading, scrutinizes a specific speech group for a definite period of time, with the aim of exploring the meaning of linguistic variation.<sup>21</sup> Despite their optimistic outlooks, there may be certain limitations to ethnographic and longitudinal studies. Observers have noted that, frequently, only small pools of subjects are taken into account, which, in turn, inhibits the representativeness of most conducted ethnographic as well as longitudinal studies.

#### **4.2.4 Preliminary Conclusions**

This chapter initially drew attention to the principal differences between the generative and variationist paradigms, followed by a short discussion of their approaches towards language variation. Inadequacies were identified and addressed in both paradigms. If this study, too, adopts the framework suggested

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<sup>21</sup> Taken from <http://www.ling.upenn.edu/~gillian/Longit.Bib.html> (3.4.2007).

by variationist sociolinguistics, and views language as being a part of social psychology rather than cognitive psychology or biology, it follows that the linguistic, social, and stylistic factors (cf. Preston 2001) causing language variation in the present study call for clarification.<sup>22</sup> In the context of second language research, however, there are a few remaining factors to be taken into consideration.

Variables, such as social class, networks, age, sex, ethnicity, style, etc., are noteworthy in both NS and NNS performance (Romaine 2003: 422). However, social class distinctions are only of limited use in second language research, as interlanguage variants rarely bear social significance for learners. In second language research, other variants are of paramount interest. Factors which are said to reinforce language variation predominantly encompass cognitive as well as aptitude aspects of the language learner, experience with the L2, and a number of individual factors. A thorough discussion of these factors follows in Section B: *Theory underlying the Analysis of 20 EFL Speakers' Vowel Quality*. Although 2.3 *Comparison to Norm* has briefly addressed the grounds on which the concept of Standard English is used in the present study, the following chapter attempts to do so in further detail.

### 4.3 Standard English as a Benchmark for Swiss English

According to the view put forward in 2.3 *Comparison to Norm*, speaking with an accent implies the production of a form of speech that deviates from a given norm, which evokes the issue of correctness in language. Bartsch asserts that perfection of a sound, i.e. phonological correctness

is not an inherent quality, but a social phenomenon: speakers who are accepted as providing the models by their pronunciation determine what the sound has to be like. A sound is perfect, or nearly perfect, if it is like the sounds of the kind produced by the people who give the models. (1987: 5)

With regard to the present study, Standard English as well as Standard American English serve as models for the EFL speakers.

2.3 *Comparison to Norm* also briefly addressed reasons as to why it is precarious to use Standard Englishes as a point of reference, a theme that the following chapter tries to expand on. At the outset, it explores definitions of Standard English, followed by an example of direct consequences of the ideologically charged notion of Standard English. Then, the section offers a discussion of the concept of Standard English from the standpoint of language variation and change. Finally, in light of these issues, sociophonetic approaches towards Standard English will be examined.

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<sup>22</sup> Cf. Figure 4.1 *Strength of factors influencing variation*.

### 4.3.1 Definitions of Standard English

The ways to define Standard English are as diverse as assessments and understandings of the ideological baggage it carries. Only the spoken standard is relevant to the present study, so the outlined definitions are restricted to spoken Standard English only. As regards the ideological baggage, specifications will follow the remarks regarding the definitions of Standard English. As noted in 2.3 *Comparison to Norm*, Lippi-Green maintains that standard language can only be conceived as an abstraction (1997: 53), an abstraction, for which *Merriam-Webster's Collegiate Dictionary* (11th ed. 2003) offers the following definition:

The English that with respect to spelling, grammar, pronunciation, and vocabulary is substantially uniform though not devoid of regional differences, that is well established by usage in the formal and informal speech and writing of the educated, and that is widely recognized as acceptable wherever English is spoken and understood.

There are limits to how far this understanding of Standard English can be taken. Lippi-Green holds that even if this definition does not entirely factor out regional differences, it certainly does not make any mention of social differences (1997: 54). She essentially objects to the subtext of the phrase “[Standard English] is widely recognized as acceptable wherever English is spoken and understood,” from which she infers that English is spoken and understood only by a particular group of people, i.e. the educated class. Consequently, *Merriam-Webster's Collegiate Dictionary* depicts Standard English as the language of the educated (1997: 54). This ordering of the social groups, so Lippi-Green, leaves it open to those with authority to determine the standard.

Lippi-Green's critical approach towards the definition and ideology of Standard English is merely one among many. Language purists, for example, who attempt to guard language from change, present a different conception. Their ideological background, so Neustupný (1989: 221), “may be of an economic nature (defense of markets), it may be ‘political’ (struggle for political independence), or perhaps of a ‘cultural character.” Yet Stalker (1986), who views Standard English as a dialect based on accommodative principles, gives a further perspective towards Standard English. According to him, speakers pursue conventional principles “to account for the demands of social relevance, register relevance, and clarity considerations” (Stalker 1986: 50). Put differently, the use of Standard English is “maximally clear, socially appropriate, and falls in some appropriate register” (Stalker 1986: 56).<sup>23</sup> Trudgill (1999) extends Stalker's affirmation of Standard English as a dialect. He explains that, from a social, intellectual, and cultural perspective, Standard English is the most important dialect in the English speaking world, yet adds the crucial

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<sup>23</sup> Stalker qualifies this argument by adding that the specific choice of lexicon or syntax is relative to a particular audience.

qualification that “[Standard English] does not have an associated accent” (Trudgill 1999: 123).

All these diverging conceptions of Standard English suggest a cautious treatment of the concept. Whenever the term *Standard English* emerges in this paper, the ongoing ideological debate should be borne in mind. If Standard English were accepted noncritically, “we reify ‘standard’ forms and silence the ideological conflicts that have given rise to the notion of ‘standard’,” as Coupland (2000: 632) eloquently puts it. After all, the concept of Standard English and its connotation as the language of the educated can have direct social and cultural consequences. The following is one instance of the implications of Standard English on society and culture.

### **4.3.2 Social and Cultural Consequences of Standard English**

In the early 70s, there was a controversy as to why black children in American inner cities showed poor educational achievement. On these grounds, Labov (1972a) and his associates decided to explore the relationship between language and educational performance of inner-city children in South Central Harlem, New York City. Several psychologists, headed by R.J. Herrnstein, claimed that the black children’s failure in school is attributable to hereditary factors rather than environmental factors. In support of their argument, they paid a great deal of attention to language. Effectively, they speculated that the black children’s means of verbal communication are limited because they receive little verbal stimulation within their speech community, including the lack of well-formed sentences. In reality, according to the psychologists, this is reflected in the fact that black children “cannot speak complete sentences, do not know the names of common objects, cannot form concepts or convey logical thought” (Labov 1972a: 2).

In sharp contrast to the psychologists’ views, Labov reasons differently. He (1972a) highlights two key points in his line of argument. Counter to what the psychologists state, he notes that long-term studies of children in New York reveal remarkably consistent children’s grammars. These consistencies, so Labov, also emerge in other major cities such as Detroit, Chicago, Philadelphia, and Washington D.C. Through thorough research he further concludes that the teachers stigmatized the children’s language as “not a language at all” and “not possessing the means for logical thought” (Labov 1972a: 12). In expanding his argument, he maintains that as soon as the teachers were confronted with copula missing phrases or phrases with negative concord, they were reinforced in their frequently pre-existing prejudice that black children are verbally deprived. As often upheld by social and educational psychologists, the teachers’ attitudes undoubtedly contribute to whether the children succeed or fail in school. The crucial question remains why the teachers perceived Black English Vernacular as a reflection of “the primitive mentality of the savage mind” (ibid: 12).

One way to answer this question is to consider an exclusive conception of Standard English, which holds that Standard English denotes stringent rules and clear-cut boundaries. According to this definition, all other variants of English not conforming to such characteristics are “uniformly classified errors” (Beaugrande 2001: 1). It is appealing to speculate that the teachers, as part of the educated class, may have thought of themselves as speaking Standard English in its exclusive conception, which consequently led them to stigmatize language that does not obey the rules of their own language as “not a language at all” (Labov 1972a: 12). Not only is Standard English a delicate topic from the perspective of language ideology, but also with regard to language variation and change.

### 4.3.3 Language Change and Standard English

More often than not, it is assumed that standard languages are definite, unmodifiable language dialects. It should go without saying, however, that language, including what is referred to as standard language, changes and varies over time. There are ongoing changes in General American and Received Pronunciation at the time of writing. Kretzschmar (1997: 321), for instance, observes that “American English in the 21<sup>st</sup> century will see a new kind of linguistic diversity” and continues that “it still will pose a challenge to the schools and to society at large.” What Kretzschmar is referring to is what Labov describes as current trends in American English sound change. These trends primarily constitute the Northern Cities Shift, the Southern Shift, and Low Back Merger. Trends towards phonological alterations are paralleled in British English.

Wells (1997) asserts that many phonological changes in British English have spread from the south of England. Over the past two decades, so Wells, Estuary English, which incorporates the middle ground between RP and London working-class speech, has given rise to copious developments in RP, as many of the processes of Estuary English are eventually integrated into RP English. Equally essential to such language-internal developments are external factors that can prompt language change. In conducting a survey of pronunciation preferences in British English, Wells (1999: 1) concluded that American has had crucial impact on British English, predominantly on the level of phonology. For example, the researchers observed that *schedule* was pronounced with an Americanized /sk-/word-initially; instead of the traditional voiceless post-alveolar fricative /ʃ-/ . Such trends in modifications of standard language offer a further reason as to why the applicability of the term *Standard English* requires caution. To attain correctness of affirmation in using the term *Standard English*, one should ascertain at what point in time this reference is made, as Standard English is continually subject to change.

Despite such ideological debates regarding the notion of Standard English and apparent drawbacks in trying to pin Standard English down as an unambiguous, unchangeable, language dialect, sociophoneticians did not refrain from researching acoustic properties of General American English and Standard Southern British English.<sup>24</sup> A few of the titles of the studies consulted directly imply the believed existence of Standard English as a definite entity, and further color it with the attributes of being located easily geographically and immune to change. What is more, the varied definitions of Standard English are addressed in none of the articles consulted. Hence, as pointed out in *4.3.1 Definitions of Standard English*, all considered articles noncritically “reify ‘standard’ forms and silence the ideological conflicts that have given rise to the notion of ‘standard’” (Coupland 2000: 632).

#### **4.3.4 Sociophonetic Approaches towards Standard English**

This section intends to shed a critical light on how sociophonetic studies, together with the two studies that function as points of reference for the measurements in the present study, address the notion of Standard English. The titles of the selected studies are as follows: “Acoustic characteristics of American English vowels” (Hillenbrand et al. 1995), “The formants of monophthong vowels in Standard Southern British English pronunciation” (Deterding: 1997), “A study of the formants of the pure vowels of British English” (Wells 1962), “Control methods used in a study of the vowels” (Peterson and Barney 1952), and “Accent in L2 – An acoustic analysis of the vowels of German speakers of English” (Liebthal 2004). With the exception of the final two studies, one can infer from the titles that their authors presuppose the findings as a “set of static formant frequencies” (Hagiwara 1997: 655). Moreover, in an article incisively criticizing Hillenbrand et al.’s (1995) line of argument, Hagiwara (ibid: 655) declares the rationale behind such studies to “[serve] as a comparator with other languages and pathological speech, target values in speech synthesis, prototype values, etc.” The following presents a discussion of how the previously mentioned scholars addressed the concept of Standard English, with a particular focus on the studies that provide the reference formant values for the present study.<sup>25</sup>

##### **4.3.4.1 Hillenbrand et al.’s (1995) Study on American English Vowels**

The foremost purpose of Hillenbrand et al.’s study “Acoustic characteristics of American English vowels” (1995) was to replicate and further extend the study of vowel acoustics by Peterson and Barney (1952), a study that attempted to

<sup>24</sup> Cf. *4.3.1 Definitions of Standard English* and *4.3.2 Social and Cultural Consequences of Standard English*.

<sup>25</sup> The two studies that provide reference points to the present study are Hillenbrand et al. (1995) and Deterding (1997).

explore acoustic characteristics of American English vowels. Although Hillenbrand et al. generally agreed with Peterson and Barney's research design, they note that Peterson and Barney did not extensively screen the subjects for dialect. To compensate for this limitation, Hillenbrand et al. used an extensive screening procedure and eventually wound up selecting 139 subjects from a large pool of potential subjects. The subjects were primarily screened for the same geographical origins, which turned out to be Michigan's Lower Peninsula. If for some reason, subjects of a specific age group or gender from that area could not be found, they chose subjects from Illinois, Wisconsin, Minnesota, northern Ohio, or northern Indiana, where General American English is said to be spoken. The research group conducted audio recordings of the subjects reading lists that included 12 English vowels in /hVd/ environments.<sup>26</sup> Each subject read the list three times.

It is certainly hard to see a plausible alternative to how Hillenbrand et al. tackled Standard English. Still, Robert Hagiwara (1997) voices a number of limitations. He fundamentally objects to Hillenbrand et al.'s study in presenting the findings of a study he conducted with subjects from southern California. Hagiwara reveals severe differences between the vowel spaces in Hillenbrand et al.'s study and those of his own study. To Hagiwara, these findings exhibit the existence of "considerable variation between contemporaneous regional variants of American English" (1997: 658). He draws the vital conclusion that "in discussions of vowel production, references to 'General American' are not as informative as references to data from *specific* dialects" (1997: 658), a statement that should be kept in mind once the findings of the present study are presented.<sup>27</sup>

#### 4.3.4.2 Deterding's (1997) Study on British English Vowels

In "The formants of monophthong vowels in Standard Southern British English pronunciation" (1997), Deterding examines formants of eleven monophthong vowels of Standard Southern British English pronunciation. The scrutinized data were retrieved from the MARSEC database.<sup>28</sup> The database consists of recorded BBC broadcast monologues from the 1980s, such as news readings and commentaries. Although only modestly, Deterding somewhat acknowledges the intricacies of the notion of Standard English by maintaining that

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<sup>26</sup> The phonetic symbols used in this paper are those of the International Phonetic Association.

<sup>27</sup> Cf. Section C: *Data Analyses*.

<sup>28</sup> The acronym MARSEC stands for *Machine Readable Spoken English Corpus*.

This style of speech might be regarded as a kind of reference speech, in the sense that it is used as a model for pronunciation in many parts of the world, though of course it may differ considerably from the sort of speech that would be uttered in ordinary conversation. (1997: 47)

Additionally, Deterding admits to the existence of variation on the part of the speakers, variation that, according to Roach et al. (1993: 48, in Deterding 1997: 48), is limited, as the dialect of MARSEC speakers is consistently RP, or occasionally near-RP variants. Contrary to Hillenbrand et al.'s vowel measurement in standardized /hVd/ environments, Deterding proceeds differently. Since the recordings are taken from a spoken corpus of speech, the latter investigates vowels of connected speech. While to some researchers this may be viewed as disadvantageous, as the varying neighboring consonants affect the target vowels, Deterding asserts that this methodology "represent[s] somewhat more natural data than the rather artificial vowels derived from specially articulated citation speech" (1997: 54). As in Hillenbrand et al.'s (1995) approach to Standard English, here too, a few shortcomings can be detected.

Although Deterding expands on some of the precautions in conducting studies with Standard English, he still "[hopes] that these measurements can serve as a reference for other researchers, and, since these data come from a standard database, it is also hoped that others will be easily able to monitor their accuracy, and build upon them in further studies" (ibid: 54). This is a statement that brings to mind Hagiwara's (1997) previously outlined critical stance towards studies that examine acoustic properties of Standard Englishes and the authority they claim for themselves. There are many other studies that endeavor to delve into the acoustic properties of Standard English.

Peterson and Barney's "Control methods used in a study of the vowels" (1952), a classic as far as studies scrutinizing Standard English in acoustic phonetics goes, primarily examines methodologies of studying vowels, rather than results from it. In the existing research design, subjects were requested to articulate monosyllabic words consisting of /hVd/ constructions, and Peterson and Barney included 10 English monophthongs. As previously explained in 4.3.4.1, the principal criticism is that Peterson and Barney's line of argument does not provide specifications as to the origins of their subjects. The only generic information given is that the majority of the speakers represented a broad regional sampling of subjects who first and foremost spoke General American English (177). GA is left undefined throughout the course of their study report.

In "A study of the formants of the pure vowels of British English" (1962), Wells explicitly designates the variety of English used by his observed speakers. Wells states the entire group spoke with what is commonly acknowledged as "speaking perfect English," or "speaking without an accent" (1962: 12). He

further maintains that “[t]here were no regional characteristics in their speech which might suggest what part of England they came from” (ibid: 12). It is tempting to suggest, however, that there are limits to how far his understanding of the phrases “speaking perfect English” or “no regional characteristics” can be taken. It might be argued that perfect English or speech that does not include regional characteristics is inexistent in reality – an idea Wells obviously avoids commenting on.

To end with, the study of Liebthal (2004), “Accent in L2 – An acoustic analysis of the vowels of German speakers of English”, too, needs to be qualified. Liebthal assesses the effect of English language experience on native German speakers’ production of English vowels. She acoustically analyzed the speech of a non-experienced and an experienced German speaker of English. She then contrasts the obtained formant values to those supplied by Deterding (1997).<sup>29</sup> The first conspicuous deficiency of her study, although hinted at in the results section, is the small explanatory power of her results, largely due to the small sample size. More importantly, Liebthal deliberately circumvents an analysis of almost all English back vowels, which is probably attributable to the back vowels’ often poor formant-tracking representation on spectrograms. It should be further noted that Deterding’s (1997) benchmark data were retrieved from connected speech, while Liebthal, quite the reverse, embeds the target vowels in carrier sentences, such as “I will say X.”<sup>30</sup> Finally, mention should be made that throughout her article Liebthal avoids commenting on possible precautions in using formant values from Standard English.

#### 4.3.5 Preliminary Conclusions

This chapter outlined a few of the facets of Standard English. Various definitions of Standard English and its ideological baggage were presented. A brief digression into Labov’s (1972a) exploration of the real causes for young black children’s poor performance in school introduced a second-hand account of social as well as cultural implications of Standard English. A few comments as to why the notion of Standard English is delicate from the viewpoint of language variation and change preceded a critical discussion of how selected sociophoneticians tackled the concept of Standard English. One may speculate as to why, after such critical discussion, the reference points for the present study are nevertheless retrieved from the previously mentioned studies.<sup>31</sup> In answer to this question, I may reiterate the view presented earlier. I do not subscribe to the conviction that standardized language exists in reality but

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<sup>29</sup> Cf. 4.3.4.2 Deterding’s (1997) *Study on British English Vowels*.

<sup>30</sup> X consists of two syllables, the initial being stressed and X being nuclear accented as well (Liebthal 2004: 33).

<sup>31</sup> Cf. 4.3.4.1 Hillenbrand (1995) *et al.’s Study of American English Vowels* and 4.3.4.2 Deterding’s (1997) *Study of British English Vowels*.

concur with Lippi-Green's view that "[standardized language] certainly does exist as an ideal in the minds of the speakers" (1997: 44). Before entering Section B: *Theory underlying the Analysis of 20 EFL Speakers' Vowel Quality*, one further issue must be outlined, i.e. that of methodological problems of measurement in acoustic phonetics.

#### 4.4 Methodological Problems in Acoustic Phonetics

The quality of a vowel is distinguished by "the position of the tongue and the lips during its articulation, but excluding such features as pitch, loudness, duration, and usually also phonation type" (Trask 1996: 384).<sup>32</sup> Acoustic properties of vowels are best analyzed by measuring vowels' formants, which are defined as concentrations of acoustic energy within a certain frequency band. Formants are represented on sound spectrograms that display the frequencies and intensities of the constituents as a function of time. The three lowest formants represented on the spectrogram, first formant (F1), second formant (F2), and third formant (F3), offer diagnostic information as to the position of the articulators for vowels. In this study, PRAAT (2005), a program geared at speech analysis and synthesis, was used for formant measurements. Before speech can be analyzed, however, sensible recordings need to be conducted. Several methodological as well as technical issues hinder the achievement of homogeneous, invariant sets of data.<sup>33</sup>

This subchapter offers an understanding of some problems of acoustic phonetic analysis. These encompass the disadvantageous effects of background noise, the subjects' voice quality, coarticulatory effects of consonants and vowels, and the segmentation of spectral analyses, among which the last two will be given most attention. All of the following spectrogram segments are adopted from the sound corpus of the present study. Prior to looking into the outlined issues, it is appropriate to briefly sketch how various manners of articulation are characterized on spectrograms, as the following subchapters assume an understanding of them.<sup>34</sup>

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<sup>32</sup> It is not clear whether nasalization and its absence contribute to the concept of vowel quality.

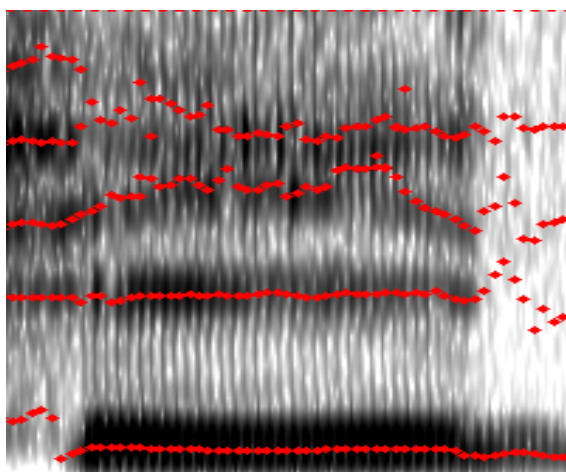
<sup>33</sup> Cf. 4.2 *Language Variation*.

<sup>34</sup> This brief overview of spectral cues relies on Ladefoged's (2003) chapter 5: *Characterizing vowels* and chapter 6: *Acoustic analysis of consonants*, as well as on Ladefoged (1993) chapter 8: *Acoustic phonetics* and chapter 9: *Vowels and Vowel-like Articulations*.

#### 4.4.1 Spectral Cues to Spectrogram Readings

Each sound has a distinctive signature on the spectrogram which leads to varied representations of formant tracks. Formant tracks, shown as dots on the spectrogram, “[Estimate] and [plot] each vocal tract resonance frequency at regular intervals along the signal, producing a continuous red trail through each formant” (Wood 2005), which is why they are frequently referred to as LPC (Linear Predictive Coding) traces.<sup>35</sup> LPC traces are beneficial in so far as hand measurements of formants, often inaccurate in nature, are no longer required.<sup>36</sup> The following is a general overview as to how monophthong vowels, diphthong vowels, approximants, nasals, fricatives, and plosives are represented on spectrograms that feature LPC traces.

**Monophthong vowels:** Monophthong vowels are characterized by heavy vertical striations that correspond to the vibrations of the vocal chords. As opposed to diphthong vowels, monophthongs only present one target vowel position, thus they do not demonstrate shifts on the y-axis of frequency.

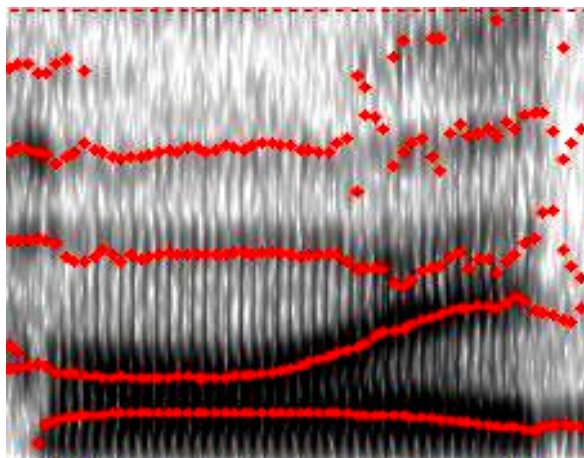


**Figure 4.2:** Spectrogram segment of the monophthong vowel /i:/ as in *heed*.

**Diphthong vowels:** Diphthongs demonstrate strong moving voicing. Unlike monophthongs, diphthongs are typified by a moving tongue; therefore, the formants move from a beginning configuration to a target configuration. Diphthongs can further be distinguished between falling, rising, closing, and opening diphthongs, all of which are accordingly represented differently on spectrograms.

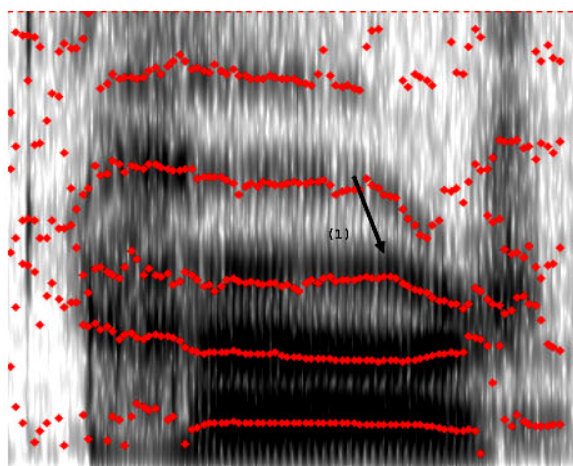
<sup>35</sup> Retrieved from <http://www.ling.lu.se/persons/Sidney/praaate/frames.html>, 3.4.2007.

<sup>36</sup> This is not to say, however, that a measurement of formants by hand is excluded as a whole. Although only on rare occasions, LPC analyses, too, are flawed due to imprecise computer measurements – an instance where hand measurement must be applied.



**Figure 4.3:** Spectrogram segment of the diphthong vowel /ɔɪ/, as in the inexistent word *hoyed*.

**Approximants:** The liquids /ɹ/ (see (1)) and /l/, and the glides /j/ and /w/, are barely represented on spectrograms, as the sound waves that produce these sounds only come across slight obstructions on exiting the vocal tract.<sup>37</sup> A specific characteristic of the phoneme /ɹ/ is that the F3 suddenly rapidly decreases, as seen in Figure 4.4 (1).

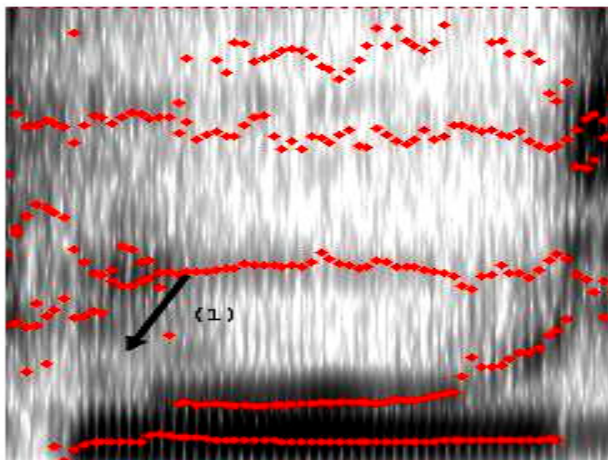


**Figure 4.4:** Spectrogram segment of the approximant /ɹ/, as in *heard*.

**Nasals:** Nasal sounds possess even less energy than approximants. The complete blocking of the oral tract makes the sound waves radiate primarily through the

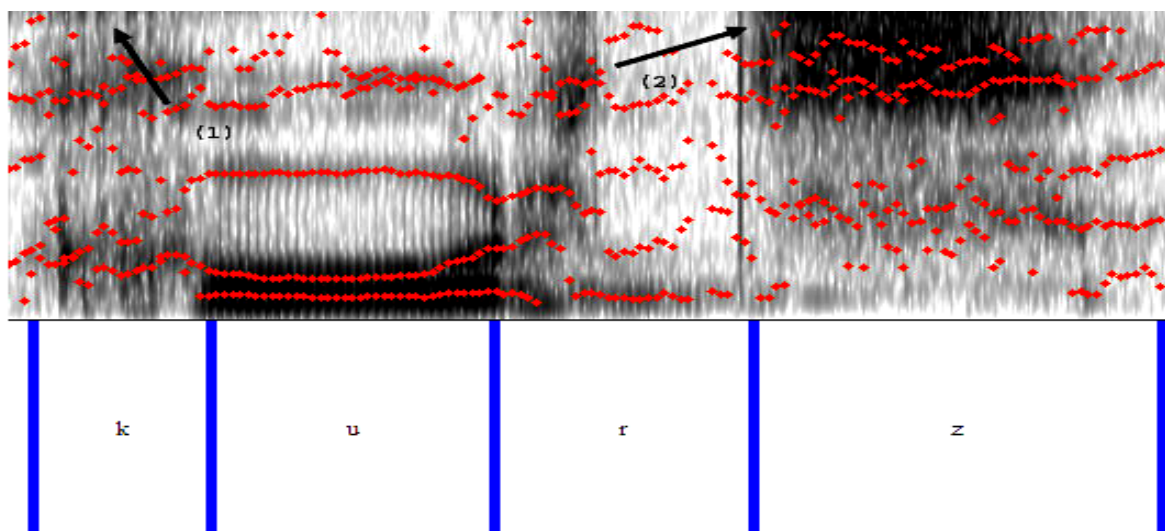
<sup>37</sup> Due to its vowel-like sound quality and its coarticulatory effects on adjoining vowels, there is some disagreement in scholarship if /ɹ/ should be viewed as a vowel or not.

nose. Due to this lack of oral resonance, the spectrogram regions where nasals occur are habitually referred to as “zeroes,” cf. (1).



**Figure 4.5:** Spectrogram segment of the Swiss German word *Maus*, as articulated by a Swiss German speaker.

**Fricatives:** By definition, fricatives involve an occlusion or obstruction in the vocal tract, which is still small enough to let air pass by and cause disturbances in the airflow. See Figure 4.6.

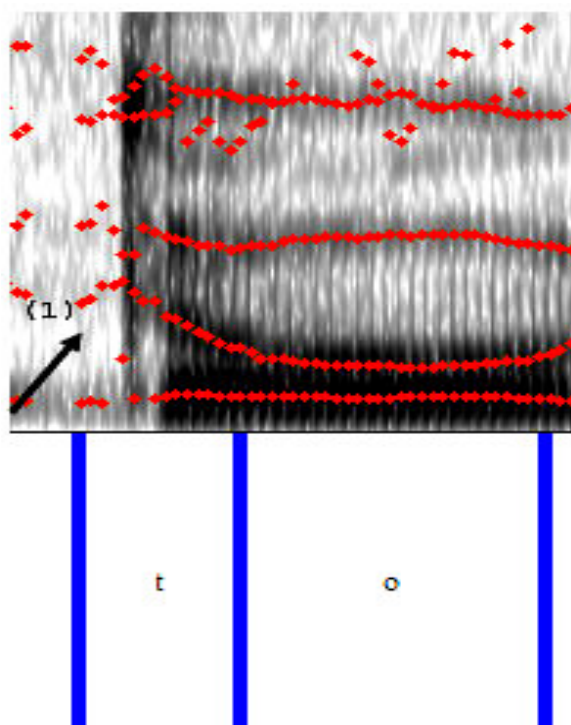


**Figure 4.6:** Spectrogram segment of the Swiss German word *kurz* /χurʦ/. Note the frication of /χ/(1), and /s/(2).

The noise produced is characterized by its frication. Fricatives are aerodynamically complex sounds, which makes them the most intricate to read off spectrograms. On spectrograms, the signature of fricatives is the occurrence

of high-energy areas in very high-frequency regions and their often random energy distribution. The sibilant /s/, for instance, is undoubtedly one of the loudest fricatives, i.e. charged with a comparatively high amount of energy in high-frequency areas; as a result, /s/ is often represented as the darkest part on the top half of the spectrograms, as seen in Figure 4.5 (2). The voiceless velar fricative /x/, in contrast, presents less energy, i.e. is smaller in amplitude, thus the gray-colored representation of it on the spectrogram is fainter than in /s/, cf. Figure 4.6 (1).

**Plosives:** Plosives involve a complete oral closure followed by an explosive burst of acoustic energy. The release is immediately followed by a short period of voicelessness. Therefore, it has been widely argued that the best cue to recognize stops on the spectrograms is the transition from the explosive burst to the following silence.



**Figure 4.7:** Spectrogram segment of the Swiss German word *Ton* as articulated by a SG speaker. Note (1): The release of the stop and the subsequent voice onset time (VOT).<sup>38</sup>

<sup>38</sup> Voice onset time denotes the time between the release of the stop and the voice onset in the following vowel; or as defined by Allen, Miller, and DeSteno “a temporal property that robustly specifies voicing in stop consonants” (2003: 544). In aspirated stops, voice onset time comes perceptibly later than the actual release of the stops, while in unaspirated stops; voice onset time is short and barely perceptible acoustically.

#### 4.4.2 Sound Signal Interference

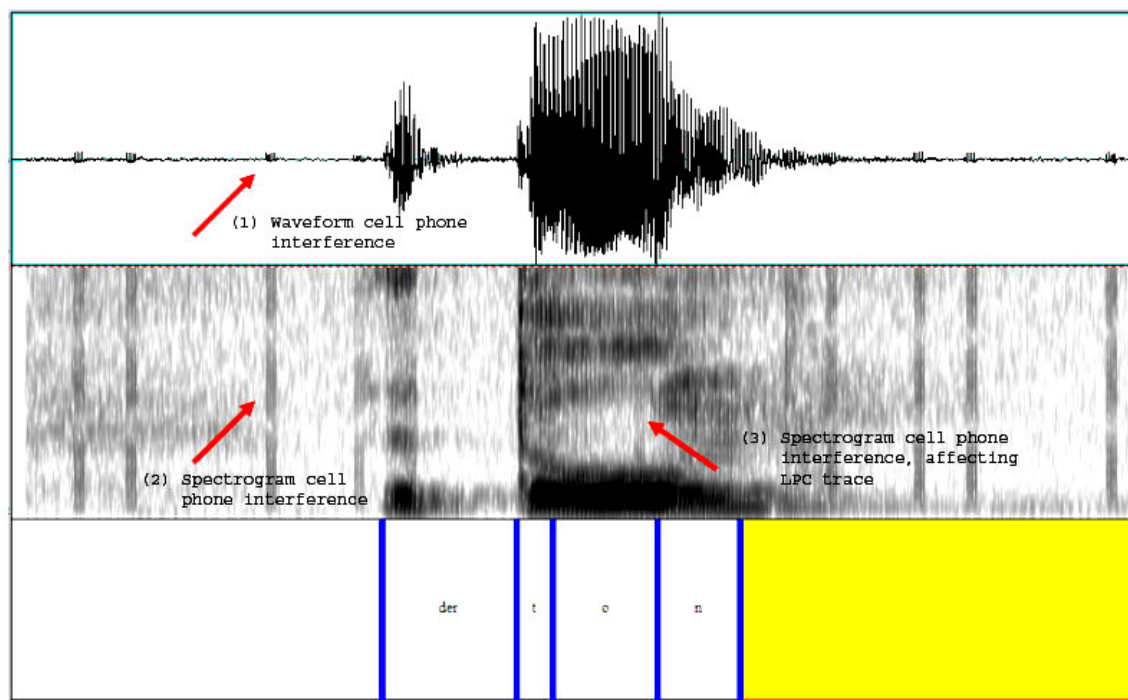
Contemporary technology markets offer a wide array of recording devices, with an increasing number of digital recording formats. For spectral analyses, most researchers prefer digital as opposed to analogue recordings, as spectrograms are fundamentally contingent on a good incoming sound signal. Ladefoged (2003: 16), a principal authority in the field of phonetic data analysis, also, advocates digital audio tape recording (DAT) and CD recorders for high quality recordings despite the easy accessibility of MP3 recorders these days. He further suggests a number of properties one wants from a recording system:

1. A good frequency response
2. A good signal/noise ratio
3. Reliability and user-friendliness
4. The possibility of using the recordings for a long time.

The first property, as Ladefoged adds, refers to the “extent to which [the recorder] faithfully records and reproduces each frequency” (2003: 17). Preferably, the sound signal should enter and exit the recording device with the same loudness, a standard nearly met by DAT recorders. The second property, the signal/noise ratio, refers to the background noise present in virtually all recordings – a property which, in fact, is closely linked to the choice of adequate microphones. As regards Ladefoged’s third criterion, reliability and user-friendliness, the former frequently increases while the latter often declines in the process of today’s technological advancements. DAT recorders, for example, are much more reliable and durable than old-fashioned tape recorders, while tape recorders are often considered more accessible than Digital Audio Tape (DAT) technology. Property 4, to be able to use the recordings for a long time, really addresses how long a specific type of recording will last and how long the system that allows the playback of the data storage device (tape, CD-Rom, DAT-tape) is available. Of these properties, the second, a good signal/noise ratio, needs to be explored in more detail.

It is assumed that the disturbing background noise is the central problem when creating high quality recordings in acoustic phonetics. A recording device’s superior signal/noise ratio notwithstanding, it is equally if not more important to wisely decide on the proper recording location as well as microphone techniques. The wrong choice of microphones or poorly-adjusted microphones as well as incorrect recording settings reinforce noise interferences on the recordings, from which it follows that spectrograms and waveforms are difficult to read and often remain hard to interpret. The disturbing effects of background noise on spectrograms can be portrayed with an incidence of the present study’s corpus.

Even if the sound signal interference in the following instance is not congruent with usual occurrences such as cars driving by or a refrigerators turning on and off, it will be seen at once how cell phones, too, unfavorably affect the sound signal representation on the spectrogram and waveform. See Figure 4.8.



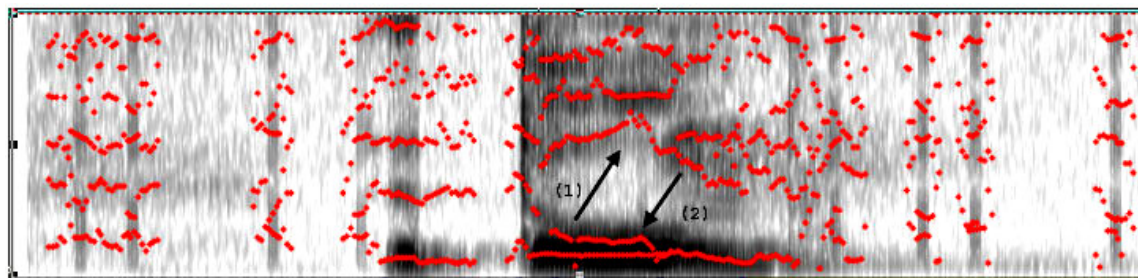
**Figure 4.8:** Cell phone radiofrequency interference as represented on the waveform (1) and spectrogram (2) of a subject articulating the Swiss German noun phrase *der Ton*. (3) points to the unfavorable interference on the target vowel /ɔ:/, see Figure 4.9.

It has been shown that cell phones have the capacity of interfering with electronic devices, as in their radio frequency's interaction with cardiac pacemakers. If cell phones are in close enough distance to the recording devices, the mobile phones' radio frequency disturbs the electromagnetic field of the recording devices. Figures 4.8 and 4.9 exemplify this phenomenon.<sup>39</sup>

In Figure 4.8, (1) points to the sound signal interference as represented on the waveform, while (2) shows the cell phone's effects of radio frequency intrusion on the spectrogram. Moreover, (3) designates the radio frequency's effects on the formant representation. Figure 4.9 shows this interference even more clearly. F1 and F2 augment at the point where the radiofrequency of the cell phone interacts with the magnetic field of the recording device. Before the formant tracks of F2 (2) and F3 (1) progressively decrease, caused by the small

<sup>39</sup> The bottom third of Figure 4.9 represents the text grid as used in PRAAT. Primarily, the text grid permits the labeling of the sounds represented on the spectrogram and waveform.

amount of energy of the adjacent nasal consonant /n/, the F2 and F3 formant tracks are visibly disturbed by the cell phone's interference.<sup>40</sup> (1) Highlights the sudden increase in the formant tracks of F3 in /o:/, while (2) points to the abrupt peak in the formant tracks of the second formant.



**Figure 4.9:** Spectrogram segment of *der Ton*. (1) and (2) point towards the cell phone's radiofrequency effect on the spectrogram's formant representation.

In recordings where the sound signal is affected such as in Figure 4.8, one can opt for hand measurement of the formants in reading off formant values at selected points. However, this imprecise technique is possible only if the sound signal is disturbed punctiformly, i.e. devoid of incessant permeating background noise. In case of a continuous buzzing of neon lights or any other disturbing source of sound, a reassessment of the spectral analysis would more than likely not yield the desired results. At any rate, it is well worthwhile to cautiously study the spectrograms prior to applying LPC analyses. Speech synthesis programs might spot background noises as vocal tract resonances, which leads to biased results due to the inaccurate formant values at the points of interference. Unfavorable background noises are only one obstacle in acoustic phonetic analyses. It should also be noted that the voice quality of speakers can equally exacerbate spectrogram readings.

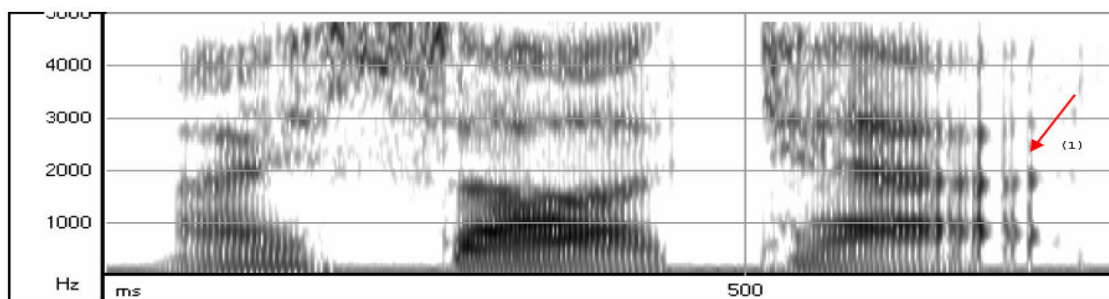
#### 4.4.3 Individual Voice Quality

Baken (1987: 379) has observed that the acoustic spectrum of speech “is the product of the vocal source signal as well as the filter characteristics of the vocal tract.” In analyzing such acoustic spectra, he furthermore states, “spectrum analysis provides a window on laryngeal function as well as on articulatory movements.” Laryngeal functions and articulatory movements are primarily responsible for the differences in voice quality between individual speakers.

There are various types of voice quality. They range from stiff vocal folds, ensuing in “creaky” or “laryngealized” sounds, to “default” vocal fold vibration, to overly relaxed, incomplete closure of the vocal folds, resulting in “breathy” voice (Epstein 2002: 1). In *Voice Quality and Prosody in English*

<sup>40</sup> Cf. Figure 4.5.

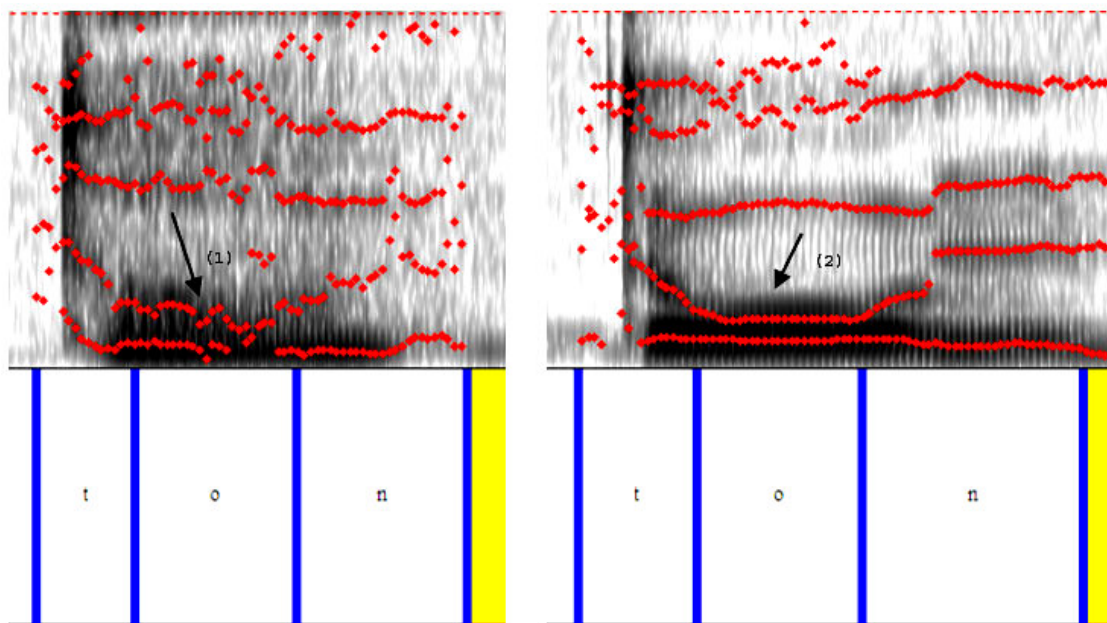
(2002), Epstein brings in an instance of creaky voice, as encountered in the sentence “I saw a cat,” cf. Figure 4.10 (1). She attributes the causes for this creak to linguistic rather than speaker-individual factors.



**Figure 4.10:** Spectrogram segment of the sentence “I saw a cat” (adapted from Epstein 2002: 2). Note the creaky voice at the end of the sentence in the word *cat*, cf. (1).

Epstein’s reasoning for this occurrence is twofold: Generally, complete and/or declarative sentences tend to be creaky in the last part, as they usually end in low pitches that are pronounced with creaky voice. Secondly, she proposes that accentuation in a sentence, as in the word *cat* in the sentence *I saw a cat*, might reinforce glottalization of /t/ in /kæt/ which in turn stimulates creaky voice. Nothing can detract from the central fact that linguistic reasons can at times be responsible for the occurrence of varying voice qualities. Yet it is important to realize that the sound of a person’s voice, according to Laver (1994: 399), is largely dependent on a speaker’s idiosyncratic vocal anatomy.

Although voice quality distinctions between individuals are not as unique as those of fingerprints, they serve as powerful identity markers (Laver 1994: 398). The implications of such varying voice qualities on acoustic phonetics, however, are less intriguing. Distinctiveness and individuality in vocal quality makes consistent spectrogram analyses more difficult. The spectrograms of a speaker with a breathy voice, for example, are very different from those of a speaker with a habitual nasal quality in speech. The corollary of such diverse voice qualities are the different representations of formants on spectrograms. Creaky and breathy voices, as opposed to default voices, for example, often present uneven, wobbly formant tracks, which in turn bias acoustic phonetic measurements. The concurrent spectrogram segments, retrieved from the body of data of the present study, make clear the effects of different voice qualities on LPC traces. Figure 4.11 illustrates the Swiss German word *Ton*, as articulated with a breathy voice (1), juxtaposed to its articulation with a default voice (2). Subsequently, Figure 4.12 demonstrates the Swiss German word *Kopf* with a creaky voice (1), next to its default voice equivalent (2).

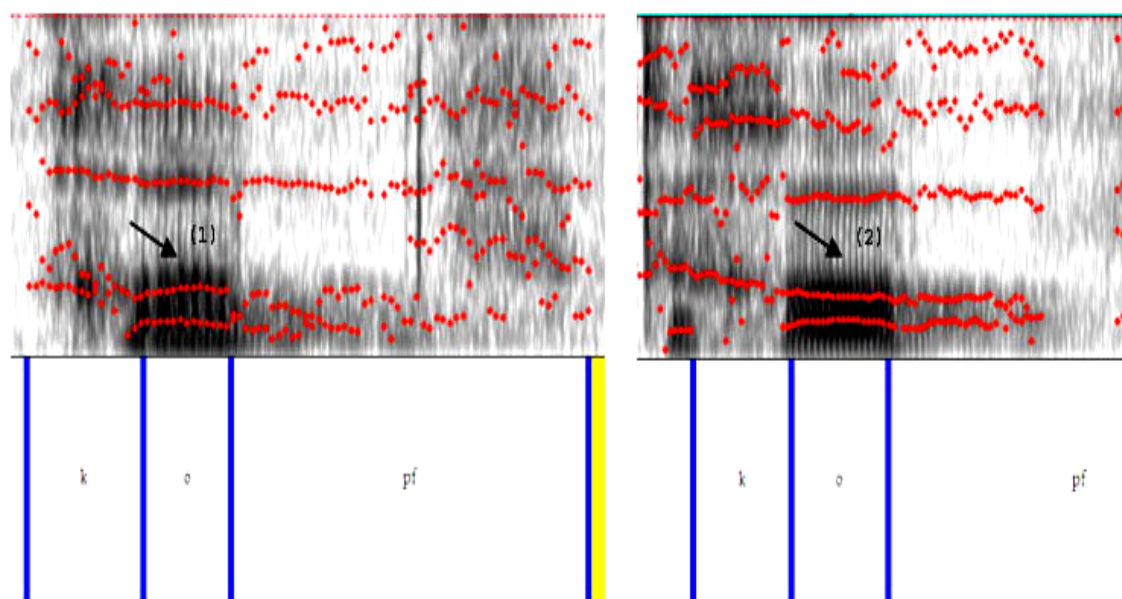


**Figure 4.11:** Spectrogram segments of the Swiss German word *Ton* with a breathy voice (1), and a default voice (2). F1, F2, and F3 are much more stable in default voice, (2), which permits a more consistent and reliable measurement.

Breathy voices emerge if the vocal folds are not fully approximated, which causes a constant flow of air throughout the period of articulation. On the spectrogram, this is represented at the very top of Figure 4.11 (1). The gray shades during the articulation of /o:/ in high frequencies suggests steady frication, as previously explained, which, in turn, interferes with the speech synthesis program's attempt at representing the vowel's LPC traces steadily.<sup>41</sup> Default voices (2), as opposed to breathy voices, often grant easily readable and measurable spectrograms, because the represented formants are devoid of interfering frication.

In contrast to speakers with a breathy voice, creaky voice speakers possess very firmly approximated vocal folds. During the articulation period, this leads to a long closing of the vocal folds before they are abruptly re-opened. This regular closing without phases of transition and opening sheds light on why the formants are unstable, as depicted in the spectrogram in Figure 4.12 (1).

<sup>41</sup> Cf. *Figure 4.6* for spectral cues of fricatives.



**Figure 4.12:** Spectrogram segments of the Swiss German word *Kopf*, articulated with a creaky voice (1) and a default voice (2). Once more, F1, F2, and F3 are steadier in the default voice (2).

As far as accurate measurement practices are concerned, the essential difference between breathy and creaky voices may be that creaky voices bear less severe consequences than breathy voices, as the LPC traces of breathy voice speakers often digress further from default voices than creaky voices do.

To this point, two problems of acoustic phonetic measurement have been addressed: the detrimental effects of background noise on the sound signal and the impact of differing voice qualities on spectrogram readings. There are, regrettably, further serious drawbacks in acoustic phonetic measurement, among which coarticulation is perhaps the most serious one. The following examination of coarticulatory effects shows how consistent measurements of vowels can nevertheless be achieved.

#### 4.4.4 Coarticulation in Acoustic Phonetic Analysis

Coarticulation refers to the overlapping of adjacent articulations, which Laver attributes to the neuromuscular planning and execution of speech. Although coarticulation is of great advantage as it considerably accelerates the flow of speech, the downside of it is its negative effect on the intelligibility and measurement of spectrograms. On spectrograms, coarticulation causes the formants of each sound to blend in with its adjacent sounds, in some instances more in others less so. On these grounds, the researcher is required to make a judgment about where the measurement of an individual sound segment begins and ends prior to analyzing the sounds. Furthermore, it is sensible for the researcher to apply the same measurement techniques consistently throughout

the course of a sound segmentation. If he fails to do so, measurements are likely to be inaccurate, which results in the invalidity of the data, an instance which will be highlighted next (cf. Ladefoged 1993, Laver 1994)

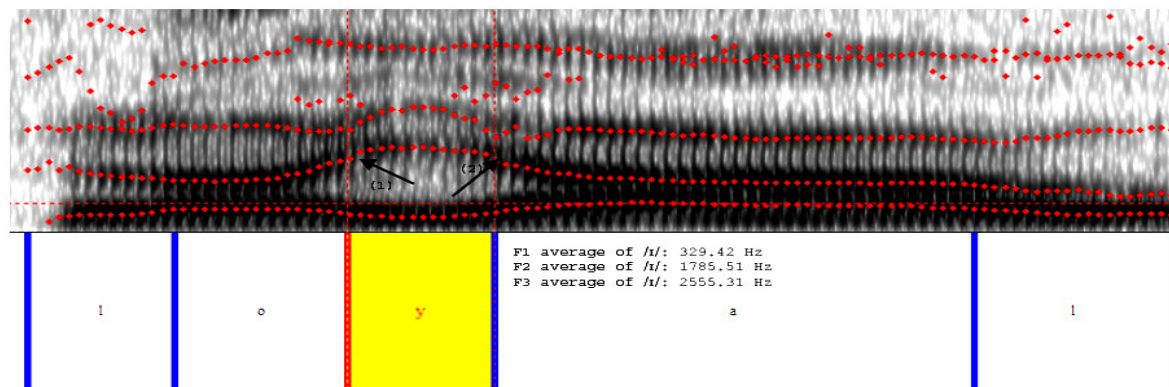
As mentioned above, monophthong vowels and diphthong vowels are represented as strong vertical striations on spectrograms, which represent the vibrations of the vocal chords.<sup>42</sup> During the course of the articulation period, diphthongs are characterized by a modification of their F1 and F2 values along the y-axis. At some stage within this shift, there is a point of transition where the sound is neither one nor the other. If one is interested in measuring only one sound segment within a diphthong, there are a number of measurement strategies to choose from.<sup>43</sup> The following figures, 4.13, 4.14, and 4.15, demonstrate identical spectrograms of the Swiss German word *loyal*, as pronounced by a SG speaker. Due to the speaker's dialect, *loyal* is articulated with the final lateral approximant vocalized. In this attempt, it has been vocalized to /lɔɪau/. Let us assume one only intends to measure the first three formants of the high front vowel /ɪ/ of /lɔɪau/. Consider how the three subsequent measurement strategies affect their outcome.

In Figure 4.13, (1) and (2) indicate at which point the interval measurement begins and ends. It starts precisely in the middle of the transition between /ɔ/ and /ɪ/, and ends in the exact middle of the transition between /ɪ/ and /a/. Figure 4.14, in contrast, demonstrates an interval measurement of /ɪ/ at F1's, F2's, and F3's most stable phase, starting at (1) and ending at (2). The application of such measurements is tricky in so far as the most stable phase of F1 is usually not the most stable phase of F2, while the most stable phase of F2 is typically not the most stable phase of F3. Lastly, Figure 4.15 shows a punctiform formant measurement at the nuclei of F1 (1), F2 (2), and F3 (3). The discrepancies of the different measurement techniques in the outcome are remarkable.

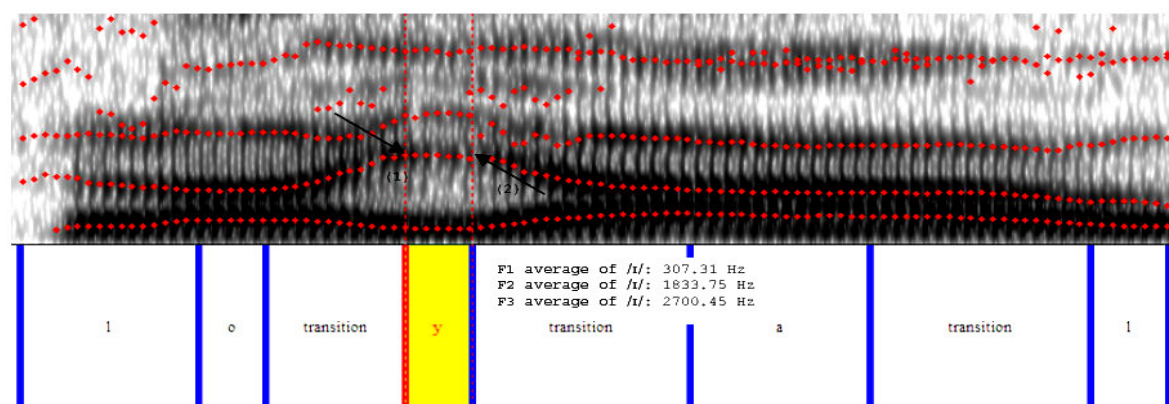
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<sup>42</sup> Cf. Figure 4.1.

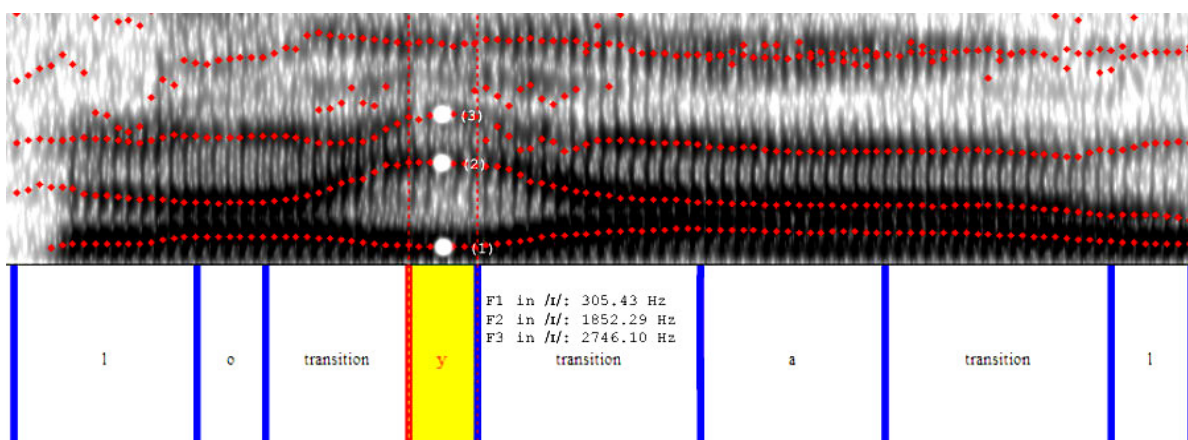
<sup>43</sup> The following does not address an acoustic phonetic measurement of diphthongs in its own right. Instead, diphthongs are viewed as a sequence of sounds, which, also, are evidence for coarticulation. Although it would be appealing to explore how diphthongs are measured as a single sound unit, it would be a much too complex venture, and, furthermore, irrelevant to the present study, as diphthongs will not be further investigated.



**Figure 4.13:** Spectrogram of *loyal*, as pronounced by a SG speaker with /ɪ/ vocalization. Interval measurement at the beginning (1) and ending (2) transition points.



**Figure 4.14:** Spectrogram of *loyal*, as pronounced by a SG speaker with /ɪ/ vocalization. Interval measurement of /ɪ/ at the most stable F1, F2, and F3 phase; starting at (1), ending at (2).



**Figure 4.15:** Spectrogram of *loyal*, as pronounced by a SG speaker with /ɪ/ vocalization. punctiform formant measurement at the nuclei of /ɪ/ F1 (1), F2 (2), and F3 (3).

The F1, F2, and F3 formant values of /ɪ/ for each measurement technique are indicated on all of the three figures. While interval measurements of the formants' most stable phases, Figure 4.14, and punctiform measurements of the formants' nuclei, Figure 4.15, provide similar results, at least as far as F1 and F2 values are concerned, results from interval measurement at transition points, Figure 4.13, behave differently. The Hertz (Hz) discrepancy in F3s for example, almost reaches 200 Hz between punctiform and transitional point measurements. These findings underline the importance of consistently applying the same strategy during the course of vowel measurements in a scientific study. Even more so, they lay emphasis on the significance of declaring one's measurement technique in study reports, as the collected formant values may serve as reference points to further studies.

The discussion of measurement techniques eventually boils down to the necessary establishment of labeling and measurement conventions in acoustic phonetics. If they were adhered to, an unbiased cross-comparison between acoustic phonetic studies would be realistic. Siebenhaar et al. (1998) tackled this issue in setting up *Conventions de segmentation pour la construction de diphtongues*, i.e. conventions according to which two juxtaposed sounds could be labeled and measured. Siebenhaar et al.'s proposition for the measurement of neighboring vowels, for example, postulates that "la segmentation s'effectue au milieu de la transition, au point de la pente la plus forte" (1998: 2). It is unclear as to whether or not such conventions are generally adhered to in acoustic phonetics. Unfortunately, authors often only give generic information as to their formant measurement strategies.<sup>44</sup>

#### 4.4.5 Preliminary Conclusions

This section has introduced some of the methodological issues in acoustic phonetic analysis. It has analyzed how background noise, differing voice qualities, and coarticulatory effects can have severe consequences on phonetic data analysis. It was shown how cell phones, for example, can interfere with the electromagnetic field of recording devices, to what extent creaky and breathy voices are difficult to analyze by way of LPC trackers, and how inconsistent approaches towards coarticulatory effects can yield diverse results. The section ended by emphasizing the importance of consistent measurements and the declaration of these in acoustic phonetics.

It was the rationale behind Section A to put forward a critical approach towards standardized sociolinguistic procedures, such as varying strategies of achieving

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<sup>44</sup> Cf. Deterding (1997), Hillenbrand et al. (1997), Liebthal (2004), Wells (1962), and Peterson & Barney (1952). For a detailed discussion how Deterding (1997) and Hillenbrand et al. (1997) treat the measurement techniques applied in their studies see 10.2.1 *Comparison to RP* and 10.2.2 *Comparison to GA*.

authenticity in language and language variation, a few means towards tackling the precarious concept of Standard English, and issues regarding measurement procedures in acoustic phonetics. Where appropriate, the discussion was not only confined to sociolinguistics in general but also narrowed down to sociophonetic research. Before the results of the present study can be demonstrated and discussed, the implications of Section A must be translated to the present study. That is to say, authenticity, or lack of it, in the researched language of 20 EFL speakers, as well as potential reasons for variation in their production of English must be asserted. At the same time, it is indispensable to declare how the acoustic phonetic measurements were carried out. First, however, a few words on the subject selection procedure and the characteristics of the tasks the subjects were asked to perform.<sup>45</sup>

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<sup>45</sup> The reason as to why Standard Englishes were selected as reference points and why this is a moot point will not be investigated further. Section A has outlined a good number of points contributing to that discussion, cf. 4.3 *Standard English as a Benchmark to Swiss English*.

## **Section B: Theory underlying the Analysis of 20 EFL Speakers' Vowel Quality**

### **5 Sampling the Subjects**

Empirical sociolinguists are faced with the issue of having to select the parameters of the object to be studied. On the one hand, they need to decide upon the size and the heterogeneity of the sample they are willing to scrutinize. On the other hand, they are confronted with matters of representativeness. As to the latter, a truthfully representative sample of the speech community, "must be based on a random sample in which each one of several million speakers has an equal chance of being selected" (Labov 2001: 38). Yet there is a dilemma in having samples that consist of hundreds of speakers, which Labov terms the Sampling Paradox: "The more confident we are that a sample represents a population, the less confident we are that the sample can explain the behavior of that population" (ibid: 40). In light of this paradox, the subjects for the present study were sampled with the objective of maintaining the group as homogeneous as possible.

The subjects were screened in accordance with their L1 background, L2 experience, age, and gender. It was assumed that such variables could best be controlled in a school setting. It turned out that the Kantonsschule of Zofingen would match the ideal profile. There are mainly two explanations why a school in the canton of Aargau was selected. In view of Zofingen's geographical location, it is worth emphasizing that the town is at the very center of the German speaking part of Switzerland. It is nestled in a valley in the midst of the largest SG-speaking cities, which all feature distinctive SG dialects: Zurich, Basel, Berne, and Lucerne. Reasons as to why it was deemed valuable to screen subjects still attending school include the fact that external variables such as L2 experience, age, and gender could be controlled fairly well.

From a pool of approximately 40 potential subjects, 29 were selected for further recordings. Undoubtedly, a larger sample of subjects would have raised the degree of representativeness and explanatory power of the results. Unfortunately, as can be inferred from Labov's supposition of the Sampling Paradox (2001: 38), the number of sociolinguistic variables increases proportionately to the augmentation in the sample size, a corollary of which is the difficulty of formulating conclusions and generalizations from the results obtained. In addition,  $n=29$  was opted for on grounds of operability. A larger sample would have gone beyond the scope of a licentiate paper and would have been unfeasible to manage, taking into consideration the already laborious and time-consuming recording and data evaluations. Unfortunately, the recordings of eight students turned out to be defective due to a malfunctioning microphone. Additionally, one of the 29 subjects called in ill on the first day of the recordings, which meant that the number of subjects decreased to 20.

## 6 Reading Tasks

Acoustic phonetic studies evolve primarily around the analysis of vowels, while the study of consonants, prosody, and voice quality has received relatively moderate attention (Thomas 2002: 168). Before vowels can be analyzed instrumentally, they must be recorded. Various types of elicitation tasks, especially word lists, have become entrenched methodologies – a trend that is grounded in advantages unique to this kind of elicitation task. Most importantly, word lists allow for an integration of vowels in identical environments. Secondly, the researcher can elicit more data with less effort during the recordings as word lists can be employed in a mechanical fashion. Finally, elicitation tasks tolerate an easier analysis and scoring of the linguistic data collected (Chaudron 2003: 772).

The subjects were asked to perform in three pre-designed reading tasks.<sup>46</sup> The purpose of the first task is to analyze SE vowel quality in standardized phonological surroundings. Therefore, the subjects were prompted to articulate word list items that included English target vowels in an /hVd/ phonological environment. The second task requested the students to read an English text that included words which are said to be articulated differently in GA and in RP. The underlying principle behind this task is to explore whether or not the subjects' performance in the first task correlates with the findings of the text readings. Finally, the subjects were asked to read word lists that contained SG target vowels in an /hVd/ phonological environment, with the rationale of testing their performance of SE for L1 interference. As will be argued in the next section, there are a range of moot points in the application of such reading tasks which need to be taken into consideration.

### 6.1 Word Lists with English Vowels

In order to achieve standardized phonological surroundings of vowels, they must be placed in identical phonological environments. For this reason, the subjects were initially asked to read word lists in which the English target vowels were embedded in an /hVd/ phonological environment. The motive for the /hVd/ environment is twofold. For one, the CVC configuration is practical as nearly all English vowels can be inserted to form existing words. In addition, the studies that provide the reference point values, too, made use of /hVd/ structures, with the exception of Deterding (1997), who analyzed vowels from connected

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<sup>46</sup> Originally, there were four different tasks. Due to time limitations, however, the third task (which constituted word lists that contained the English target vowels in a /pVp/ phonological environment, geared at testing the modifications in Swiss English vowel quality in a phonological environment other than /hVd/) was ultimately discarded during the data analysis process.

speech.<sup>47</sup> Based on Ladefoged's (2001: 27) conception of English vowels, ten English monophthong vowels along with six English diphthong vowels were inserted in /hVd/ constructions. See Figure 6.1.

|                                    |                               |                               |
|------------------------------------|-------------------------------|-------------------------------|
| <b>heed</b> ( <u>need</u> ) /i:/   | <b>hawd</b> ( <u>saw</u> )    | <b>who'd</b> ( <u>noon</u> )  |
| <b>hid</b> ( <u>bit</u> ) /ɪ/      | <b>hayed</b> ( <u>May</u> )   | <b>had</b> ( <u>sad</u> )     |
| <b>head</b> ( <u>bed</u> ) /ɛ/     | <b>had</b> ( <u>sad</u> )     | <b>hide</b> ( <u>guide</u> )  |
| <b>had</b> ( <u>sad</u> ) /æ/      | <b>heard</b> ( <u>bird</u> )  | <b>heed</b> ( <u>need</u> )   |
| <b>hod</b> ( <u>God</u> ) /ɑ:/     | <b>hide</b> ( <u>guide</u> )  | <b>hud</b> ( <u>but</u> )     |
| <b>hawd</b> ( <u>saw</u> ) /ɔ:/    | <b>hid</b> ( <u>bit</u> )     | <b>hoyed</b> ( <u>boy</u> )   |
| <b>hood</b> ( <u>good</u> ) /ʊ/    | <b>hud</b> ( <u>but</u> )     | <b>hid</b> ( <u>bit</u> )     |
| <b>who'd</b> ( <u>noon</u> ) /u:/  | <b>head</b> ( <u>bed</u> )    | <b>hod</b> ( <u>God</u> )     |
| <b>hud</b> ( <u>but</u> ) /ʌ/      | <b>heed</b> ( <u>need</u> )   | <b>hewed</b> ( <u>view</u> )  |
| <b>heard</b> ( <u>bird</u> ) /ɜ:/  | <b>who'd</b> ( <u>noon</u> )  | <b>hood</b> ( <u>good</u> )   |
| <b>hayed</b> ( <u>May</u> ) /eɪ/   | <b>hoed</b> ( <u>rope</u> )   | <b>head</b> ( <u>bed</u> )    |
| <b>hoyed</b> ( <u>boy</u> ) /ɔɪ/   | <b>hood</b> ( <u>good</u> )   | <b>hayed</b> ( <u>May</u> )   |
| <b>hoed</b> ( <u>rope</u> ) /ɔʊ/   | <b>hod</b> ( <u>God</u> )     | <b>hoed</b> ( <u>rope</u> )   |
| <b>how'd</b> ( <u>power</u> ) /aʊ/ | <b>hoyed</b> ( <u>boy</u> )   | <b>heard</b> ( <u>bird</u> )  |
| <b>hide</b> ( <u>guide</u> ) /aɪ/  | <b>how'd</b> ( <u>power</u> ) | <b>how'd</b> ( <u>power</u> ) |
| <b>hewed</b> ( <u>view</u> ) /jʊ/  | <b>hewed</b> ( <u>view</u> )  | <b>hawd</b> ( <u>saw</u> )    |
| <b>heed</b> ( <u>need</u> ) /i:/   | <b>hawd</b> ( <u>saw</u> )    | <b>who'd</b> ( <u>noon</u> )  |

**Figure 6.1:** Word lists containing English target vowels.<sup>48</sup>

In designing these word lists, a few well-known methodological precautions in acoustic phonetics were taken. With the intention of accumulating as much usable phonetic data as possible, the instructions asked the subjects to read the first column silently before articulating the words aloud. Had the subjects proceeded differently, chances are that much of the collected data could not have been evaluated due to likely mispronunciations on the part of the subjects. Furthermore, as is evident in Figure 6.1, the second and third columns replicated the identical items in random order. The purpose of this tactic is to prevent subject memory effects. Additionally, the word lists are designed such that the first item always re-occurs at the end of each column. In 4.3.3 *Individual Voice Quality*, it was mentioned how complete and/or declarative sentences have a tendency to be creaky in the last part because they typically end in low pitches (Epstein 2002: 2). The reading of word lists functions accordingly. To prevent

<sup>47</sup> Although Wells (1962), also, analyzed RP English by means of /hVd/ constructions, Deterding's (1997) formant values of connected speech were preferred as a point of reference for this study because Wells' study dates back 40 years.

<sup>48</sup> Cf. *Appendix A 1 Word Lists and Texts*. The complete appendices can be downloaded at <http://www.isw.unibe.ch/metanav/pers/aleemann.html>.

pitch differences due to the position in the word lists - a rising intonation contour is expected for all items except the last one – a dummy word was added at the end of the list. As a final point, the subjects were offered cues as to the articulation of the target vowels in adjacent parentheses. This measurement was also taken to reduce the number of mispronounced word list items. There are, regrettably, a number of difficulties in the application of such methodologies.

As discussed in 4.1.4.5 *Word Lists, Reading Passages, and Minimal Pairs*, the fundamental weakness in using word lists is that they stimulate the subject to highly monitor his speech. Put it in a different way, the recorded words and evaluated vowels are not retrieved from natural but from unnatural, inauthentic speech. Furthermore, words such as *hawd*, *hayed*, *hoyed*, *hoed*, *hud*, and *hewed* do not carry semantic contents and are as such inexistent in the English language. Therefore, it may well be maintained that the students' articulation of words never heard, seen, or read before, are, also, inauthentic in nature. Moreover, the words in parentheses, essentially serving as an aid to articulate the target vowels properly exert an influence on the pronunciation of the English target vowel. Finally, in order to evade outliers that skew statistical tendencies, there are mispronounced items that need to be discarded prior to in depth analyses. This addresses the issue as to when a mispronounced item is discarded. Generally, items were discarded if they clearly deviated from the additional two attempts. In case subjects mispronounced an item in each of the three attempts, each item was categorized as a missing value. In case of doubt, items were also discarded.

## 6.2 Text with RP and GA Tokens

The subjects were further asked to read a short, artificially constructed text.<sup>49</sup> It included tokens that, in their pronunciation, are said to differ in GA and RP. The text aspires to examine if the subjects' pronunciation of the words in the text shows a relationship with the results acquired in the word lists with English vowels.<sup>50</sup> The choice of the inserted words relies on the often-cited chief differences between GA and RP pronunciation: RP /ju:/ vs. GA /u:/ (3 tokens), RP /ɑ:/ vs. GA /æ/ (3 tokens), RP /ɔ/ vs. GA /α/ (2 tokens), and RP /t/ vs. GA /r/ (3 tokens). See Figure 6.2.<sup>51</sup>

<sup>49</sup> Cf. *Appendix A 1 Word Lists and Texts*.

<sup>50</sup> Cf. 6.1 *Word Lists with English Vowels*.

<sup>51</sup> Some of these pronunciation differences are retrieved from Crystal's *Encyclopedia of the English Language* (2000: 307).

“His **new car** broke down in the morning. 120 miles an hour **on** the road - cars cannot function if they are **mistreated**, even if they’re **new**. Dave drove too **fast**. Maybe **not** even half the speed and the car would still **work** today. He took a **chance**; he didn’t care to listen to his brother.

Dave has several **temporary jobs**: **tuning** guitars for bands that come to the **city**, helping his father’s over-worked **secretaries** at the **centre**, thinking of **advertisement** options for his father’s firm and working in the **garage**. Having so many jobs, he put up a **schedule** next to the **vase** in their living-room. You **can’t** have only one job - it’s **better** to have **four** jobs.”

**Figure 6.2:** The artificially constructed text the subjects were asked to read. Tokens with allegedly different GA or RP pronunciation are in bold print.<sup>52</sup>

In addition, by way of inserting 3 tokens where GA speakers are said to rhotacize, the text explores whether the Swiss speaker uses rhotacism or not. It was also of interest to consider how Swiss speakers articulate entire words that are supposedly pronounced differently in GA and RP, such as *secretaries*, *garage*, *centre*, *advertisement*, *schedule*, and *vase* (each token occurring once). The recordings of each of the 22 tokens were narrowly transcribed, followed by a comparison of the SE transcriptions to the transcription of the target words’ GA and RP form, as indicated in Wells’ (2000) *Longman Pronunciation Dictionary*.<sup>53</sup> For the statistical analyses, here, too, transcriptions were discarded in case of doubt.

### 6.3 Word Lists with Swiss German Vowels

In order to detect the extent to which the subjects’ first language, which in virtually all cases is some form of SG, has an effect on their pronunciation of English, their SG vowels needed to be recorded and analyzed as well. It was deemed useful to apply the exact same strategy as introduced in 6.1 *Word Lists with English Tokens*, except that the items conveying the target vowels were printed in High German which the subjects were asked to read in their own dialect.<sup>54</sup> As far as the application of word list strategies is concerned, the previously outlined methodologies were applied: Re-occurring of the first item at the end of the list and item randomization. See Figure 6.3.

<sup>52</sup> The text that the subjects read did not include any writing in bold print.

<sup>53</sup> Cf. *Appendix A 2 Transcription of Text with RP and GA Tokens*.

<sup>54</sup> Of course, it would have been more sensible to write down the words in Swiss German. The problem is, however, that this would have led to biased results as each speaker’s written representation of Swiss German parallels his spoken dialect. Additionally, Swiss German writing is often heavily dependent on the writer’s idiosyncratic style.

|                         |                   |                   |
|-------------------------|-------------------|-------------------|
| <b>der Reis</b> /i: /   | <b>blau</b>       | <b>Jupiter</b>    |
| <b>kurz</b> /ʊ /        | <b>die Ratte</b>  | <b>der Reis</b>   |
| <b>der Ritter</b> /ɪ /  | <b>der Reis</b>   | <b>blau</b>       |
| <b>die Rebe</b> /æ /    | <b>kehren</b>     | <b>die Maus</b>   |
| <b>retten</b> /ɛ /      | <b>die Frauen</b> | <b>kehren</b>     |
| <b>der Säbel</b> /æ: /  | <b>der Kopf</b>   | <b>die Frauen</b> |
| <b>kehren</b> /ə /      | <b>hey</b>        | <b>die Rebe</b>   |
| <b>der Kopf</b> /ɔ /    | <b>der Ritter</b> | <b>der Kopf</b>   |
| <b>der Staat</b> /ɑ: /  | <b>die Maus</b>   | <b>der Ritter</b> |
| <b>Jupiter</b> /jʊ /    | <b>kurz</b>       | <b>hey</b>        |
| <b>der Vater</b> /ʌ /   | <b>der Vater</b>  | <b>der Staat</b>  |
| <b>loyal</b> /ɔɪ /      | <b>Jupiter</b>    | <b>die Ratte</b>  |
| <b>die Maus</b> /u: /   | <b>retten</b>     | <b>kurz</b>       |
| <b>hey</b> /eɪ /        | <b>die Rebe</b>   | <b>der Vater</b>  |
| <b>der Ton</b> /o: /    | <b>loyal</b>      | <b>der Ton</b>    |
| <b>der Riese</b> /ɪ /   | <b>der Säbel</b>  | <b>retten</b>     |
| <b>die Frauen</b> /ɔʊ / | <b>der Staat</b>  | <b>der Riese</b>  |
| <b>blau</b> /aʊ /       | <b>der Ton</b>    | <b>loyal</b>      |
| <b>die Ratte</b> /ʌ /   | <b>der Riese</b>  | <b>der Säbel</b>  |
| <b>der Reis</b> /i: /   | <b>blau</b>       | <b>Jupiter</b>    |

**Figure 6.3:** Word lists containing SG vowels. Although printed in High German, the subjects were prompted to read them in their idiosyncratic SG dialect.<sup>55</sup>

To give an example: The student read the word *der Reis*, which he articulated in SG dialect as [də ri:s].<sup>56</sup> In spite of the palpable deficiency that this word list, also, elicits unnatural rather than authentic speech, there is a further serious methodological flaw grounded in the direct semantic and phonological interference from High German to SG. To take yet another example, consider the following line of argument: If a subject from Zofingen is asked to translate the English noun phrase *the giant* into SG, he will, with the utmost probability, say [də ri:s]. If, however, he is requested to read the Swiss German noun phrase *der Riese*, which in High German is transcribed as [dœ ʁi:zə], the Swiss is likely to say [də ri:s], notice the use of /i/ in place of /ɪ/. It could be argued that this modification from /i/ to /ɪ/ came about through phonological transfer from High German [dœ ʁi:zə] to SG [də ri:s]. Finally, in keeping with the discarding conventions introduced in 6.1 *Word Lists with English Tokens*, here, too, items were disposed of if they deviated from the other two attempts.

<sup>55</sup> Phonetic symbols as well as vowel underlining were of course not present on the word lists the students read.

<sup>56</sup> Definite articles were deliberately inserted to avoid potential disambiguation problems which might again have caused an increase in the number of mispronounced items.

## 7 Approaching Authenticity in Language and the *Observer's Paradox*

### 7.1 Authenticity, or Lack of it, in the Recorded Data

Before the degree of authenticity in the recorded data can be assessed, it is essential to delineate factors which may have induced inauthenticity in the subjects' English language production. These include their awareness of the general purpose of the study, of the recording devices, and finally of the unnatural settings. This section endeavors to explore the effects of these factors on language authenticity in further depth.

Dimmendaal (2001: 58) observes that prior to recording the subjects, it is reasonable for the researchers to calculate ample time to adjust to the field situation and to establish rapport with the people at the field site. Accordingly, an English teacher at the *Kantonsschule Zofingen* was contacted in an early phase of the research procedure.<sup>57</sup> Following a handful of email correspondences it was settled that matters should be talked over in further detail at the field site. During the discussions, both parties were in agreement on keeping the students' knowledge of the intentions behind the present study to a minimum, with the objective of being able to gain maximum access to unbiased, authentic speech. It was further concluded that the board of the *Kantonsschule Zofingen*, too, ought to be informed about the research project.

Notifying the principal and co-principal of the school about the general framework of this study did not pose a problem. It was problematic, however, that the school board demanded that the subjects be informed about the goals of the project, even if only to a minimal degree.<sup>58</sup> As a means to avoid arousing the prospective subjects' interests, consensus was reached to dub this study as *English Language Research Project* - a title that would soon after be applied in the project announcement letters, distributed to the participating students, the school board, and the English teachers.<sup>59</sup> Apart from discussing predominantly administrative matters, several recording settings were screened for suitability at the field site.

Because the school board chose to allocate only 2 hours for the recording of 20 students further rooms as well as additional fieldworkers were needed. As to the rooms, Maddieson (2001: 221) suggests recording at locations where background noises as well as reverberations can be minimized. Although background noises such as neon lights, cars driving by, people socializing in the hallways etc. generally overload the sound signal, this is not further problematic

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<sup>57</sup> For the sake of anonymity and upon request of the teacher his name will not be revealed.

<sup>58</sup> The rationale given behind the school board's decision was the extra-curricular nature of these recording sessions.

<sup>59</sup> Cf. *Appendix B 1 Letter to the School Board* and *Appendix B 2 Letter to the Students*.

for the transcription of texts. As far as acoustic analyses are concerned, however, background noises severely affect their validity and reliability. Taking heed of these criteria, three detached rooms in the school library, provided by the library personnel and the school janitor, and one classroom, were selected as suitable recording locations. As for the extra fieldworkers, four friends of mine were glad to provide assistance.

On the day of the first recording, the agitated group of students was gathered in the classroom which later served as one of the four recording locations, and they were instructed about the further proceedings of this project. At that point, the subjects took notice of the previously installed, clearly visible recording devices. They realized that they were going to be recorded in an unnatural setting, as classrooms and library rooms obviously define artificial settings. Having introduced which factors affect language authenticity in the present study let us draw the connections between these findings and the theoretical approaches towards language authenticity introduced in *4.1 Authenticity in Researched Language*.

As previously highlighted, it is one of the intentions of this study to work with natural, authentic language, which is congruent with Coupland's definition of authentic language 2, "naturally occurring language" (2001b: 415).<sup>60</sup> As explained in *4.1 Authenticity in Researched Language*, authentic language exists only if an authentic speaker is situated in an authentic context. Deplorably, the speakers of the present study do not match the profile of Eckert's (2003: 392) definition of the authentic, "spontaneous speaker of the pure vernacular." Neither do they provide "pure vernacular" (ibid: 392) speech, as the subjects were primarily asked to read off pre-designed, often inexistent, word list tokens.<sup>61</sup> In this respect, it is significant to highlight one example of the direct repercussions of this inauthentic, read-off speech on the present study's acoustic phonetic analysis. Vowel quality in spontaneous speech is often different from vowel quality in read speech. Strik, Cucchiaroni, and Binnenpoorte (2000: 4) observe that the former often features pauses and a slow rate in speaking which permits the natural articulation of vowels. Fast, read speech, on the contrary, is characterized by its vowel centralizations – a phenomenon, which, also, affected the authenticity of the present study's recordings. Finally, the subjects were not recorded in authentic contexts, as will be argued in the following section.

## 7.2 Approaching the Observer's Paradox

The implications of the theory behind the *Observer's Paradox*, as presented in the theory-driven Section A, must be discussed in light of the present study.<sup>62</sup> It

<sup>60</sup> Cf. *4.1.1 Sociolinguistic Authenticities*.

<sup>61</sup> Cf. *6 Reading Tasks*.

<sup>62</sup> Cf. *4.1.3 Authentic Context* and *4.1.4 The Observer's Paradox*.

was emphasized in 4.1.4.5 *Word Lists, Reading Passages, and Minimal Pairs* that the reading of word lists involuntarily puts the speaker in inauthentic contexts. In view of Labov's *Attention-to-Speech Continuum* (1972c: 209), such contexts induce highly monitored speech because word lists constitute the stylistic end of careful speech. In consideration of Bell's (1984: 159) *Audience Design* model, it could be argued that the fieldworker in the present study assumed the role of the addressee.<sup>63</sup> Accordingly, the subjects designed their speech for the addressee, from which it follows that the interview situation is subject to addressee effect, which, as opposed to eavesdropper effect, for example, is fairly extensive. The corollary of addressee effect is again the high degree of unnaturalness in language. All things considered, it can be maintained that the subjects were situated in inauthentic contexts, which accordingly induces inauthentic speech; a fact that is corroborated by an account of the subjects' demeanor during the recording sessions.

Throughout the recordings the entire group of researchers noticed that the majority of the subjects showed moderate cooperation and behaved in a visibly anxious and unnaturally shy. Other peculiar behavior patterns included the following: While performing the elicitation tasks, a few students needed to be reminded to speak up, as otherwise the recordings would have been too weak in amplitude. Furthermore, a number of students were urged to start over because they were passing through the tasks at excessive speed.<sup>64</sup> Finally, the real subjects performed unguardedly and much more quickly as opposed to the subjects who participated in the trial runs prior to the recording sessions. It is understood that such behavior patterns reflect unnatural behavior.

The fundamental repercussion of these social psychological phenomena on language is that spectral representations of emotionally charged speech, be it anxiety, fear, boredom, etc., are very different from those of neutral speech. In "Acoustical analysis of spectral and temporal changes in emotional speech", Kienast and Sendlmeier (2000) studied the vocal expressions of anger, happiness, fear, boredom, and sadness in speech. With regard to vowel formant analyses their findings are remarkable, namely that "sentences expressing fear, sadness or boredom are characterized by a formant shift towards a centralized position in all different vowels" (Kienast and Sendlmeier 2000: 3), a shift that parallels the vowel centralization emerging in fast, read speech.<sup>65</sup>

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<sup>63</sup> Remember that the fieldworker can choose to occupy any of 5 suggested audience roles.

<sup>64</sup> As noted in 7.1 *Authenticity, or Lack of it, in the Recorded Data*, fast speech often induces vowel centralization, which, of course, would have been unfavorable.

<sup>65</sup> Cf. 7.1 *Authenticity, or lack thereof, in the Recorded Data*.

### 7.3 Preliminary Conclusions

This section has construed the issue of authenticity and the effects of the *Observer's Paradox* in connection with the language produced by the 20 EFL speakers. It is difficult to escape the conclusion that there is a lack of authenticity in the researched speech due to inauthentic speakers and inauthentic contexts. The inauthenticity in the language produced can be traced back to the subjects' awareness of the general principles behind the study, of the recording devices, and of the unnatural settings. Their demeanor during the recording sessions reinforced the assumption of a low degree of authenticity, given that the subjects behaved in an overly shy and anxious way. The negative ramifications of such behavior patterns on acoustic phonetic analysis are altered spectrographic representations as a result of emotionally affected language.

## 8 Approaching Language Variation

Gaetano Berruto claims that “for most linguists, variation and variety are in fact a *crux*” (Berruto 2004: 293), a *crux* that, unfortunately, often leads to the disregarding or camouflaging of critical procedural and analytical decisions within the data analysis process, with the researcher’s objective of making the analysis seem much clearer than it is (Wolfram 1993). Thus, “presentations of elegant-appearing summaries of results found in published versions of variation studies are often far removed from the laboratory in which such analyses are conducted” (ibid: 194). It is one of the goals of this paper to identify and clarify as many language internal as well as language external variables as possible and to give emphasis to, rather than to camouflage, critical procedural and analytical decisions.

Variation in SLA exists at all levels of language. Preston’s conception of language variation in an L1, as introduced in 4.2 *Language Variation*, also applies to SLA – yet there is need for qualification. While linguistic and stylistic factors for language variation in SLA follow along similar lines as in a general sociolinguistic framework, external factors, or in Preston’s terminology, “social factors” (2001: 208), must be viewed somewhat differently. Major (2001: 66) observes that the vast majority of external factors in SLA are dependent on the individual himself, while from a general sociolinguistic standpoint external factors are largely contingent upon social class. Romaine (2003: 422) reinforces Major’s (2001: 66) stance in ascertaining that interlanguage variation seldom has social significance for foreign language learners. This raises the issue of which methodologies best serve the detection of individual factors.

Surveys and questionnaires are two of the most prominently featured techniques in the social sciences. The construction of a survey is an art, as contents, format, positioning, and precise wording significantly influence the outcome of a study. The present study, also, decided to work with questionnaires, the contents of which are examined in the following chapter.<sup>66</sup> The questionnaire concerns the most significant external factors for variation in the 20 EFL speakers’ production of English. Causes for internal and stylistic variation will be discussed finally.

### 8.1 External Variation

External factors that are said to cause variation in SLA are psychologically and socially complex in nature. Commonly, they are subsumed as under “personality of the individual” factors (Major 2001: 66). As early as the late 1970s, Schumann portrayed factors he estimated as affecting the process of SLA as

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<sup>66</sup> Cf. *Appendix C Questionnaire* for the whole document in the form in which it was handed out to the subjects.

mostly neurological, affective, cognitive, instructional, and social in nature (Schumann 1978: 10). See Table 8.1.

26 years later, Moyer (2004: 14) claims that many of the factors listed in this catalogue have become verified as significant. Therefore, it was decided to select and scrutinize some of Schumann's postulated external variables for further analysis in the present study - a selection process which is, as Wolfram (1993: 204) predicts, ultimately subjective and qualitative in nature.<sup>67</sup> In reconsideration of the proper goal of this study, i.e. to compare the vowel quality of 20 EFL speakers to the vowel quality of RP and GA, it was deemed useful to specifically delve into the issue of what Schumann refers to as "Social factors: Attitudes towards target language (dominant) group" (1978: 10). The questionnaire, functioning as an instrument to explore the significance of a few subjectively selected external variables, was divided into 3 sections: Demographics, L2 background, and language attitudes.

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<sup>67</sup> Given the time and space limitations of this study, the scrutiny of Schumann's entire taxonomy would not have been feasible.

|                                                                      |
|----------------------------------------------------------------------|
| <b>Neurological factors</b>                                          |
| Lateralization                                                       |
| <b>Affective and personality factors</b>                             |
| Motivation                                                           |
| Ego permeability                                                     |
| Tolerance for ambiguity                                              |
| Sensitivity to rejection and self-esteem                             |
| Introversion/extroversion                                            |
| Culture Shock                                                        |
| <b>Cognitive and aptitude factors</b>                                |
| Cognitive maturity process (imitation, generalization, memorization) |
| Strategies and styles                                                |
| Intelligence/IQ                                                      |
| Interference from L1                                                 |
| Field dependence /independence                                       |
| <b>Instruction and input factors</b>                                 |
| Teacher and class dynamics and reaction to feedback                  |
| Curriculum: Method, texts, approach                                  |
| Intensity and duration of instruction                                |
| Saliency                                                             |
| <b>Social factors</b>                                                |
| <i>Group Community Level</i>                                         |
| Dominant versus subordinate status                                   |
| Assimilation and acculturation                                       |
| Preservation of ethnic and cultural identity                         |
| Degree of enclosure, size, cohesiveness of community                 |
| Attitudes towards target language (dominant) group                   |
| Intended length of residence                                         |
| <i>Personal level</i>                                                |
| Transition anxiety                                                   |
| Social strategies                                                    |
| Linguistic shock                                                     |

**Table 8.1:** Taxonomy of factors affecting SLA (adapted from Schumann 1978: 10).

### 8.1.1 External Variables I: Demographics

#### 8.1.1.1 Sex

This study is restricted to an investigation of 20 male speakers. This is primarily due to the fact that two studies that offer benchmark values of RP and GA, also, examined predominantly men. Deterding (1997) studied the speech of five male

and five female BBC broadcasters, retrieved from the MARSEC database, while Hillenbrand et al. (1995) acoustically analyzed vowels of 45 men, 48 women, 27 boys, and 19 girls, most of whom were raised in Michigan's (US) lower peninsula (3099).<sup>68</sup> Apart from the apparent difference in voice quality between male and female voices, it has been convincingly argued that there are tendencies of sex-preferential phonological variation (Cheshire 2002: 425). On the other hand, it should be emphasized that there have only been few reports that demonstrate an exclusive use of phonological variants of either females or males. Labov subsumes these tendencies in postulating three principles:

**Principle I:** In stable sociolinguistic stratification, men use a higher frequency of non-standard forms than women (1990: 210).

**Principle Ia:** In change from above, women favor the incoming prestige forms more than men (1990: 213).

**Principle II:** In change from below, women are most often the innovators (1990: 215).

It is tempting to hypothesize whether or not in stable sociolinguistic stratification women who learn a second language favor prestige forms more than men, cf. Principle Ia (Labov 1990: 213), which in the context of SLA would translate to a preference to native-like forms. In this respect, it is important to note that the results of the present study, which will be presented in Section C: *Data Analyses*, are to be understood with some precaution. It is very possible that an examination of the vowel quality of females would have turned out differently, as “women tend to use a higher proportion of the standard variants than men” (Cheshire 2002: 426), if the noun phrase “the standard variants” is understood in its SLA denotation as “native-like forms.”

### 8.1.1.2 Age

To begin with, the questionnaire prompted the students to tick off their age within a range of age 16 to age 23. Age, pertinent to cognitive and aptitude factors, is said to occupy a crucial role in the acquisition process of a second language. An important point often brought up is that age is a particularly unsatisfying variable for its continuity on the one hand, and its coincidence “with the increasing significance of so many other variables” (2004: 20) on the other. Consequently, age can neither be isolated nor can it be falsified - only tendencies as to whether or not there are maturational constraints in SLA can be

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<sup>68</sup> Cf. 4.3.4.1 Hillenbrand et al.'s (1995) *Study on American English Vowels* and 4.3.4.2 Deterding's (1997) *Study on British English Vowels*.

established (ibid: 20).<sup>69</sup> Closely interconnected with the general debate over age is the perennial concept of the critical period hypothesis (CP), which holds that “a person must be exposed to a language during a certain period of time (also called a *sensitive period* for those more sensitive to the term critical) in order to acquire that language natively” (Major 2001: 7). Studies suggest, however, that its existence is a moot point; a claim supported by Flege, Schirro, and MacKay, who, also, maintain that additional research is required “to clarify the role of age-related variations” (2002: 236). All the same, CP and age in general as a factor affecting SLA have been dealt with extensively by various authors.

Numerous studies have been conducted to shed light on this gray area, a few of which are briefly outlined. Birdsong and Molis (2001) investigated the speech of 61 Spanish speakers who were in the process of learning English. The rationale behind their study was to explore whether there are maturational constraints in the learning of an L2. They concluded with a trend, i.e. “L2 attainment negatively correlates with age of learning if learning commences after the presumed end of the critical period” (Birdsong and Molis 2001: 235), a trend that reinforces the credibility of the CP hypothesis. Quite the reverse was shown in a study conducted by Flege, Munro, and MacKay (1997). 240 native Italian subjects who started learning English in Canada were asked to produce sentences which were rated according to perceived foreign accent by native speakers of English. The results turned out to be extraordinary in one respect. According to the native-speaker judges, many native Italian subjects who had lived in Canada longest demonstrated the most obvious foreign accent while a few subjects who began learning English after the age of 15 were perceived as almost-native like. Conversely, in a further study, MacKay and Flege (2004) endeavored to give an explanation for why late bilingual Italians who immigrated to Canada produce longer sentences in English than early bilinguals do. Flege et al. attribute these findings to the reality that late bilinguals “needed to expend more resources to suppress their Italian subsystem than the early bilinguals” (MacKay and Flege 2004: 373), which would then, in contrast to Flege, Munro, and MacKay’s (1997) previously outlined study, underscore the existence of maturational constraints.

Flege, Munro, and MacKay’s (1997) and MacKay and Flege’s (2004) briefly summarized studies as well as Birdsong and Molis’ (2001) results have shown that “there is no conclusive support for the existence of a critical period for human speech learning” (Flege 1987a: 162). Regardless of the current pondering about the link between age and SLA and despite the fact that age is an

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<sup>69</sup> Maturational constraints in second language acquisition demonstrate similar patterns as those developmental processes that affect a child’s L1 speech production (Flege and Davidian 1984: 323).

external variable that is neither falsifiable nor isolatable, it is interesting to look into the issue as to what degree age may exert an influence on SE vowel quality.

### 8.1.1.3 L1 Interference

L1 interference pertains to the category of cognitive and affective factors. L1 interference, also referred to as *transfer from L1 to L2* or simply as *transfer*, is one of the most thoroughly studied areas in second language phonology (Major 2001: 31). Transfer on a phonological level of language consists of sound substitution, phonological processes, underdifferentiation, overdifferentiation, reinterpretation of distinctions, phonotactic interference, and prosodic interference. The present study carefully pursues sound substitution which denotes the linguistic phenomenon of “an L2 learner us[ing] the nearest equivalent in the L1” (ibid: 31-32). To test SE for L1 interference the students needed to provide information as to their native language(s).<sup>70</sup>

While the answer “Swiss German” would have been too generic in nature, they were asked to note down which SG dialect they believe they are speaking. Barry (1974) has argued that not only do different languages interfere with second language production but also different dialects of a particular language. In “Language background and the perception of foreign accent” (1974), Barry investigated German speakers acquiring English. He concludes that specific acoustic properties in foreign accented speech vary systematically according to the dialect background of the speaker (Barry 1974: 87). The setback in asking about the SG dialects is that the answers rely on self-reporting by the students.

In gathering information for the Longman Pronunciation Dictionary, J.C. Wells (2003) was also faced with the methodological dilemma of self-reporting. Wells argues that sociolinguists “are well aware that informants are not able to give an accurate account of their own pronunciation” (2003: 3), which is why he aims at an alternative solution by offering the informants a number of variant pronunciations for which they are asked to state their preferences. There is no doubt that this is a valuable strategy; however, it could not have been applied to the present study as there are too many specific and independent dialect regions, often varying from one town to another.

## 8.1.2 External Variables II: Language Background

### 8.1.2.1 L2 Experience

Moyer depicts the factor of language experience in second language learning by equating it with a “black box” (2004: 3). Many of the mechanisms of the effects of experience on second language learning are unobservable, which makes them

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<sup>70</sup> To be able to account for L1 interference even more exhaustively, the subjects were further asked to indicate their fathers’ and their mothers’ dialects as well as to provide information regarding the place where they had been raised.

exceptionally intricate to explore (McAllister 2001: 117). Further adding to the difficulty of attempting to establish a link between L2 experience and second language performance is that the learner's experience with the foreign language in a non-classroom setting accrues in a random manner. The corollary of this is that isolated statistical analyses are virtually impossible, which in all probability accounts for the lack of empirical approaches towards the multivariate factor L2 experience in second language research (Moyer 2004: 61). Notwithstanding such obvious intricacies, Moyer postulates a distinction between L2 experiences: L2 classroom experience and residence in the target language country experience. The latter is often expressed as length of residence (LOR) (ibid: 61).

As regards an IL phonology, LOR most often positively correlates with a move towards native-like pronunciation. In the technical literature, this claim has been substantiated through numerous studies like Flege and Liu's (2001) as well as Flege, Bohn, and Jang's (1997). Flege and Liu (2001) observed groups of Chinese adults living in the U.S., but the subjects differed from one another with respect to LOR. With the aim of assessing the role of input in adults' naturalistic acquisition of an L2 through grammaticality judgment tests and comprehension tests (ibid: 527), Flege and Liu (2001) ended on the note that students with a relatively long LOR obtained significantly higher scores than subjects who had lived in the U.S. with a shorter LOR (ibid: 550). In a further study, Flege, Bohn, and Jang (1997) investigated 90 non-native speakers who had recently moved to the U.S. They tested the NNSs' ability to produce and perceive the vowels /i/, /ɪ/, /ɛ/, and /æ/. The speech was examined by native-speaker listeners who attempted to identify the vowels that had been articulated. Flege et al. conclude that the experienced non-native subjects, i.e. those with a long LOR, produced and perceived vowels much more accurately than the inexperienced, i.e. subjects with a short LOR, did. On the grounds of the positive correlation illustrated in these studies, the students were asked if they had been to an English speaking country and, in case of a positive answer, they were requested to specify the LOR checking any one of the boxes that indicated time spans (the options were 1-4 weeks, 1-3 months, 3-6 months, or 6 months or more). In addition, if they answered yes, they were asked to specify where and when this experience in an English-speaking country had taken place.

L2 classroom experience, too, is said to assume a central role in the process of second language learning. Various studies propose that an instructional focus on communication and interaction in the classroom exerts a positive influence on a student's IL phonology. One study conducted in 1970 found that students who receive systematic training in articulatory phonetics perform much better in articulating new sounds than students who are merely given discrimination training by listening to the new speech sounds (Catford and Pisoni 1970). Whereas Catford and Pisoni focused on the effects of pronunciation training on the production of speech, Matthews (1997)

contemplates the effects of pronunciation training on the perception of second language contrasts in a more recent study. In evaluating the results, Matthews learnt that explicit instruction in pronunciation training “contributes to the development of novel segmental categories which can then be used to discriminate members of the novel category from members of other categories perceptually” (Matthews 1997: 229). By virtue of these influential arguments, the students were asked to mark with a cross whether their English teacher paid particular attention to pronunciation, vocabulary, syntax, or interactive competence during classroom instruction. Further appertaining to L2 classroom experience is the teacher’s spoken English, which, also, leaves its marks on the students.

The students in the present study were asked to point out whether they were taught British English or American English in secondary school,<sup>71</sup> even though it is not clear whether the students are sufficiently aware of phonological differences between British and American English to give an honest answer to the question.<sup>72</sup> Stephan (1997) specifically observed this moot point. In using Gymnasium German EFL students as subjects he tested the ability to identify varieties of English. He concluded that only half the students were equipped to tell an American speaker from a British speaker and only a quarter of the students could discern someone from the North of Britain from someone from the South. Superficially, such results are not very encouraging. Yet Stephan claims that they are astonishingly good, considering that, after researching the Gymnasium’s school curricula, the students have never been explicitly taught to discriminate one variety of English from another. He essentially ascribes these findings to travel, personal contact, youth culture, popular stereotypes, and folklore (1997: 106).

Aside from LOR and L2 experience, the learner is frequently exposed passively to the L2, particularly if the L2 is English. The questionnaire attempts to address such variables, which largely cover verbal communication in the L2, reading in the L2, and listening to the L2. Flege, Frieda, and Nozawa (1996) scrutinized Italian speakers with the objective of exploring whether variation in amount of native language use influences second language production accuracy. They concluded that subjects who frequently talk in Italian with their fellow Italians demonstrate stronger foreign accents in English than those who speak less Italian. If the converse of the implications of these findings is assumed, it could be argued that the corollary of speaking extensively in the foreign language reduces the foreign accent. On these grounds, each student was asked

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<sup>71</sup> I.e. Bezirksschule in the canton of Aargau, Sekundarschule in most of the rest of Swiss German-speaking Switzerland.

<sup>72</sup> The question is further problematic in so far as some of the students may have had more than one secondary school English teacher in the course of their education, due to changing of the school, having substitute teachers, etc.

to specify the amount of verbal communication he had with a native English speaker over the past month.<sup>73</sup> Options were 0 minutes, 1-30 minutes, 30-60 minutes, and 1 hour to 5 hours. In case the subjects answered positively they were asked to write down the geographical origins of the native-speaker interlocutor and to indicate whether this verbal exchange happens on a regular basis. The collection of the information regarding the amount of reading in the L2 and the amount of listening to the L2 was handled analogously, except that in listening to the L2 the period given was one week as opposed to 1 month.

Music and movie consumption unquestionably contribute to the L2 listening experience, which particularly applies to English as a foreign language and to adolescents and young adults. Due to the English language's ubiquitous presence in everyday Swiss life it was decided to design a question that addresses L2 listening experience through music. Accordingly, each student was asked to indicate the extent to which he enjoys music containing English lyrics to the non-English lyrics of his overall music consumption. Within the resulting percentage of consumption of English lyrics, the student is further prompted to differentiate between British and American English music. One may wonder as to the rationale behind inquiring about music consumption.

In studying the effect of music on the acquisition of English vocabulary in a group of second grade limited-English proficient children, Medina (1993) argues that music indirectly affects SLA by providing a pathway for language acquisition. Medina ends on the following note: "If music is a viable vehicle for SLA to the same extent as other nonmusical means, then songs can no longer be regarded as recreational devices, having little instructional value" (Medina 1993: 8). Even though Medina does not analyze the likely impact of music consumption on pronunciation in its own right it might be argued that, analogous to the expansion of the lexicon, music could positively affect pronunciation. Similarly, one could speculate whether L2 experience through watching movies in English positively correlates with L2 performance. Thus, the students in this study were asked whether they watch movies in English or in their non-English, often dubbed, version. In case of an affirmative answer, the students were asked to indicate the number of movies they watch in English per month.

It must not be overlooked, however, that the questions regarding movie and music consumption indirectly address language perception rather than language production, the latter being the object under scrutiny in the present study. This brings about the discussion of whether it is legitimate to evaluate the answers given to the speech perception questions according to the same criteria as those given to language production questions. On an even more fundamental level, it

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<sup>73</sup> One month was chosen as a suitable time period as it is likely that the impact of the native speaker on the language learner is still present.

raises the issue of whether the two notions, i.e. L2 perception and L2 production, can be separated as such. Among those who deny the separation of language production and language perception is Major, who claims that “adult language learners do not invent their L2 pronunciation in a vacuum with no input; rather, their pronunciation is in part the result of how they perceive the L2, which is often in terms of the L1 perceptual system” (2001: 52). This is not an isolated view. Llisterri (1995: 1) maintains that “the perception of a new phonetic contrast must necessarily precede its production.” He sketches out the complexity of the relationship between speech production and speech perception, noticing the following trends.

- It does not seem possible to infer production abilities from perceptual ones and vice-versa.

- Stage in the acquisition of L2, experience with the language, degree of exposure, and age of acquisition seem to play a major role in the interaction between production and perception in L2.

- The relation between production and perception might differ according to the class of sounds, to the acoustic and perceptual correlates of these classes and to contextual effects.

- Similarity between L1 and L2 sounds might also have an effect on the interplay between production and perception.

- Social factors such as pressure to improve production may provide an explanation for cases in which production precedes perception. (1995: 5)

In view of Llisterri's (1995) penultimate trend that “Similarity between L1 and L2 sounds might also have an effect on the interplay between production and perception” (ibid: 5), Strange et al. (2004) examined cross-language spectral similarity between North German and American English vowels. The subjects were asked to produce vowels in hVC(a) (di)syllables and in isolated hVC syllables, which were embedded in a short sentence. By means of discriminant analyses, “the extent to which acoustic similarity was predictive of perceptual similarity patterns” (Strange et al. 2004: 1791) was examined. Strange et al. conclude that “neither context-dependent comparisons of vowel production (as specified by spectral similarity patterns) nor context-independent impressionistic descriptions of vowel inventories capture all the relevant information necessary to account for perceived similarities of vowels by non-native listeners” (Strange et al. 2004: 1805), a line of argument that underscores the apparent gray area of the link between language production and language perception. Although

various studies suggest that L2 perception and production do not develop separately from each other (Jacewicz 2002: 330), the complexity of the interplay between these two modes is yet to be explored in further depth.

The repercussions of this gray area on the present study are noteworthy. Even though only three questions in the questionnaire attend to speech perception rather than to speech production, the answers provided by the students are to be taken with great caution as the relationship between L2 perception and L2 production is yet to be determined.

### 8.1.2.2 Affective Factors

One's native language is part of one's identity and personality (Major 2001: 66).<sup>74</sup> It is frequently assumed that identity and personality can affect the strength of a perceived foreign accent in an L2. In the case of the personality trait motivation, for example, some studies suggest that motivation and success reinforce each other in learning a second language (ibid: 66). It is also possible, however, that an anxious learner, or a learner who possesses a low ego-permeability, can be highly motivated, be it integrative or instrumental motivation, and still not attain close to native-like linguistic competence.<sup>75</sup> These findings mean that affective and personality factors have the power to prevent or boost success in learning a foreign language (Major 2001: 67). Consider the following illustration of how ego-permeability can positively affect L2 performance: Guiora et al. (1972) were interested in exploring whether the lowering of inhibitions in students makes them psychologically more receptive to language learning. The students were given ingestions of small amounts of alcohol. Guiora et al. concluded that students who were given 1 ounce of alcohol showed improved pronunciation while students who were given slightly more alcohol demonstrated L2 pronunciation that deviated further from the target forms than the deviations of the speakers in a sober state. In minimally raising the boundaries of inhibition or anxiety students were able to show more ego-permeability and thus exhibit a refinement in pronunciation. What this case shows is that affective factors have a significant impact on L2 production. Motivation, as a further affective factor is also widely accepted as exerting influence on L2 performance.

A high degree of motivation correlates with spending more time in learning an aspect of the L2 (Spolsky 1989: 148), which in turn results in an improvement on various levels of the L2. Gardner and Lambert (1972)

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<sup>74</sup> While L2 experience evidently pertains to the generic taxonomy of L2 background, cf. 8.1.2.1 *L2 Experience*, affective as well as cognitive aptitude factors should have been grouped as taxonomies in their own rights. For the sake of clarity in the presentation of external variables, it was deemed practical to include affective and cognitive aptitude factors within the category of L2 background.

<sup>75</sup> Instrumental and integrative motivation as defined by Gardner and Lambert (1972).

introduced the notions of integrative and instrumental motivation which have become widely recognized and embedded in introductory SLA textbooks. The latter concept refers to the learning of a language for utilitarian purposes. The former, in contrast, reflects the learner's desire to acquire a language to successfully integrate in a language community. On these grounds, the questionnaire in this study prompted the students to grade their degree of motivation on a scale from -1 to +5.<sup>76</sup> It is important in this context to note that motivation in learning a foreign language is closely linked to other personality factors such as anxiety, lack of empathy or ego-permeability, all of which are capable of exerting an inhibiting effect on a motivated student (Major 2001: 67).

Apart from affective factors, cognitive and aptitude factors are also highly pertinent to second language speech production. Although neither the questionnaire nor the reading tasks of the present study touch upon these issues, it is worth briefly turning our attention to the intricacies of cognitive and aptitude factors and commenting on why this study refrains from delving into them.

### 8.1.2.3 Cognitive and Aptitude Factors

Moyer (2004: 33) points out that not much research has been devoted to cognitive and aptitude factors within the framework of SLA over the past 20 years. There are good reasons why little attention has been paid to these factors as there are several apparent problems involved in the assessment of aptitude and cognition. Results may vary inter-individually and between tasks. To circumvent the scrutiny of the speaker's aptitude one could ask the subject about his learning style preferences, which in turn is problematic since an answer relies exclusively on the self-reporting of the student.<sup>77</sup> Additionally, Moyer observes that a specific language-learning situation may not address a learner's idiosyncratic aptitude, a line of argument that has been further developed by Spolsky (1989). He judges that learners favor certain learning styles and learning modes over others (Spolsky 1989: 110).

Learners vary (both individually and according to such characteristics as age, level and cultural origin) in their preference for learning style (visual, auditory, kinesthetic, and tactile) and mode (group or individual); as a result, learning is best when the learning opportunity matches the learner's performance. (Spolsky 1989: 110)

The corollary of such individual learning styles is that a collective labeling of learners' aptitudes is invalid due to the high number of learning styles (ibid:

<sup>76</sup> The question deliberately refuses to address whether the students are driven by instrumental or integrative motivation. It was chosen to resort to a simple question-answer pattern in order to avoid the accumulation of even further variables.

<sup>77</sup> Cf. 8.1.1.3 *L1 Interference* and Wells 2003.

110). Finally, the study of cognitive and aptitude factors in SLA is problematic because of the uncertainty of whether a student consistently applies the same language learning strategy in performing different tasks (Moyer 2004: 34). For all these reasons, cognitive and aptitude factors need to be treated with caution, which is why the present study avoids them altogether.

### **8.1.3 External Variables III: Language Attitudes**

Schumann (1978: 10) classifies language attitudes as social factors on the community level.<sup>78</sup> The subjects' information about attitudes towards the target language is particularly pertinent to the present study as SE is directly contrasted to the target dialects RP and GA. Yet the importance of language attitudes must not be overstated as Lasagabaster et al. convincingly argue that "although it is usually taken for granted that attitudes predict social behavior, there seems to be a gap between what people say (their expressed attitudes) and what they do (their actual behavior)" (2004: 401). This section of the questionnaire attempts to explore how and if the Swiss speakers' attitudes towards English affects their performance.

#### **8.1.3.1 Attitudes towards English**

In the social sciences, attitudes are often thought to elucidate human behavior. In SLA research it is widely acknowledged that the students' attitudes towards the foreign language and its culture are central to language learning success. However, there is a core difficulty in studying language attitudes as they are "political as much as educational, sociolinguistic no less than economic, ethical even more than they are cultural" (Lasagabaster et al. 2004: 403), which makes it tricky to pinpoint them. Since the 1960s, studies on language attitudes have been gaining ground, which, in the meantime, has led to an extensive recognition that "people express definite attitudes towards speakers representing different speech styles" (Giles et al. 1987: 585).

The question remains as to why people form speech style judgments at all. Despite the common assumption that languages cannot be portrayed as better or worse nor be assessed in terms of aesthetic differences such as qualities of beauty, people have not refrained from judging languages according to such criteria (cf. Trudgill and Bauer 1998). It is true that in the present study, too, students are likely to express attitudes towards English anchored in aesthetic judgments. Yet it is equally plausible that their evaluations towards English reflect social and ideological issues. As the following examples illustrate, language attitudes can be altered on political and ideological grounds.

As Giles et al. (1987: 587) report, Bourhis (1983) has shown how a change in the political climate of Quebec has caused modifications in attitudes towards the use of French and English in the area. Analogously, it is probably

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<sup>78</sup> Cf. *Table 8.1*.

off the mark to argue that the atrocities of September 11, 2001 and the subsequent anti-American sentiment in Europe, also, induced negative, politically charged attitudes towards the English language, particularly American English. Aside from politically induced modifications of language attitudes, Philippon observes that the worldwide expansion of the English language causes especially English as a second language (ESL) speakers gradually to become verbal about what they perceive as “unequal communication rights” (2000: 96). As Philippon reports, this point has been emphasized by Ammon (2000), who argues that Germans do not relish the pervasive use of English for scientific purposes. What these examples transparently display is the difficulty in studying attitudes towards language as they can be political, educational, sociolinguistic, economic, and ethical in nature, which, as noted earlier, makes them virtually impossible to isolate. The following section discusses how the questionnaire nevertheless tries its best to assess the nature of the Swiss EFL speakers’ attitudes towards the English language.

### 8.1.3.2 Researching Language Attitudes with the Questionnaire

The questionnaire asked the students to answer the following set of questions:<sup>79</sup>

*1. Do you prefer American English or British English?*

It is the aim of this question to detect potential correlations between the speakers’ preferences of a variety of English and their actual L2 performance. There are several benefits and at the same time difficulties in this formulation of the question. The students are not given the option to tick off “I prefer neither,” which forces them to make a judgment. This strategy might backfire as the students may mark either answer arbitrarily in defiance of not being provided with an alternative answer. Additionally, the answers do not elicit specifics on the students’ understanding of RP and GA. That is to say, a student indicating a preference for RP may inwardly think, “I prefer British English, but only the British English Robbie Williams speaks.” Analogously, the same pattern may occur if the student opts for American English.

*2. Which of the following people do you believe speaks English nicely? Tony Blair, George W. Bush, Al Pacino, Kevin Spacey, Bridget Jones, Queen Elizabeth, CNN Presenters, BBC Presenters, Robbie Williams, Madonna, Ozzy Osbourne, MTV Dismissed/MTV Real World, Eminem/50 Cent, or others.*

This question evokes Lasagabaster et al’s (2004: 403) statement that language attitudes are charged on various levels, from which it follows, with high probability, that the students do not judge these people by their linguistic output, but rather by what they represent in terms of ideology, politics, ethics or

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<sup>79</sup> Cf. Appendix C Questionnaire.

economics. Although it is virtually impossible to eliminate such bias, it would still be intuitively appealing to explore whether or not the subjects' choice in any way correlates with their pronunciation patterns.

*3. Do you feel embarrassed speaking with an American, with a Briton, or with a Swiss?*

The rationale behind this question is to consider if embarrassment in conversing with a Swiss, Briton, or American bears any repercussions on the pronunciation patterns of a speaker. This question primarily evokes the issue of language learner anxiety, which holds that the more anxious the learner is the more embarrassed he feels when speaking in the target language (Horwitz and Gregersen 2002: 562). Anxiety is reinforced if the subjects are "highly concerned about the impressions that others form of them," therefore, they "tend to behave in ways that minimize the possibility of unfavorable evaluations" (ibid: 562), i.e. they try to avoid conversing in the L2 or with a particular accent.

*4. Do you like speaking in English?*

Again, this question intends to test the correlation between finding pleasure in speaking the target language and the quality of the verbal behavior. The reasons for liking to converse in the target language cannot usually be traced to any one specific factor. Rather, it is a medley of factors, as illustrated in Schumann's (1978: 10) taxonomy, that is accountable for whether or not one finds pleasure in conversing in the target language.<sup>80</sup> It is worth noting that linguistic factors such as similarity in language, also, contribute to positive or negative feelings towards speaking in the target language. The downside of this question is its neglect in revealing the reasons behind the subjects' selections.

*5. Do you like the pronunciation of the people on classroom audiotapes?*

The purpose of this question is to find out whether people who like the pronunciation of classroom audiotope speakers articulate English vowels differently than those who are not fond of them. The students were offered pre-formulated answers: Yes, but only the pronunciation of speakers with British English; yes, but only speakers from America; no, neither of them. There are limits to how far data retrieved from this question can be taken. Primarily, it is not detectable what dialects the British, American or other speakers on the tapes have. In consequence, answers provided to this question remain vague and, unfortunately, largely unusable.

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<sup>80</sup> Cf. Table 8.1.

6. *Do you consciously try to attain the same pronunciation as the majority of the people that you hear on classroom audiotapes?*

Evidently, questions 5 and 6 are closely linked. If 5 is affirmed positively for either variety, it is likely that 6 is answered likewise. The pre-suggested answers to 6 replicate the same answering pattern as those for question 5.

7. *Do you attempt to attain the same pronunciation as your current English teacher?*

This question is again geared at examining correlations between the subjects' responses and their L2 performance. Once more, the students had to select from pre-designed answers: "Yes, but only during English lessons," "Yes, during English lessons and outside English lessons," and "No, I do not try to speak like my English teacher at all."<sup>81</sup> The question as well as the pre-designed answers are in need of thorough qualification, for it is likely that the answers given are subject to bias.

Before handing out the questionnaires, the students were unequivocally notified that their information would be handled with the utmost discretion. Despite this information, students may have had the idea that their teachers would nevertheless catch sight of the answers to this question. Consequently, they may have thought their grades in English class might be compromised if they opt for "No, I do not try to speak like my English teacher at all." Although highly speculative, bias in the answers to this question can be presupposed, a hypothesis further corroborated in the following qualifications.

As opposed to questions 1-4, questions 5-7 attend to language attitudes in the context of in-classroom situations. This difference in context may arouse feelings of uneasiness within the student, which, again, might affect his responses to questions 5-7. Furthermore, it could be argued that factors such as liking or disliking the teacher, good or bad experiences a student may have had with a teacher, or simply whether the teacher is regarded as an authority also exert an influence on the student's likeliness want to speak like his teacher. With respect to the latter point, Ray (1971) conceptualizes authority in three content areas, which, if positively stated, are 1. View of the leader as a guide and director rather than an executive of democratic decisions, 2. Approval of institutions and practices notable for the exercise of authority, and 3. Preference for regulation versus libertarianism (Ray 1971: 1). If the teacher is understood as the authority and a student negates any of these axioms, e.g. he disapproves of school as an institution notable for the implementation of authority, it is probable that the student will persistently refuse to attain the same pronunciation

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<sup>81</sup> The inside of class / outside of class distinction was considered useful as some students may purposely try to accommodate to the teacher's pronunciation within the classroom, with the objective of making a positive impression on the teacher.

as his teacher because he prefers to form his pronunciation according to his own mindset.<sup>82</sup>

*8. Does the tendency of anti-Americanism have an impact on your pronunciation of English?*

This question directly addresses the issue raised by Bourhis (1983), introduced earlier in this subchapter, that a change in the political, social or cultural climate is capable of inducing modifications in attitudes towards languages. Likewise, as attitudes towards Americans may have shifted due to post-September 11 anti-Americanism, students may consciously avoid speaking what they perceive as American English.

*9. Do you try to speak with an American English or British English accent?*

The students were provided with the answers “American English,” “British English,” and “Neither of them.” The same qualifications highlighted in the first question apply to this question. In case he opts for the third answer it remains undetermined on what grounds this choice was made. In addition, it may be suggested that the student again only wants to speak a certain kind of American English, e.g. he may desire to sound like Al Pacino, who speaks with a middle-class New York accent, whereas to the student Al Pacino might represent standard American English.<sup>83</sup> The underlying principle behind this decisive question is to learn whether correlations exist between what the subjects say they wish to sound like and what they sound like in reality.

Despite this array of influences on the subjects, one crucial variable has not yet received adequate attention. While *8.1.2.1 L2 Experience* briefly explained the impact of varying instruction methods on the ultimate phonological attainment of the student (see Catford and Pisoni (1970) and Matthews (1997)), the effects of the teacher’s verbal input, i.e. specifically the teacher’s pronunciation, need further consideration. One could argue that teacher pronunciation input also exerts an obvious impact on the student’s pronunciation patterns, particularly if pronunciation is taught prescriptively. Regrettably, this hypothesis can only be speculated about because the English teachers of the *Kantonsschule Zofingen* unanimously refused to be recorded.

The teachers’ justifications for their decision were somewhat controversial. Without going into much detail as to the teachers’ line of reasoning, their answer to the question of whether they would be so kind as to also take part in the recordings can be subsumed by saying that they collectively

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<sup>82</sup> An example that is, in fact, not so far-fetched, considering that the questioning of authority is often typical for students aged between 16 and 23.

<sup>83</sup> Unfortunately, a questionnaire of this format does not have the capacity to research such technicalities.

decided not to have their speech recorded as this was not part of the arrangement when this study was first instigated. Even after reiterating the imperative need for these data, positive answers could not be generated. This was admittedly a relatively awkward reaction considering the astounding cooperativeness the teachers offered prior to being asked to contribute samples of their own speech. The following section nevertheless endeavors to present the significance of the teachers' pronunciation for the pronunciation of the students, and further attempts to identify potential grounds for the teachers' non-participation.

#### **8.1.4 The Significance of Pronunciation Input**

Notwithstanding the fact that many studies exploring classroom techniques have been conducted, little is known about how input and teacher instruction affect ultimate attainment in a second language learner (Moyer 2004: 35). In "Effectiveness of L2 instruction: A research synthesis and quantitative meta-analysis", Norris and Ortega (2000) summarize findings from experimental and quasi-experimental studies conducted between 1980 and 1998 on the value of L2 instruction. Unfortunately, Norris and Ortega do not directly refer to the effectiveness and relevance of the teacher's pronunciation on that of the student. They do, however, conclude that, generally, a particular focus on one aspect of the L2 induces substantial target-oriented change.<sup>84</sup> Additionally, Norris and Ortega stress that the effects of this process are durable and only marginally decrease over time (2000: 524).

Preceding Norris and Ortega's (2000) conclusion that instruction is beneficial and therefore effective is Michael Long's (1983) research on the issue as to whether second language instruction makes a difference in a learner's ultimate attainment. Long agrees with Norris and Ortega's claim that the instruction of a specific aspect of an L2 leaves its marks on the language learner, regardless of age and level of learning. He especially highlights the impact of instruction on learners at early stages of second language learning (1983: 379). Unfortunately, Long, too, omits explicit comments on the effects of the teachers' pronunciation on the students; rather, he is concerned with instruction effects from a wider perspective. Despite this dubiety, it may well be argued that the teacher's pronunciation affects that of the student, particularly if pronunciation is taught in a prescriptive way.

##### **8.1.4.1 Fear of Evaluation as a Cause for Non-Participation**

In conducting an interview study with eight foreign language learners, four anxious and four non-anxious learners, Horwitz and Gregersen (2002) investigated the relationship between foreign language anxiety and perfectionism. The findings of their study are pertinent to the teachers' refusal to being recorded in the present study. Horwitz and Gregersen conclude that

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<sup>84</sup> Cf. Catford and Pisoni (1970) and Matthews (1997).

anxious second language learners strongly manifest communication apprehension, test anxiety, and fear of being evaluated where fear is either academic or personal in nature. Horwitz and Gregersen draw the connection between perfectionism and foreign language anxiety in claiming that “[l]ike anxious foreign language learners, perfectionists set excessively high standards for performance accompanied by overly critical self-evaluations” (Horwitz and Gregersen 2002: 563). Furthermore, they observe that “[p]erfectionist students would not be satisfied with merely communicating in their target language – they would want to speak it flawlessly, with no grammatical or pronunciation errors, and as easily as a native speaker” (ibid: 563). It is tempting to suggest that the English teachers’ refusal to be recorded might parallel the anxiety reactions observed in anxious language learners. This is a logical conclusion, considering that, in a very real sense, non-native language teachers are still language learners, albeit very advanced ones (Horwitz 1996: 366).

Having outlined a few subjectively selected variables from a pool of potential variables, nothing can detract from the central fact that the high number of variables render problematic subsequent correlations with the findings of the vowel quality analyses. Therefore, gaining further understanding in the field of SLA well deserves to be called “the correlational challenge,” as Dörnyei and Skehan state at the beginning of “Individual differences in second language learning” (2003). That said, this qualitative selection of variables ultimately provides the foundations for the subsequent quantitative analyses. Before entering this discussion, however, linguistic as well as stylistic variables in the context of the present study need to be outlined.

## 8.2 Internal Variation

There are several ways of approaching internal variation in SLA.<sup>85</sup> Spolsky (1989: 117) suggests that internal factors for variation in second language learning mostly rely on the relatedness of L1 and L2. The closer the languages are related from either a genetic or typological perspective the easier it is for a learner to acquire linguistic competence in that language. On these grounds, he postulates the *shared feature condition*: “When two languages share a feature, learning is facilitated” (1989: 117). Major (2001: 41) alternatively argues that markedness, as a component of language universals, equally explains internal variation in SLA. The following discussion attends to how similarity in language and markedness universals contribute to internal variation. Finally, stylistic variation, as a component of language universals (Major 2001: 41), will be addressed.

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<sup>85</sup> Cf. 4.2.2 *The Variationist Paradigm and Language Variation*.

### 8.2.1 Similarity in Language

The distinction between similar and dissimilar forms is easier to define on the level of phonology than on other linguistic levels such as semantics or discourse. This is largely due to the fact that descriptions on acoustic, articulatory, perceptual, and structural levels of language are often unambiguous, which facilitates a classification of sounds as similar or dissimilar (Major 2001: 36). From a psycholinguistic standpoint, causes for the difficulty of acquiring new sounds are grounded in the phenomenon that “gross differences are likely to be noticed, due to perceptual saliency, whereas minimal differences are less likely to be noticed, resulting in non-learning” (Major 2001: 37). This hypothesis is extended by Flege (1995), who claims that similar sounds are more difficult to acquire because the speaker classifies them as equivalent, which, in turn, does not allow for the establishment of a new phonetic category.

A new phonetic category can be established for an L2 sound that differs phonetically from the closest L1 sounds if bilinguals discern at least some of the phonetic differences between the L1 and L2 sounds. (Flege 1995: 239)

Accordingly, dissimilar L2 sounds are easier to acquire as the speaker perceives the differences and therefore forms new categories.

Category formation for an L2 sound may be blocked by the mechanism of equivalence classification. When this happens, a single phonetic category will be used to process perceptually like L1 and L2 sounds (diaphones). Eventually, the diaphones will resemble one another in production. (ibid: 239)

Such theses are well substantiated. In “The production of “new” and “similar” phones in a foreign language: Evidence for the effect of equivalence classification”, for instance, Flege (1987b) investigated Americans learning French. He concluded that advanced learners pronounced /ʏ/ as in *tu* authentically while /u/ as in *nous* was produced less authentically, the latter being the similar, equivalent sound.<sup>86</sup> The translation of these findings onto the present study raises the question as to how similar the SG vowel system and the American, respectively British English vowel systems are.

One could, of course, consider comparing vowel formant values of the dialect of Zofingen with those of RP and GA. The problem is, however, that formant values of GA and RP are easily available while those of the Zofingen dialect are not. Although the *Sprachatlas der Deutschen Schweiz 1962 – 2003* (1962) offers meticulously specific phonetic transcriptions, it does not provide formant values. It can be presupposed, however, that the vowel inventories of RP, GA and the dialect of Zofingen are not very different from each other. With all

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<sup>86</sup> Similar trends are found in a further study by Flege and his collaborators (cf. Flege 1993).

certainly, only GA and RP /ɜ/ is dissimilar to any vowel found in SG vowel inventories. In consideration of Flege's (1995) category formation hypothesis, the Swiss students should therefore articulate RP or GA /ɜ/ most accurately because of its dissimilarity, while the rest of the vowels should be produced less target-like.

### 8.2.2 Markedness Universals

Eckman was the first to introduce the concept of markedness universals to SLA research. He defines markedness in SLA as follows: "A phenomenon A in some languages is more marked than B if the presence of A in a language implies the presence of B; but the presence of B does *not* imply the presence of A" (1977: 320). He further elaborates this thesis by postulating the *Markedness Differential Hypothesis* (MDH) constituting of three axioms, all of which should be understood in reference to the relative difficulty in SLA:

Those areas of the target language which differ from the native language and are more marked than the native language will be difficult.

The relative degree of difficulty of the areas of difference of the target language which are more marked than the native language will correspond to the relative degree of markedness.

Those areas of the target language which are different from the native language, but are not more marked than the native language will not be difficult. (1977: 321)

Two of the corollaries of the MDH are that unmarked phenomena are acquired before marked phenomena and that not all differences between the native language and the target language will be equally difficult to learn. It could be argued that difficulty in acquisition reinforces variation as the articulation of difficult sounds could cause varying degrees of uncertainty on the part of the speaker. This, in turn, has an influence on his pronunciation and thus may bring about variation.

It should be emphasized that statistical frequencies equally provide information as to the markedness of a linguistic element. In view of the present study, this means that words such as *hawd*, *hayed*, *hoyed*, *hoed*, *hud*, and *hewed* are marked as their statistical frequencies in the L2 approximates zero.<sup>87</sup> In addition, statistical frequencies of spoken English sounds indicate that the vowel /ɜ/ is the most infrequent vowel used in RP English. Due to its markedness in the L2, the SG learners should consequently find /ɜ/ the most difficult vowel to learn, cf. Table 8.1.<sup>88</sup> According to Table 8.2, the subjects would be expected to pronounce /ɪ/ more RP-like as it is the most frequently occurring, i.e. unmarked, vowel in spoken RP. The vowels /ɑ:/, /ʊ/, and /u:/,

<sup>87</sup> Cf. 6. *Reading Tasks*.

<sup>88</sup> Cf. 8.2.2 *Markedness Universals*.

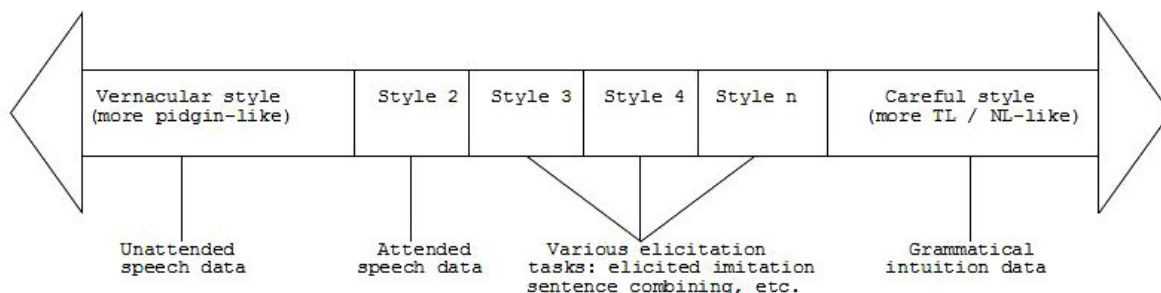
conversely, should be pronounced least like RP because they occur most infrequently.

| <i>Vowel /<br/>Diphtho<br/>ng</i> | <i>Frequency<br/>of vowel in<br/>%</i> | <i>Corresp<br/>onding<br/>wordlist<br/>token</i> |
|-----------------------------------|----------------------------------------|--------------------------------------------------|
| /ɪ/                               | 8.33                                   | hid                                              |
| /ɛ/                               | 2.97                                   | head                                             |
| /aɪ/                              | 1.83                                   | hide                                             |
| /ʌ/                               | 1.75                                   | hud                                              |
| /eɪ/                              | 1.71                                   | hayed                                            |
| /i:/                              | 1.65                                   | heed                                             |
| /oʊ/                              | 1.51                                   | hoed                                             |
| /æ/                               | 1.45                                   | had                                              |
| /o:/                              | 1.24                                   | hawd                                             |
| /u:/                              | 1.13                                   | who'd                                            |
| /ʊ/                               | 0.86                                   | hood                                             |
| /ɑ:/                              | 0.79                                   | hod                                              |
| /aʊ/                              | 0.61                                   | how'd                                            |
| /ɜ/                               | 0.52                                   | heard                                            |
| /oɪ/                              | 0.14                                   | hoyed                                            |

**Table 8.2:** Vowel frequency of conversational RP (Crystal 2000: 239).

### 8.3 Stylistic Variation

Suzanne Romaine understands stylistic variation in second language acquisition “in terms of amount of attention paid to speech in different situations or while performing different tasks” (2003: 422). Major attributes an increase of speech monitoring in a formal context to the decreased amount of L1 transfer (Major 2004: 170). In order to graphically capture speech monitoring and the resulting shift in style within the framework of SLA, Tarone (1988) conceptualized an interlanguage continuum, cf. Figure 8.1.



**Figure 8.1:** Interlanguage continuum (adapted from Tarone (1988: 41)).

While Figure 8.1 on the one hand illustrates different styles, according to whether the speech is monitored or unmonitored, it further represents a progression from zero to full command of a foreign language (1988: 41). In the context of this study, one could argue that the students' styles approximate the formal end of the interlanguage continuum as their speech was highly monitored.<sup>89</sup> There are, however, two possible shortcomings in Tarone's interlanguage continuum. First, it assumes that formality and attention paid to speech can be measured as definite entities. Secondly, the continuum ignores the fact that foreign language learners allegedly respond differently to specific tasks, which in turn might affect their style shift. In an attempt to respond to the first point of criticism, I would like to refer the reader to subchapter 4.1 *Authenticity in Researched Language*. As for the second criticism, Tarone (1988: 40) believes that tasks centered on grammaticality judgments evoke a more formal style, whereas tasks, such as the narration of a story or sentence-combining tasks tap into a more casual style. As noted in 8.1.2.1 *L2 experience*, each student reacts differently to tasks. One could argue that these varied responses equally affect shifts in style, an instance that is not captured in Tarone's interlanguage continuum.

## 8.4 Preliminary Conclusions

These sections have outlined a variety of factors, internal, external, and stylistic in nature, which are believed to cause variation in SLA. This catalogue of variables is by no means exhaustive. It merely represents an array of subjectively selected variables that are hoped to underline the complexity of language variation in SLA. It is quite reasonable to suggest that within the framework of SLA and the present study, the outlined variables symbolize the nexus around which other, even more specific, variables evolve. Yet in reality, one should not forget that second language acquisition is essentially an individual process. Each learner experiences his learning context differently

<sup>89</sup> Cf. 4.1.4.5 *Word Lists, Reading Passages, and Minimal Pairs*.

(Moyer 2004: 43), a fact that often receives too little attention. Before the results of the present study are presented, a few words should be devoted to how the present study tackled audio segmentation and acoustic phonetic measurements. Furthermore, the technical equipment used will be discussed.

## 9 Acoustic Phonetic Measurement

### 9.1 Recording Devices

For the recordings, four DAT (Digital Audio Tape) devices were used. The DAT format was preferred as it records and converts higher sampling rates than CD or MiniDisc recorders, and, unlike MiniDisc recorders, DAT is not based on lossy data compression. Since DAT recorders are expensive and not many people use them, researchers from varied departments at the Universities of Berne and Zurich had to be contacted. Beat Siebenhaar from the German Department at the University of Berne generously offered his own private DAT recorder, a SONY DTC-57 ES, and a preamplifier of the type Behringer Ultragain Pro Mic 2200. The Phonogram Archive and the Phonetics Laboratory at the University of Zurich kindly provided three portable DAT recorders of the type SONY TCD-D7, SONY TCD-D8, and SONY TCD-D100. Three microphones of the type SONY ECM-MS907 and two highly sensitive condenser microphones of the type Behringer B2 were used with a directive angle of 90°. The speech was recorded with a sampling frequency of 48 kHz.

### 9.2 Audio Segmentation

The data on the DAT tapes needed to be digitized and segmented into separate word units in order to acoustically analyze them with PRAAT.<sup>90</sup> The DAT recorder was interfaced with an Apple Macintosh G5 computer, which allowed digitization of the data through optic inputs and outputs. The recordings were burnt onto DVDs with a sampling rate of 48 kHz. Subsequently, the sound files were segmented to the level of word units with Adobe Audition at the same sampling rate. Altogether the recordings comprise 320 minutes of recorded speech. The total of single segmented word units is 2660,<sup>91</sup> of which 2220 were analyzed with PRAAT and 440 were narrowly transcribed. Each sound unit that

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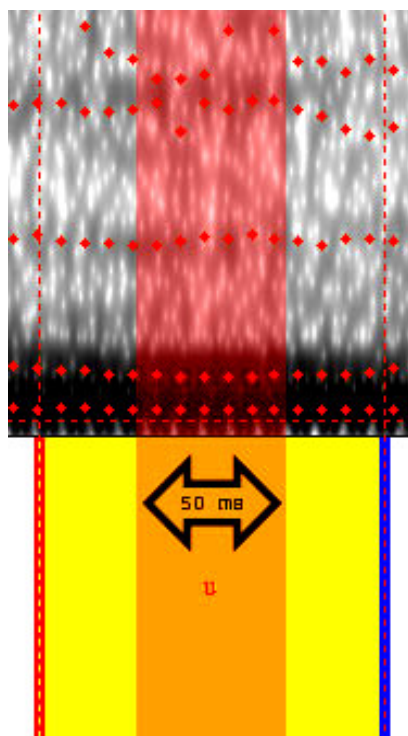
<sup>90</sup> Once the recordings were on DAT tapes, the devices needed to be returned to their owners, which meant that none of the DAT devices were available for the digitization process. Therefore, a DAT recorder of the type SONY DTC-27 was purchased.

<sup>91</sup> 51 tokens from the word lists with English vowels, cf. 6.1 *Word Lists with English Vowels*, 22 tokens from the English text, cf. 6.2 *Text with RP and GA Tokens*, and 60 tokens from word lists with Swiss German Vowels, cf. 6.3 *Word Lists with Swiss German Vowels*; multiplied by the number of subjects, i.e. 20 (subjects)\*133 = 2660.

was analyzed with PRAAT resulted in an F1 and an F2 value, which adds up to a total of 4440 measurements points.

### 9.3 Formant Measurement with PRAAT Script

4.4.4 *Coarticulation in Acoustic Phonetic Analysis* emphasized the importance of consistently applying the same measurement technique. This study opts for an interval measurement within the F1's and F2's most stable phases.<sup>92</sup> Once the spectrograms were labeled at the points where the formants were visibly most stable, interval measurement was conducted with a PRAAT script, kindly provided by Beat Siebenhaar.<sup>93</sup> Siebenhaar's script automatically measures the labeled vowel formants 10 milliseconds (ms) after the beginning of the vowel, 10 ms before the end of the vowel, in the middle of the vowel, at an interval of 50 ms around the middle of the vowel including its standard deviation, and at an interval of the total of the vowel minus 10 ms at the beginning and minus 10 ms at the end of the vowel including its standard deviation. For the final analyses, only the interval measurement of 50 ms around the vowel nucleus including its standard deviation were used, cf. Figure 9.1. Formant measurements were rechecked by hand and by observing the output standard deviations in Microsoft Excel.

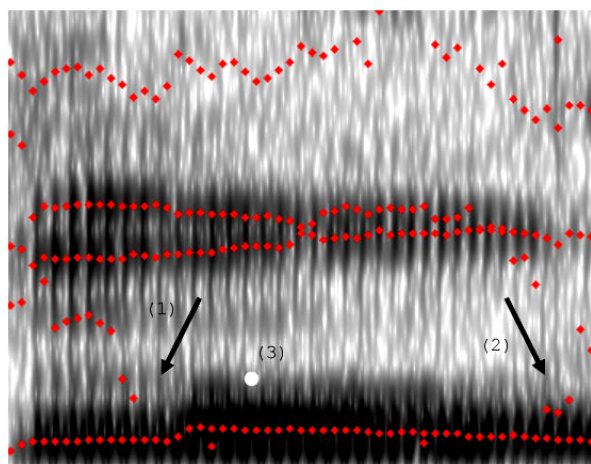


**Figure 9.1:** For this study, an interval measurement of 50 ms around the nucleus of the vowel was applied.

<sup>92</sup> Cf. Figure 4.14.

<sup>93</sup> Retrieved from <http://www.germanistik.unibe.ch/siebenhaar/subfolder/Aussprache/PraatscriptFormant.zip>, 2.4.2007.

If the measurement with this PRAAT script was not workable due to unfavorable spectrograms or because PRAAT refused to function properly for some other reason, formants were measured by hand according to their LPC traces at the temporal midpoint of the nucleus of the target formant. Where LPC (Linear Predictive Coding) traces were missing, the formants were measured by their gray scales. In Figure 9.2,<sup>94</sup> the F2 falls into F1 entirely at (1), caused by the low energy of the preceding nasal /m/.<sup>95</sup> Note also how F2 re-emerges at (2) due to the following sibilant /s/. In these cases, hand-measurement was conducted according to the formants' gray scales, cf. (3).



**Figure 9.2:** This spectrogram segment shows /u:/ as in the SG pronunciation of *Maus*.

If, however, neither PRAAT nor the eye could decipher formant values by examining the spectrograms' gray scales, the formant values were categorized as missing values.<sup>96</sup> The same procedure was applied in cases where the PRAAT script could not measure the vowel due to a formant's stable phase of less than 50 ms.

<sup>94</sup> Cf. 4.4.1 *Spectral Cues to Spectrogram Readings*.

<sup>95</sup> Cf. 4.4.1 *Spectral Cues to Spectrogram Readings*.

<sup>96</sup> For a detailed outline of values that were re-measured, measured by gray scales, or simply considered indecipherable, cf. *Appendix D Re-measured Tokens*.

## Section C: Data Analyses

We have come a good way into this paper without considering the actual results of this study. Initially, the findings of the word lists with the English vowels will be presented.<sup>97</sup> The analyses begin with a discussion of how the students articulated the English vowels differently during the three attempts. Secondly, the SE vowels are compared to the formant values of GA and RP. This accounts for the main part of this section. Thirdly, the SE vowels are contrasted with those of SG, and finally, the results of the comparison between SE and RP and between SE and GA, as well as the results of the comparison between SE and SG, are tested for correlations with information gained via the questionnaire.<sup>98</sup> The paper ends with final conclusions and with a brief outlook as to how findings and methodologies discussed in this paper could be applied in future EFL or ESL classroom settings.

## 10 Results Swiss English

### 10.1 Analysis of Repeated Measures

During the recordings, the five fieldworkers simultaneously observed that the subjects performed like native speakers to the greatest extent when they attempted to read the word lists for the third time. A general explanation for this is difficult to find. It could be argued that this improvement is attributable to learning effects. These learning effects are evidence for variation in the speakers' knowledge of English in that the words were pronounced differently in relation to the three attempts. This raises the issue of whether a comparison of the Swiss group's English to GA and RP is at all possible, given that extensive variation would put the reliability and subsequent use of the data into question. The following subchapter endeavors to verify any such learning effects and to discuss their respective implications for this study.

As a means of finding proof for variation, the number of words that were pronounced wrongly in progression from the first to the last attempt will be scrutinized. It will be shown that the number of missing values decreases, which corroborates the learning effect hypothesis. The downside of such an ad hoc analysis, however, is its lack of statistical power. Therefore, an ANOVA is further applied to explore whether or not variation is statistically significant.<sup>99</sup>

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<sup>97</sup> Cf. 6.1 *Word Lists with English Vowels*.

<sup>98</sup> Data analyses were conducted with SPSS 12.0 Advanced Version and vowel plots were made with jPlot Formants, [http://www.linguistics.ucla.edu/people/grads/billerey/JPlotFormants\\_1\\_4.zip](http://www.linguistics.ucla.edu/people/grads/billerey/JPlotFormants_1_4.zip), 3.4.2007.

<sup>99</sup> Henceforth simply referred to as significant.

### 10.1.1 Descriptives Missing Values

The students were asked to read every word carrying the target vowels three times. F1 and F2 were measured for each target vowel.<sup>100</sup> Descriptive statistics of these data shed light on the number of missing values.<sup>101</sup> Most often, missing values are words that were pronounced wrongly and therefore discarded from the analyses to avoid skewing of the data.<sup>102</sup> Every now and again, some vowels could not be analyzed on the grounds of spectrogram-related problems. For the back vowels /u:/, /ʊ/, and /ɔ:/, for instance, formant measurements with PRAAT were often difficult.<sup>103</sup> In the case of these vowels, the first two formants are in near-parallel distance which frequently results in a clashing of F1 and F2 and a subsequent loss of either F1's or F2's LPC traces.<sup>104</sup> Furthermore, missing values could also have been caused by poor spectrogram representations due to a speaker's creaky or breathy voice.<sup>105</sup> Unfavorable background noises, which led to the discarding of measured values twice.<sup>106</sup> The English tokens' F1 and F2 values and their corresponding frequencies of occurrence are listed in Table 10.1.<sup>107</sup>

<sup>100</sup> For a detailed presentation of Swiss English vowel formants see *Appendix E 1 F1s and F2s of Swiss English Vowels (Hz)*.

<sup>101</sup> Descriptives statistics is henceforth referred to as descriptives. Cf. *Appendix E 2 Descriptives to F1s and F2s of Swiss English Vowels*.

<sup>102</sup> Cf. *Appendix E 3 Transcription of Mispronounced Tokens in Swiss English and Swiss German Word Lists*.

<sup>103</sup> It could be argued that this may be the reason why Liebthal (2004) did not analyze high back vowels in her study "Accent in L2 – An acoustic analysis of the vowels of German speakers of English, cf. 4.3.4 Sociophonetic Approaches towards Standard English."

<sup>104</sup> Cf. *Figure 9.2*.

<sup>105</sup> Cf. 4.4.3 *Individual Voice Quality*.

<sup>106</sup> Cf. 4.4.2 *Sound Signal Interference*.

<sup>107</sup> Note: The tokens always occur in pairs. If, for example, F1 for *hudl* could not be measured, consequently neither could F2 for *hudl*.

| <i>Token</i> | <i>N</i> | <i>Token</i> | <i>N</i> | <i>Token</i> | <i>N</i> |
|--------------|----------|--------------|----------|--------------|----------|
| had1f1       | 20       | hid3f1       | 20       | hud3f1       | 17       |
| had1f2       | 20       | hid3f2       | 20       | hud3f2       | 17       |
| had2f1       | 20       | hid2f1       | 19       | whod1f1      | 17       |
| had2f2       | 20       | hid2f2       | 19       | whod1f2      | 17       |
| had3f1       | 20       | hod3f1       | 19       | whod2f1      | 17       |
| had3f2       | 20       | hod3f2       | 19       | whod2f2      | 17       |
| head1f1      | 20       | head3f1      | 18       | whod3f1      | 17       |
| head1f2      | 20       | head3f2      | 18       | whod3f2      | 17       |
| heard2f1     | 20       | hod2f1       | 18       | heard1f1     | 16       |
| heard2f2     | 20       | hod2f2       | 18       | heard1f2     | 16       |
| heard3f1     | 20       | hood1f1      | 18       | hawd1f1      | 15       |
| heard3f2     | 20       | hood1f2      | 18       | hawd1f2      | 15       |
| heed1f1      | 20       | hood2f1      | 18       | hawd2f1      | 15       |
| heed1f2      | 20       | hood2f2      | 18       | hawd2f2      | 15       |
| heed2f1      | 20       | head2f1      | 17       | hud2f1       | 15       |
| heed2f2      | 20       | head2f2      | 17       | hud2f2       | 15       |
| heed3f1      | 20       | hod1f1       | 17       | hawd3f1      | 14       |
| heed3f2      | 20       | hod1f2       | 17       | hawd3f2      | 14       |
| hid1f1       | 20       | hood3f1      | 17       | hud1f1       | 13       |
| hid1f2       | 20       | hood3f2      | 17       | hud1f2       | 13       |

**Table 10.1:** Each English /hVd/ token and its frequency of occurrence.

As table 10.1 shows, words such as *hud*, *hawd*, *whod*, *hood*, and *hod* demonstrate the greatest number of missing values while *had* is remarkably consistent in all 6 values,<sup>108</sup> and *head*, *heard*, *heed* and *hid* show almost no missing values. If Table 10.1 is simplified by adding together the missing values of each token for the three attempts,<sup>109</sup> we obtain the following results (added missing values are in parentheses): *had* (0), *heed* (0), *hid* (1), *heard* (4), *head* (5), *hod* (6), *hood* (7), *whod* (9), *hud* (15), and *hawd* (16). These figures suggest that out of 60 possible measurements of *hawd*, for example, only 44 could be measured.

There are, however, a number of difficulties with this table. On the one hand, it omits specific indications as to why tokens were removed.<sup>110</sup> It also fails to provide details about whether the missing value originated from the first, second, or third attempt. Therefore, we need to add Table 10.2 which denotes the number of missing values (MV), the total number of mispronunciations (MP), and the number of mispronunciations present in the first, second and third

<sup>108</sup> The six values of *had1f1*, *had1f2*, *had2f1*, *had2f2*, *had3f1*, *had3f2*, or, translated to a more general syntax: (token)(attempt)(formant).

<sup>109</sup> Regardless of the token's separation into F1 and F2.

<sup>110</sup> Either the tokens were discarded due to mispronunciations on the part of the subjects or due to spectrogram-related reasons.

attempts.<sup>111</sup> To show how Table 10.2 ought to be read, the example of *hawd* will be commented on briefly: The token *hawd* has 16 missing values (MV), 13 of which are due to mispronunciation (MP), four of which were pronounced incorrectly in the first attempt (1<sup>st</sup>), four in the second attempt (2<sup>nd</sup>), and five in the third attempt (3<sup>rd</sup>).

| <i>Token</i> | <i>MV</i> | <i>MP</i> | <i>1<sup>st</sup></i> | <i>2<sup>nd</sup></i> | <i>3<sup>rd</sup></i> |
|--------------|-----------|-----------|-----------------------|-----------------------|-----------------------|
| hawd         | 16        | 13        | 4                     | 4                     | 5                     |
| hud          | 15        | 15        | 7                     | 5                     | 3                     |
| whod         | 9         | 4         | 2                     | 1                     | 1                     |
| hood         | 7         | 4         | 1                     | 2                     | 1                     |
| hod          | 6         | 6         | 3                     | 2                     | 1                     |
| head         | 5         | 5         | 0                     | 3                     | 2                     |
| heard        | 4         | 3         | 3                     | 0                     | 0                     |
| hid          | 1         | 1         | 0                     | 1                     | 0                     |

**Table 10.2:** Discarded Swiss English tokens due to mispronunciations on the part of the subjects.

These figures clearly show that, with a few exceptions, nearly all the missing values for *hawd* can be attributed to mispronunciations rather than to spectrogram-related problems. In addition, Table 10.2 points towards the fact that the number of missing values decreases from the first to the third attempt. The total of missing values in the first attempt adds up to 20, in the second attempt to 18, and in the third to 13. These findings vaguely suggest an improvement in articulation as the subjects proceeded from one attempt to the next. They thus provide evidence for variation. Whether this variation is statistically significant will be discussed in the following subchapter.

### 10.1.2 Repeated Measures ANOVA

Before examining the results, it is essential to briefly acquaint the reader with some of the universalities regarding statistical tests and statistical conventions, as the consecutive subchapters rely on a basic understanding of them. Whereas z- and t-tests are useful to assess the significance of the differences between the means of two independent samples, ANOVAs allow for a comparison of the overall means between more than two samples (Butler 1985: 129). A repeated measures ANOVA is specifically used in situations when each member of a random sample is measured in different conditions. In the case of the present study, these varied conditions are represented by the first, second, and third

<sup>111</sup> For an in-depth presentation of the tokens that were mispronounced, see *Appendix E 3* Transcription of Mispronounced Tokens in Swiss English and Swiss German Word Lists.

attempt. The entire group of these tests provides significance values. Johnstone (2000) conceives of statistical significance as follows:

To be significant for the purposes of many researchers, for example, research results have to be quantified – turned into numbers – and then undergo statistical tests that show that the results have less than some predefined chance – less than 5% or 1% - of having occurred randomly. (Johnstone 2000: 58-59)

The varying predefined chances that Johnstone alludes to are those of the natural and social sciences. In physics, the significance level is often  $p \leq .01$  or even  $p \leq .001$ , which results from the usually high degree of accuracy in measurements. In the social sciences, including linguistics, a level of  $p \leq .05$  is standard (Butler 1985: 71). This study equally adheres to the convention of a level of significance ( $\alpha$ ) to  $p \leq .05$ , yet it treats  $p$  values  $.05 < p \leq .2$  as approximations to this  $\alpha$  value. Before the first results are delved into, however, a few words should also be devoted to the interpretation of  $p$  values.

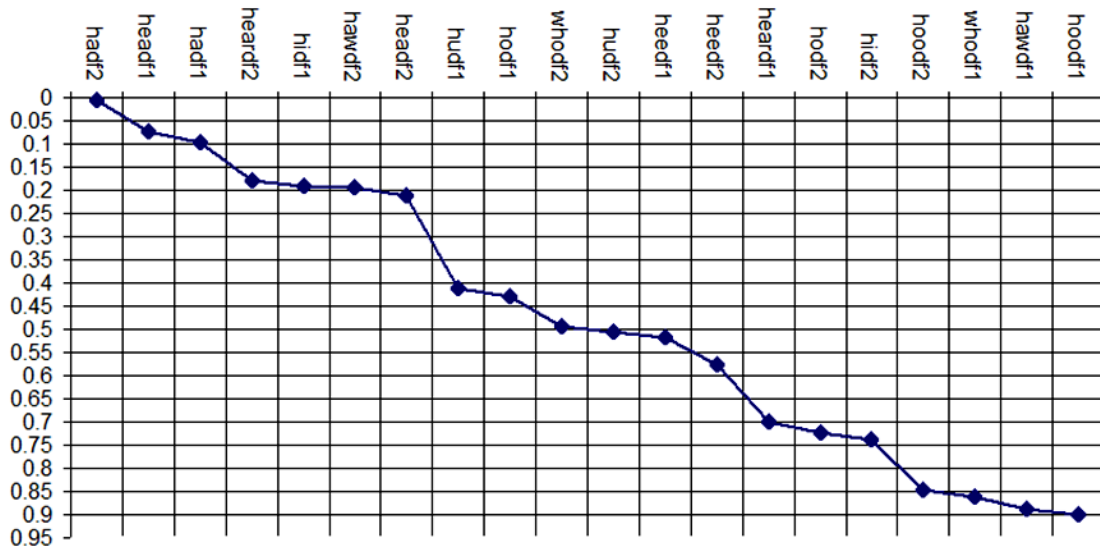
There are mainly two approaches of how  $p$  values can be understood. On the one hand, there are those statisticians who argue that once a threshold  $p$  value is set, every result is statistically significant or statistically not significant. On the other hand, one intuitively distinguishes between  $p$  values such as  $p=.001$  or  $p=.00001$  etc. and categorizes them as “very significant” or “extremely significant” etc. respectively. However, there does not seem to be a consensus among statisticians whether or not this finer distinction of  $p$  values is legitimate. The present paper adheres to the second approach as discussed in Bühl’s and Zöfel’s *SPSS 11* (2002). They state that results which are afflicted with a probability value  $p \leq .05$  are called significant, those with a  $p$  value of  $p \leq .01$  very significant, and those with a value of  $p \leq .001$  are referred to as extremely significant (111).

A repeated measures ANOVA of the SE F1 and F2 values factored by the three attempts provides the  $p$  values illustrated in Figure 10.1. Figure 10.1 shows a graphic representation of the  $p$  values for each target English vowel in their F1s and F2s in ascending order.<sup>112</sup> Probability values were calculated with the Huynh-Feldt test in SPSS. *hadf2* demonstrates the lowest  $p$  value,<sup>113</sup> whereas *hoodf1* demonstrates the highest  $p$  value.<sup>114</sup> *hadf2* is the only token with a  $p$  value below the one per cent level. *headf1* and *hadf1* slightly exceed a  $p$  value of .05. *headf1*, *hadf1*, *heardf2*, *hidf1*, *hawdf2*, and *headf2* all show  $p$  values  $\leq .2$  or around .2. These findings suggest that the F2 of /æ/ is articulated significantly different between the three attempts, as opposed to all other F1s and F2s of the rest of the vowels which demonstrate  $p$  values  $< .05$ .

<sup>112</sup> Cf. Appendix E 4 Repeated Measures ANOVA of Swiss English Vowels.

<sup>113</sup> With a  $p$  value close to zero.

<sup>114</sup> With a  $p$  value close to one.



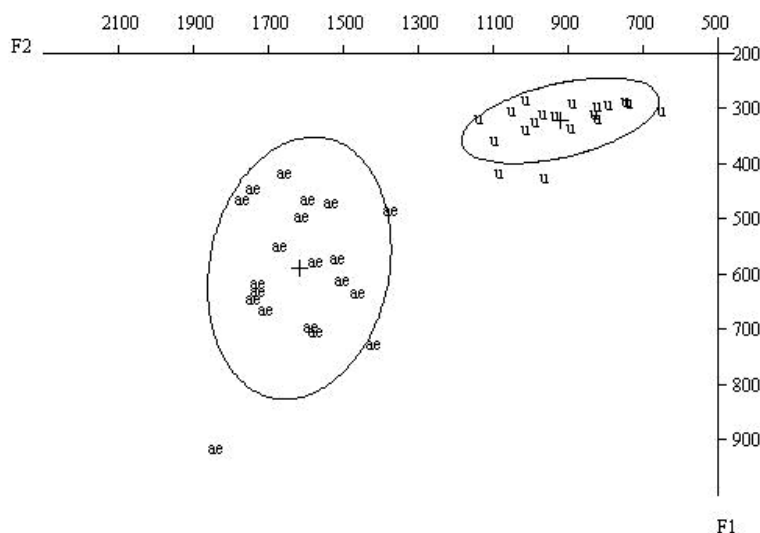
**Figure 10.1:** Line graph of p values of SE, obtained by a Huynh-Feldt test through a repeated measures ANOVA.

It is important to realize that although there are a few p values approximating zero, this one instance of a p value below .05 does not constitute adequate evidence for the hypothesized variation. The findings of the repeated measures ANOVA does, however, present one other interesting aspect. If the graph is viewed from the perspective of vowel frontedness and backness, it becomes clear how the graph's downward shape parallels a shift in the vowels from front to a back position.<sup>115</sup> The most obvious instances for this shift from frontedness to backness are perhaps visible in the p values for /æ/ and /ɛ/ versus those of /u:/, /ʊ/, and /ɑ:/. The p value for /æ/ F1 is  $.05 < p \leq .2$ . Analogously, the mid-low front vowel /ɛ/ equally shows p values approximating zero. In sharp contrast, back vowels /u:/, /ʊ/, and /ɑ:/ represent the lower end of the graph as they exhibit p values closely approximating one.

These findings essentially propose that front vowels, displaying p values near zero, present significant variation among the three attempts, while back vowels, demonstrating p values near one, are articulated without extensive variation. In order to understand the implications of this, we need to look at interpretations of F1 and F2 values. The lower the F1 value the closer the tongue to the roof of the mouth, or, put differently, the lower the F1 values, the higher the vowel. The F2 value, conversely, is proportional to the frontedness or backness of the highest part of the tongue during the vowel production. That is to say, the higher the F2 value, the more fronted the vowel (Ladefoged 1993: 217). Consider the following example. The high degree of variation on both axes, high-low (F1)

<sup>115</sup> Regardless of the vowels' F1s and F2s.

and front-back (F2), in /æ/, for instance, may be due to the Swiss speakers' articulation of *had* as /hæd/, /hɛd/, /had/, and /hɛd/, indicating variation on the x-axis (F1); or pronunciation of *had* as /hɑ:d/, which would be evidence for variation on the y-axis (F2). This type of variation can best be visualized with vowel plots, as in Figure 10.2.



**Figure 10.2:** Vowel plot of SE /æ/ and /ʊ/.

The ellipse around *ae* represents the amount of variation found in /æ/, while the ellipse around *u* denotes the small amount of variation of /ʊ/.<sup>116</sup> /ʊ/ is fairly stable on the F1 axis, but there is visible, yet insignificant, variation on the F2 axis.<sup>117</sup>

### 10.1.3 Discussion

This subchapter accounts for the above findings by discussing markedness, similarity in language, and stylistic factors as outlined in 8.2 *External Variables*. It was noted earlier that the English tokens *hawd* and *hud* would be the most difficult for the Swiss to pronounce as they do not convey any semantic contents.<sup>118</sup> Applying Eckman's MDH to these cases (1977: 321), such words are marked due to their statistical infrequency, from which it follows that the words are more difficult to acquire (ibid: 321). The previous analysis verifies

<sup>116</sup> Unfortunately, jPlotFormants (2005) does not allow the use of IPA symbols. Therefore, *ae* = /æ/ and *u* = /ʊ/.

<sup>117</sup> Cf. Figure 10.2.

<sup>118</sup> Cf. 6.1 *Word Lists with English Vowels*.

this hypothesis: It is exactly the marked forms *hawd* and *hud* which show the greatest number of missing tokens.<sup>119</sup>

Moreover, in 8.2.2 *Markedness Universals* it was argued that statistical frequencies of spoken English sounds could equally provide information concerning the markedness of a linguistic element. Accordingly, the front vowels /ɪ/ and /ɛ/ should be acquired most readily because they are two of the most frequently occurring vowels in the English language. /ɜ/, /ɑ:/, /ʊ/, and /u:/, in contrast, should be most difficult to acquire due to their infrequency. It may well be argued that difficulty in acquisition reinforces variation. Articulating difficult sounds could cause varying degrees of uncertainty on the part of the speaker, which in turn affects his pronunciation and thus induces variation.

However, the results presented in Figure 10.1 suggest the following: The vowel /ɛ/ manifests p values close to zero in both its F1 and F2, although /ɛ/ is the second most frequent vowel in spoken RP and therefore highly unmarked.<sup>120</sup> Also, if marked forms are theoretically more difficult to acquire, the vowels /ʊ/ and /u:/, statistically infrequent vowels in spoken RP, should vary extensively in SE, which, according to the p values shown in Figure 10.1, this is not the case. Judged by these criteria, the unmarked/marked frequency count analysis provides neither valid nor confirmatory evidence that variation in SE is due to the markedness of certain phonemes.

With regard to similarity in language, Flege (1995) considers similar sounds as being more difficult to acquire and thus prone to variation, because the speaker classifies the sounds as equivalent.<sup>121</sup> Dissimilar sounds, on the other hand, are often produced more authentically, with less variation, as the speaker perceives the differences and creates a new phonemic category. Evidence from the present study is, however, ambiguous. One could suggest that Flege's (1995) hypothesis explains why the SE vowels /æ/ and /ɛ/ both show p values  $.05 < p \leq .2$ . The SG counterparts of /æ/ and /ɛ/ may be very similar. On the other hand, the F2 value of /ɜ/, a dissimilar sound, equally displays a rather low p value.<sup>122</sup> There is no doubt that stylistic factors also contribute to the probability of variation in the results. However, it is too much to claim that one could give specific examples of SE tokens that were allegedly affected by stylistic variation. *Stylistic variation* is too broad and vague a concept to allow for such attributions.

If stylistic factors are viewed in light of the discussion on missing values,<sup>123</sup> however, a few speculative remarks can be made. As noted already,

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<sup>119</sup> Cf. Table 10.2.

<sup>120</sup> Cf. Table 8.3.

<sup>121</sup> Cf. 8.2.1 *Similarity in Language*.

<sup>122</sup> Cf. Figure 10.1.

<sup>123</sup> Cf. 10.1.1 *Descriptives Missing Values*.

Romaine's understanding of stylistic variation in second language learning is that different styles are contingent on the amount of attention paid to speech (2003: 422). It is interesting to speculate whether the decrease of missing values from the first to the third attempt could perhaps be attributed to the students' feeling less anxious and observed while progressing through the experimental situation. The corollary of this decrease in anxiety may in turn have induced a reduction in speech monitoring on the speakers' part, which subsequently led them to articulate each token as though they were not being observed. If this shift in style were viewed according to Tarone's interlanguage continuum (1988: 41),<sup>124</sup> the speakers may have performed a shift from Style 4, highly formal, to Style 2, rather casual.

#### 10.1.4 Preliminary Conclusions

It is difficult to escape the conclusion that although linguistic factors may account for some of the reasons for the probability of variation between the three attempts, they ought to be approached critically. The assumed learning effects were confirmed by observing how the missing values exhibit a decline in amount from the first to the third attempt. Yet, by means of a repeated measures ANOVA, it was added that this proof for differences in the results is not statistically significant, which means that the data set can be used for further analyses, i.e. it is reliable. It was also noted that SE front vowels typically show  $p$  values  $.05 < p \leq .2$ , while SE back vowels demonstrate  $p$  values near one. The scrutiny of markedness in the context of the number of missing values has shown how words, such as *hawd* or *hud*, that are inexistent and thus marked, indeed posed the greatest problem for the SG speakers. Additionally, the analyses indicated that statistical frequencies of phonemes in spoken RP do not provide confirmatory explanations for variation in SE. Finally, it was stressed how social-psychological factors such as anxiety are likely to play a crucial role in contributing to variation.

### 10.2 Comparison to Reference Points

#### 10.2.1 Comparison to RP

Deterding (1997) offers the benchmark RP values for this study. Deterding measured formant values of five women and five men, whose speech he collected from the MARSEC database.<sup>125</sup> There are a number of difficulties in comparing the present study's values with those of Deterding's (1997) as there are various differences in the data collection procedure. Primarily, it needs to be reiterated that Deterding's values are retrieved from conversational speech whereas the findings of this study are largely derived from phonologically

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<sup>124</sup> Cf. *Figure 8.1*.

<sup>125</sup> Cf. 4.3.4.2 *Deterding's (1997) Study on British English Vowels*.

standardized /hVd/ constructions. The major problem is that coarticulatory effects are ubiquitous and varied in conversational speech, while in isolated word list items they are controlled and reduced to a minimum.<sup>126</sup> Additionally, it should not be forgotten that Deterding admits some degree of variation among his values (Deterding 1997: 48).

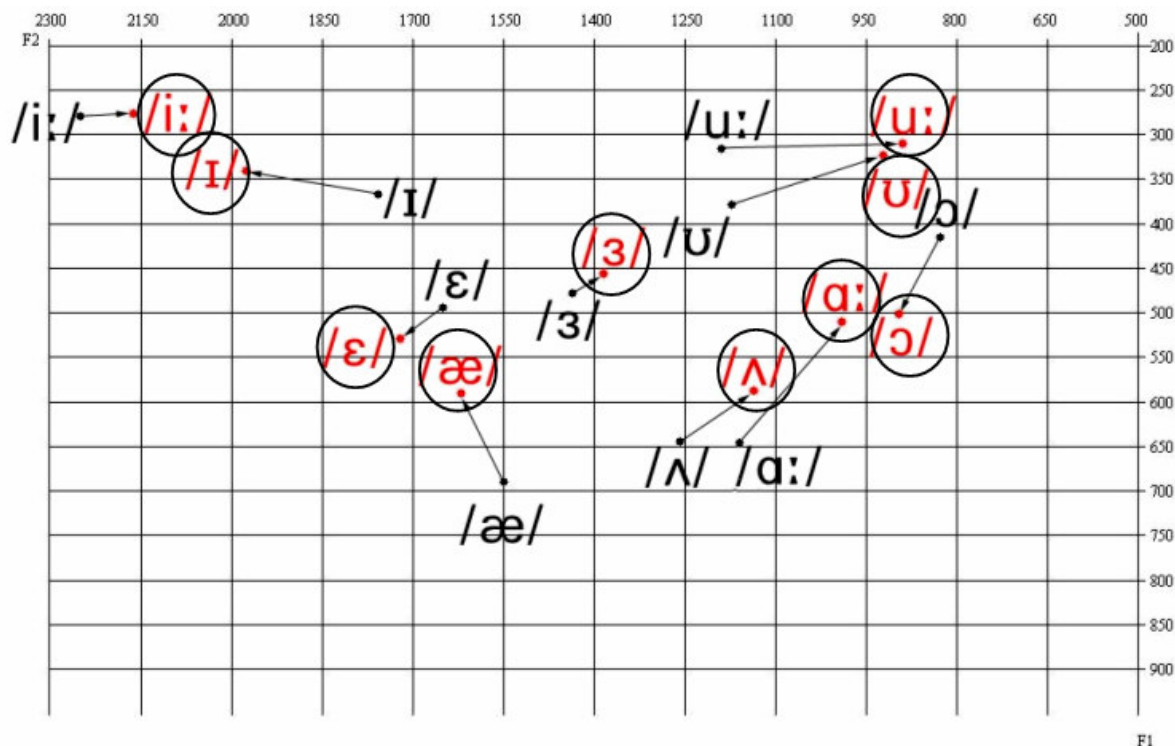
In terms of measurement techniques, Deterding's explanations seem a bit reserved. He maintains that "[c]lear instances of each vowel were identified by listening, and then digital spectrograms were derived, with overlaid linear-prediction-based [i.e. LPC] tracks, using a pre-emphasis coefficient of 0.9" (ibid: 48). Deterding circumvents specifically commenting on how formants were measured, which is why when viewing the subsequent results it needs to be remembered that it is likely that a different measurement technique was used in Deterding's study.<sup>127</sup> For all these reasons, the consecutive results should be treated with caution.

Prior to delving into a detailed assessment of F1 and F2 differences between SE and RP, consider the vowel plot of SE and RP vowels in Figure 10.3. It offers a general idea as to the differences in pronunciation between SE vowels and RP benchmark values. Deterding's (1997) RP values are represented in black, SE values are circled. In Figure 10.3, the arrows indicate the extent to which and in what direction the SE vowel is articulated differently than its RP counterpart.

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<sup>126</sup> Deterding did not scrutinize the vowels that occurred after /j/, /w/, and /r/ or before /l/ because these approximants "would have severe coarticulatory effects on the locations of the first three formants" (1997: 49).

<sup>127</sup> It is worth highlighting that differences in measurement techniques can affect the outcomes, cf. 4.4.4 *Coarticulatory Effects in Acoustic Phonetic Analysis*.



**Figure 10.3:** Vowel plot of SE (circled) and RP English (black).

It's clearly visible that the SE vowels /ʊ/, /u:/, /ɑ:/, and /ɪ/ are farther away from the target RP forms than the SE vowels /i:/, /ɛ/, /ɜ/, /æ/, /ʌ/ are. In order to endorse such visible trends with statistics, one-sample t tests were run with the SE F1 and F2 values and RP F1 and F2 values for each vowel.

#### 10.2.1.1 One-Sample T Test: SE F1 Values Tested by RP F1 Values

A one-sample t test comes in use for a comparison between a single sample, i.e. an RP value, and a population value, i.e. SE values, whereby the mean values are tested. One-sample t tests, just like ANOVAs, offer a p value. The same  $\alpha$  values conventions which were used in the repeated measures ANOVA were applied here.<sup>128</sup> This implies that if the p value of this one-sample t test is less than .05, there is significant difference between the mean value of the Swiss speakers' F1s and the corresponding single-sample F1 value of RP. The p values are presented in ascending order in Table 10.3.<sup>129</sup>

Table 10.3 suggests that virtually all SE F1 values (indicating vowel height) are articulated significantly different from RP, except for the F1s of /i:/

<sup>128</sup> Cf. 10.1.2 Repeated Measures ANOVA.

<sup>129</sup> Cf. Appendix E 5 One-Sample T Test of SE F1s and F2s Tested by RP Values.

and /u:/. This implies that the heights of the SE vowels /i:/ and /u:/ are probably very similar to those of RP. For an even more evident illustration of this nearly identical height, one only needs to look at the visualization of it in Figure 10.3.<sup>130</sup>

| Vowel  | <i>p</i> |
|--------|----------|
| /ɑ:/fl | 4.06E-07 |
| /u/fl  | 8.92E-06 |
| /ɔ:/fl | 0.00011  |
| /ɪ/fl  | 0.00048  |
| /æ/fl  | 0.00168  |
| /ɜ/fl  | 0.00866  |
| /ʌ/fl  | 0.03607  |
| /ɛ/fl  | 0.05107  |
| /i:/fl | 0.35879  |
| /u:/fl | 0.35992  |

**Table 10.3:** T-Test significance values of SE vowels tested by the F1 of RP.

Figure 10.3 shows that nearly every SE vowel is pronounced higher than in RP. This concerns the vowels /ɑ:/, /ɪ/, /æ/, /ɜ/, and /ʌ/, with *p* values < .05, and the vowels /i:/ and /u:/. The last two vowels are not articulated *significantly* higher, however, as they present *p* values that exceed .2. The vowels /ɛ/ and /ɔ/, in contrast, are articulated significantly lower.

#### 10.2.1.2 One-Sample T Test: SE F2 Values Tested by RP F2 Values

Analogous to the previous statistical test, the same one-sample *t* test procedure can be applied to the F2 values.<sup>131</sup> Table 10.4 shows the calculated *p* values. According to these statistics, there are many significant differences between SE and RP in terms of vowel backness (F2), as each *p* value is below .05. If these results are again contemplated in the context of Figure 10.3, it is evident that six out of ten observed vowels, /i:/, /ɜ/, /ʌ/, /ɑ:/, /u/, and /u:/, are articulated significantly further back than in RP, spearheaded by /u:/, which constitutes the most extreme case. The reverse is visible when the vowels, /ɪ/, /ɛ/, /ɔ/, and /æ/ are analyzed as these are articulated at a more fronted position, with /ɪ/ spearheading. One of the most important points to observe is that all SE back vowels, with the exception of /ɔ:/, are pronounced further back than the RP back vowels. Analogously, all SE front vowels, except for /i:/, are pronounced more fronted than the RP front vowels.

<sup>130</sup> Note the minor variation on the x-axis (indicating vowel height).

<sup>131</sup> Cf. *Appendix E 5 One-Sample T Test of SE F1s and F2s Tested by RP Values*.

| Vowel  | <i>p</i> |
|--------|----------|
| /u:/f2 | 5.99E-09 |
| /ɪ/f2  | 1.55E-08 |
| /ʊ/f2  | 2.23E-07 |
| /ɑ/f2  | 3.73E-06 |
| /ʌ/f2  | 3.80E-06 |
| /i:/f2 | 0.00497  |
| /ɜ/f2  | 0.00691  |
| /ɛ/f2  | 0.01046  |
| /ɔ:/f2 | 0.01165  |
| /æ/f2  | 0.02262  |

**Table 10.4:** T-Test significance values of SE vowels tested by the F2 of RP.

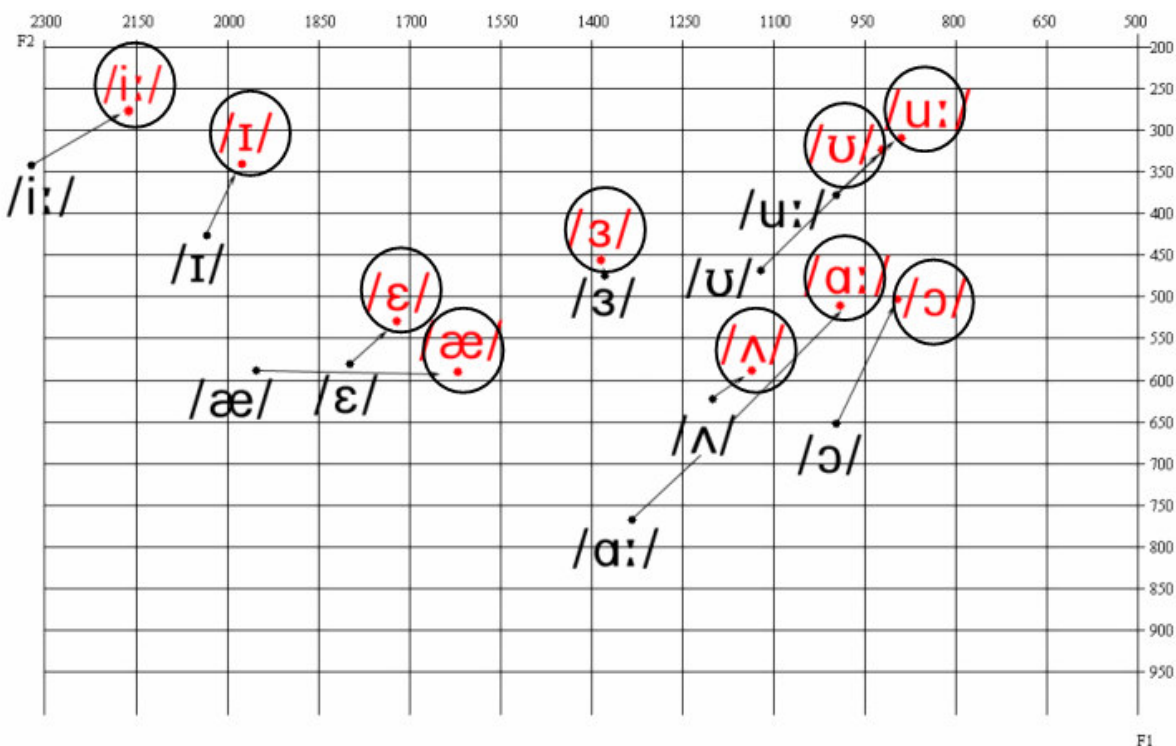
### 10.2.2 Comparison to GA

Hillenbrand et al. (1995) offers the benchmark GA values for this study. With the aim of replicating Peterson and Barney's (1952) study, they recorded 45 men, 48 women, and 46 children. Before the results are discussed, a number of qualifications should again be added. Hillenbrand et al. rely on the same methodological framework as the one presented in this study, the use of embedded target vowels in /hVd/ syllables. This similarity is advantageous, yet there are differences as Hillenbrand et al. note that "editing and analysis decisions were based on an examination of the LPC peak display overlaid on a gray-scale spectrogram and, in some cases, on an examination of individual LPC or Fourier spectral slices" (1995: 3101). In this study on the other hand, measurements were conducted at an interval of 50ms around the vowel nucleus.<sup>132</sup> This difference in measurement techniques warns us that results must be viewed with some critical distance.

Figure 10.4 shows the vowel plot of SE and GA vowels. The plot demonstrates clearly that /ɛ/, /ɜ/, and /ʌ/ are pronounced similar to GA, while all other vowels present greater discrepancies in either their F1 or F2 values. It is further conspicuous how especially SE /ɑ:/, /i:/, and /ɔ/ are articulated most differently. Once more, to verify such visible trends statistically, a one-sample t test was carried out where the SE F1 and F2 values for each vowel were tested by the corresponding GA F1 and F2 value.<sup>133</sup>

<sup>132</sup> Cf. 9.3 Formant Measurement with PRAAT Script.

<sup>133</sup> Cf. Appendix E 6 One-Sample T Test of SE F1s and F2s Tested by GA Values.



**Figure 10.4:** Vowel plot of SE (circled) and GA English (black).

#### 10.2.2.1 One-Sample T Test: SE F1 Values Tested by GA F1 Values

Table 10.5 summarizes the obtained p values of the one-sample t test. The results suggest that in terms of vowel height, all vowels except for /æ/ ( $p = .929$ ) and /ʌ/ ( $p = .175$ ) are articulated significantly different from GA, with p values that are below .0001. One can further infer that /i:/, /u:/, /ɑ:/, and /ɪ/ differ the most, with p values close to 0, whereas /ɜ:/, /ʌ/, and /æ/ diverge the least, with p values close to 1. If these findings are added to the ones in Figure 10.4, it is evident that the Swiss group pronounces all the vowels higher than GA, except for /æ/ which is pronounced at an almost equal level to the General American /æ/ in its F1. Whether the exploration of the SE F2 values in contrast to the F2 values of GA brings about equally unambiguous results will be discussed next.

| <b>Vowel</b> | <b><i>p</i></b> |
|--------------|-----------------|
| /i:/f1       | 7.02E-14        |
| /u/f1        | 2.19E-13        |
| /ɑ:/f1       | 1.26E-11        |
| /ɪ/f1        | 2.08E-11        |
| /u:/f1       | 2.70E-08        |
| /ɔ:/f1       | 1.55E-07        |
| /ɛ/f1        | 0.00703         |
| /ɜ/f1        | 0.02666         |
| /ʌ/f1        | 0.17455         |
| /æ/f1        | 0.92818         |

**Table 10.5:** T-Test significance values of SE vowels tested by the F1 of GA.

### 10.2.2.2 One-Sample T Test: SE F2 Values Tested by GA F2 Values

The same test was run for the F2 values.<sup>134</sup> Table 10.6 looks at the *p* values obtained, and again shows that in effect all *p* values are below the  $\alpha$  value of .05. In fact, half of them are below .00001. The only statistically not significant value is found in the F2 of /ɜ/, which, according to its *p* value, is pronounced nearly the same way as the F2 GA counterpart. In the context of Figure 10.4 it becomes visible that the Swiss under scrutiny articulate all vowels further towards the back, except for /ɜ/. This is especially the case for the vowels /ɑ:/ and /æ/. Events show that the examination of the F2 values offers similar clear-cut results as the checking of the F1 values.

| <b>Vowel</b> | <b><i>p</i></b> |
|--------------|-----------------|
| /ɑ:/f2       | 3.78E-11        |
| /æ/f2        | 3.31E-10        |
| /u/f2        | 4.95E-06        |
| /i:/f2       | 1.24E-05        |
| /ɔ:/f2       | 0.00106         |
| /u:/f2       | 0.00114         |
| /ʌ/f2        | 0.0028          |
| /ɛ/f2        | 0.00461         |
| /ɪ/f2        | 0.01957         |
| /ɜ/f2        | 0.76001         |

**Table 10.6:** T-Test significance values of SE vowels tested by the F2 of GA.

### 10.2.2.3 Discussion

This subsection aims to synthesize the findings of the reference point comparisons into a coherent whole in that it attempts to demonstrate which SE vowels are articulated closer to RP or closer to GA. In order to draw such

<sup>134</sup> Cf. *Appendix E 6 One-Sample T Test of SE F1s and F2s Tested by GA Values*.

conclusions, each SE vowel and its corresponding p values, which have been obtained through the tests between SE & RP and SE & GA, will be scrutinized. For the purposes of this study, the difference in p values was understood as a greater distance between vowels in the vowel space. Table 10.7 pools together the p values derived from Tables 10.3 – 10.6. The last column shows whether the Swiss are closer to RP or GA. The more the p value approximates zero, the greater the differences in articulation between the groups.

| <b>VOWEL</b> | <b>F1/F2</b> | <b>p RP</b> | <b>p GA</b> | <b>Swiss closer to</b> |
|--------------|--------------|-------------|-------------|------------------------|
| /i:/         | /i:/f1       | 0.35879     | 7.02E-14    | RP                     |
|              | /i:/f2       | 0.00497     | 1.24E-05    | RP                     |
| /ɪ/          | /ɪ/f1        | 0.00048     | 2.08E-11    | RP                     |
|              | /ɪ/f2        | 1.55E-08    | 0.01957     | GA                     |
| /ɛ/          | /ɛ/f1        | 0.05107     | 0.00703     | RP                     |
|              | /ɛ/f2        | 0.01046     | 0.00461     | RP                     |
| /ɜ/          | /ɜ/f1        | 0.00866     | 0.02666     | GA                     |
|              | /ɜ/f2        | 0.00691     | 0.76001     | GA                     |
| /æ/          | /æ/f1        | 0.00168     | 0.92818     | GA                     |
|              | /æ/f2        | 0.02262     | 3.31E-10    | RP                     |
| /ɑ:/         | /ɑ:/f1       | 4.06E-07    | 1.26E-11    | RP                     |
|              | /ɑ:/f2       | 3.73E-06    | 3.78E-11    | RP                     |
| /ɔ:/         | /ɔ:/f1       | 0.00011     | 1.55E-07    | RP                     |
|              | /ɔ:/f2       | 0.01165     | 0.00106     | RP                     |
| /ʊ/          | /ʊ/f1        | 8.92E-06    | 2.19E-13    | RP                     |
|              | /ʊ/f2        | 2.23E-07    | 4.95E-06    | GA                     |
| /ʌ/          | /ʌ/f1        | 0.03607     | 0.17455     | GA                     |
|              | /ʌ/f2        | 3.80E-06    | 0.0028      | GA                     |
| /u:/         | /u:/f1       | 0.35992     | 2.70E-08    | RP                     |
|              | /u:/f2       | 5.99E-09    | 0.00114     | GA                     |

**Table 10.7:** P values comparison, cf. Tables 10.3 – 10.6. The last column indicates whether the Swiss are closer to RP or GA.

Table 10.7 highlights how /i:/, in both its F1 and F2 values, is articulated more RP-like. In its F1, the front vowel /ɪ/ is also pronounced more like RP, in terms of /ɪ/'s backness, however, the Swiss group is closer to GA. The front vowel /ɛ/, too, sounds more RP-like, yet the p values for RP F1 and F2 are only minimally greater than those of GA. /ɜ/ is articulated more GA-like, especially in its F2 value. In terms of its F1 value, the low front vowel /æ/ is nearly identical to that of GA. As regards the F2 value of /æ/, conversely, the

Swiss are closer to RP. Their vowel /ɑ:/ was the most different vowel from the perspective of how it deviates from RP and GA. Despite these great discrepancies between SE and GA as well as between SE and RP, /ɑ:/ is articulated more RP-like in both its frontedness-backness (F2) and its height (F1). The pronunciation /ɔ:/ is more similar to RP than to GA. /ʊ/ demonstrates the same p value patterns as /ɑ:/ in that the vowel is pronounced very differently from GA and from RP. With regard to the first formant of /ʊ/, the Swiss approximate RP pronunciation, while, for the second formant, they lean towards GA. The mid-low back vowel /ʌ/ has an Americanized pronunciation in both its F1 and its F2. Finally, the vowel /u:/ sounds more RP-like in its height, in its frontedness-backness its articulation is closer to GA.

All values considered, this examination suggests that in terms of vowel height, seven out of ten vowels have a more RP-like articulation, while three are pronounced rather GA-like. With respect to vowel backness and frontedness, the results are balanced in that five vowels are articulated in GA fashion and five in RP fashion. Although the interconnectedness between the F1 and F2 values challenges the formulation of generalizations, one could speculate that the Swiss group leans towards RP rather than towards GA as far as vowels are concerned.

### 10.2.3 Preliminary Conclusions

A one-sample t test of the Swiss' average F1 and F2 values with the RP and GA test values has shown the following results: In comparison with the F1 values of RP, the Swiss pronounce /ɑ:/, /ʊ/, /ɪ/, /æ/, /ɜ:/, /ʌ/, /i:/, /u:/ (the last two of this set are not statistically significant) significantly higher than an RP speaker does, while the vowels /ɛ/ and /ɔ/ are articulated significantly lower than in RP. With regard to vowel backness and frontedness, the Swiss articulate /i:/, /ɜ:/, /ʌ/, /ɑ:/, /ʊ/, and /u:/ significantly further back than RP and /ɪ/, /ɛ/, /æ/, and /ɔ/ significantly further fronted. The comparison with GA provided different results: With the exception of /ʌ/ and /æ/ the Swiss articulate all English vowels significantly higher than GA. /ʌ/ is pronounced at the same level as GA, and /æ/ (not statistically significant) is pronounced slightly lower. In terms of the F2 values, they articulated each vowel significantly further back, except for /ɜ/ (not statistically significant), which is minimally more fronted. This chapter ends on the note that one could suggest a trend in the Swiss group to pronounce English vowels more RP-like than GA-like, particularly in terms of the vowels' frontedness and backness.

The crucial question remains as to how these findings can be accounted for. There are various approaches to this issue, some of which have been outlined in chapter 8. *Approaching Language Variation*. Most modern research assumes that external variables such as sex, age, experience, speech perception,

motivation, attitudes, input, L1 interference, etc., as well as internal variables such as markedness, similarity in language, stylistic factors, etc., affect foreign language production. While internal variables have been extensively discussed in the investigation of variation among the three attempts,<sup>135</sup> external variables have not yet received adequate attention. The subsequent sections try to detect potential correlations between external variables and the previously outlined findings of *10.1 Analysis of Repeated Measures* and *10.2 Comparison to Reference Points*.

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<sup>135</sup> Cf. *10.1 Analysis of Repeated Measures*.

## 11 Seeking Explanations

### 11.1 External Variables I: Demographics

#### 11.1.1 L1 Interference

The purpose of this subchapter is twofold. It endeavors to demonstrate whether the subjects' L1 affects their pronunciation of English and to identify the extent to which this interference takes place. First, the means of SE and SG are compared using a paired-samples t test, followed by an examination of the impact of L1 dialect background on SE production. Before doing so, however, it is important to briefly look at SG vowel production in consideration of the number of missing values and the amount of variation among the three attempts.<sup>136</sup> The findings of these analyses will either corroborate or reject the hypothesis of alleged learning effects noted in *10.1 Analysis of Repeated Measures*. Furthermore, they will contribute to the discussion of variation among the three attempts.

##### 11.1.1.1 Analysis of Repeated Measures

The SG vowels were measured in their F1 and F2 values.<sup>137</sup> Descriptives of these data elicit the number of missing values,<sup>138</sup> which are much less in number than those from the SE analyses. Altogether 14 SG vowels could not be measured instrumentally with PRAAT, which was largely due to the same reasons various SE vowels could not be measured.<sup>139</sup> A few words needed to be discarded due to mispronunciations: *der Reis* was pronounced as [dɐ 'rais] one time, *Rebe* was articulated as [rɛpə] twice and as [ræpə] once, and *retten* was articulated as [rætə] three times and [ritə] once. It seems clear that these few values do not suggest learning effects nor do they provide fertile grounds for further investigation.

The values also underwent a repeated measures ANOVA to explore whether differences in the results obtained from the three attempts at reading the SG tokens are significant. The outputs in SPSS indicate that there are virtually no significant differences in articulation between the first, the second, and the third attempt at pronouncing the SG vowels, except for the F2 of the front vowel /i:/, which demonstrates a p value of .046.<sup>140</sup> Judged by these analyses, one of the intriguing results of the SE repeated measures ANOVA, i.e. SE front vowels

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<sup>136</sup> Cf. *10 Results Swiss English*.

<sup>137</sup> Cf. *Appendix F 1 F1s and F2s of Swiss German Vowels (Hz)*.

<sup>138</sup> Cf. *Appendix F 2 Descriptives to F1s and F2s of Swiss German Vowels*.

<sup>139</sup> Cf. *Appendix D 9.1 Re-measured Tokens and 10.1.1 Descriptives Missing Values*.

<sup>140</sup> Cf. *Appendix F 3 Repeated Measures ANOVA of Swiss German Vowels*. Significance was again calculated with the Huynh-Feldt test in SPSS.

being inclined to variation, cannot be corroborated with the results obtained from the repeated measures ANOVA of SG vowels.

#### 11.1.1.2 Paired-Samples T Test: SE Values Tested by SG Values

Paired-samples t tests compare the means of two linked variables (Woods, Fletcher, and Hughes 1986: 187). In the present study, they compare the means of the collected SG vowels in their F1s and F2s with the corresponding means of the SE vowels. The following pairs were compared: SG *der Säbel* /æ/ was compared to SE *had* /æ/; *hawd* /ɔ/ – *Kopf* /ɔ/, *head* /ɛ/ – *retten* /ɛ/, *heed* /i:/ – *der Reis* /i:/, *hid* /ɪ/ – *der Ritter* /ɪ/, *hod* /ɑ:/ – *Staat* /ɑ:/, *hood* /ʊ/ – *kurz* /ʊ/, *hud* /ʌ/ – *der Vater* /ʌ/, and *whod* /u:/ – *die Maus* /u:/. The only vowel which could not be compared was /ɜ/. SPSS computes the difference between the two variables for every case and concludes with a significance value.<sup>141</sup> p values ≤ .05 suggest that differences in articulation between SE and SG have not occurred randomly, i.e. there is a significant difference in articulation between SE and SG.

Table 11.1 shows the compared pairs in ascending order by its p values. The paired differences of the mean values show that particularly the F2 of /æ/ and the F1s of /ɑ:/ and /ɔ:/ show the greatest discrepancy between SE and SG, as their p values are close to 0. One can further infer that the vowels /i:/ F2, /ɛ/ F2, /i:/ F1, /ʌ/ F1, and /ɛ/ F1 are similarly articulated in SE and SG as their p values approximate one.

<sup>141</sup> Cf. Appendix F 4 Paired-Samples T Test of SE Values Tested by SG values.

| <i>PAIR</i>         | <i>p</i> |                                            |
|---------------------|----------|--------------------------------------------|
| hoodf1m - kurzf1m   | 9E-08    |                                            |
| hodf1m - staatf1m   | 1E-06    |                                            |
| hadf2m - säbelf2m   | 2E-06    |                                            |
| hoodf2m - kurzf2m   | 4E-06    |                                            |
| hawdf2m - kurzf2m   | 6E-05    |                                            |
| hodf2m - staatf2m   | 9E-05    | <i>Significantly<br/>different</i>         |
| hadf1m - säbelf1m   | 5E-04    |                                            |
| hidf1m - ritterf1m  | 0.002    |                                            |
| hidf2m - ritterf2m  | 0.004    |                                            |
| whodf1m - mausf1m   | 0.011    |                                            |
| hudf2m - vaterf2m   | 0.012    |                                            |
| whodf2m - mausf2m   | 0.013    |                                            |
| hawdf1m - kopff1m   | 0.045    |                                            |
| headf1m - rettenf1m | 0.053    |                                            |
| hudf1 - vaterf1m    | 0.308    | <i>Not<br/>significantly<br/>different</i> |
| heedf1m - reisf1m   | 0.783    |                                            |
| headf2m - rettenf2m | 0.826    |                                            |
| heedf2m - reisf2m   | 0.993    |                                            |

**Table 11.1:** P values of SE and SG in ascending order.

There are two aspects to the explanation for why SE and SG /*ɛ*/ and /*i*:/ are very similar. On the one hand, this may suggest that perhaps the Swiss transfer their SG /*ɛ*/ and /*i*:/ vowels into SE. On the other hand, one could also argue that SG /*ɛ*/ and /*i*:/ are in reality very similar to GA and RP /*ɛ*/ and /*i*:/ . In order to examine this issue, one can contrast the p values gained from the one-sample t test of SE tested by RP and the p values from the one-sample t test of SE tested by GA with the p values collected from the paired-sample t test of SG and SE.<sup>142</sup> The p value closest to 1 indicates the probability that the differences between these 2 groups are random, i.e. not statistically significant. Thus, if it turns out that a p value of the paired-samples t test between SE and SG is higher than both the p values of the t test of SE tested by GA and of the t test of SE tested by RP, it is statistically verified that L1 interference has taken place, cf. Table 11.2.

<sup>142</sup> Cf. 10.2.1 Comparison to RP, and 10.2.2 Comparison to GA.

| <b>F1/F2</b> | <b>p SE&amp;RP</b> | <b>p SE&amp;GA</b> | <b>p SE&amp;SG</b> | <b>Closer to SG</b> |
|--------------|--------------------|--------------------|--------------------|---------------------|
| /i:/f1       | 0.35879            | 7.02E-14           | 0.78314            | ✓                   |
| /i:/f2       | 0.00497            | 1.24E-05           | 0.82642            | ✓                   |
| /ɪ/f1        | 0.00048            | 2.08E-11           | 0.00226            | ✓                   |
| /ɪ/f2        | 1.55E-08           | 0.01957            | 0.00431            | ⊗                   |
| /e/f1        | 0.05107            | 0.00703            | 0.05319            | ⊗                   |
| /e/f2        | 0.01046            | 0.00461            | 0.82642            | ✓                   |
| /æ/f1        | 0.00168            | 0.92818            | 0.00045            | ⊗                   |
| /æ/f2        | 0.02262            | 3.31E-10           | 2.2E-06            | ⊗                   |
| /ɑ:/f1       | 4.06E-07           | 1.26E-11           | 1.1E-06            | ✓                   |
| /ɑ:/f2       | 3.73E-06           | 3.78E-11           | 9.1E-05            | ✓                   |
| /ɔ:/f1       | 0.00011            | 1.55E-07           | 1.1E-06            | ⊗                   |
| /ɔ:/f2       | 0.01165            | 0.00106            | 6.5E-05            | ⊗                   |
| /ʊ/f1        | 8.92E-06           | 2.19E-13           | 8.8E-08            | ⊗                   |
| /ʊ/f2        | 2.23E-07           | 4.95E-06           | 3.8E-06            | ✓                   |
| /ʌ/f1        | 0.03607            | 0.17455            | 0.30798            | ✓                   |
| /ʌ/f2        | 3.80E-06           | 0.0028             | 0.01224            | ✓                   |
| /u:/f1       | 0.35992            | 2.70E-08           | 0.01126            | ⊗                   |
| /u:/f2       | 5.99E-09           | 0.00114            | 0.01256            | ✓                   |

**Table 11.2:** P values comparison from t tests of SE & RP, SE & GA, and SE & SG. The gray, i.e. the ticked off, values demonstrate least differences in articulation between SE and SG.

The cells that are colored gray and ticked off at the right in Table 11.2 indicate that SE pronunciation is closest to that of SG. One could argue that L1 interference has occurred extensively with respect to both F1 and F2, particularly in the SE vowels /i:/, /ɑ:/, and /ʌ/, as well as in the F1 value of /ɪ/ and in the F2s of /e/, /ʊ/ and /u:/. Table 11.2 further shows that the SE vowels /æ/ and /ɔ:/ are articulated without much L1 interference in both the vowels' F1s and F2s.

### 11.1.1.3 Discussion L1 Interference

This subchapter seeks to account for these results by discussing the roles of similarity in language and markedness, as outlined in 8.2 *External Variables*. It was noted earlier how Flege (1995) has been among the supporters of the view that similarity or dissimilarity of sounds essentially contributes to the accuracy in articulation of a new phonetic contrast.<sup>143</sup> According to this view, the Swiss perceive the English vowels /i:/, /ɑ:/, and /ʌ/ the same way as they

<sup>143</sup> Cf. 8.2.1 *Similarity in Language*.

perceive Swiss German /i:/, /ɑ:/, and /ʌ/. For this reason, the speakers do not form new categories and therefore articulate the English vowels similar to their SG counterparts. In contrast, with the application of Flege's (1995) principles to the vowels /æ/ and /ɔ:/, the Swiss perceive a dissimilarity between their own native /æ/ and /ɔ:/ and English /æ/ and /ɔ:/, which leads them to form a new category and subsequently to articulate the vowels with less L1 interference.

In virtue of Eckman's MDH (1977), the SE vowels /ɪ/, /ɛ/, /ʌ/, /i:/, and /æ/ should be articulated more RP or GA-like, rather than approximating SG articulation, as these are the most frequently occurring, thus unmarked, vowels in spoken RP English (Crystal 2000: 239).<sup>144</sup> As Table 11.2 shows, the Swiss students' articulation of these vowels contradicts Eckman's principles. It is precisely the SE vowels /ɪ/, /ɛ/, /ʌ/, /i:/, and /æ/ which in either one of their F1s or F2s or in both their F1s and F2s are closer to SG, i.e. demonstrating a p value that approximates one, than they are to GA or RP.

Although this line of argumentation has some merit and undoubtedly revealed interesting results, it tends to overlook that the Swiss group demonstrates a heterogeneous dialect background. This, in turn, means that conclusions of the Swiss articulating the English vowels /i:/, /ɑ:/, and /ʌ/ with their own SG vowels and /æ/ and /ɔ:/ with little L1 interference, are to be understood with some provisions, as the group's own SG vowel inventories differ. The degree to which heterogeneity in language backgrounds of the students affects their articulation of SE is examined next.

#### 11.1.1.4 Correlations between L1 Background and Swiss English

The students were requested to provide information as to the region where they grew up, their father's dialect, their mother's dialect, and their own SG dialect. In this respect, it should be reiterated that informants are not very well equipped to give a precise account of their own pronunciation (Wells 2003);<sup>145</sup> thus the results ought to be taken critically. An ANOVA of SE means factored by the place of growing up shows that significant differences in SE pronunciation between speaker-groups that were raised at different places cannot be established, except for the F1 value of /ɛ/, which, however, is likely to be attributable to arbitrariness.<sup>146</sup> The term arbitrary is meant in the sense that the small group of subjects within each dialect group, usually not consisting of more

<sup>144</sup> Cf. 8.2.2 *Markedness Universals*.

<sup>145</sup> Page numbers were not available, cf. *References*.

<sup>146</sup> Cf. *Appendix F 5.1.1 Dialect Influence Frequencies*, *Appendix F 5.1.2 Descriptives to ANOVA of SE Means Factored by Place of Growing up*, and *Appendix F 5.1.3 ANOVA of SE Means Factored by Place of Growing up*.

than four or five subjects,<sup>147</sup> may accidentally have produced deviating vowels, which in turn led to significant differences. The disadvantage of having small groups is that differences only need to be small for statistical tests to categorize them as significant. Were the sample size doubled or tripled, it is likely that such minimal differences would not be classified as significant.

The ANOVA of the father's dialect factored by the SE means presents a similar picture: only the F1 value of /ʊ/ demonstrates a  $p \leq .05$ .<sup>148</sup> This would entail that a subject who has a father who speaks Bernese dialect, for example, articulates /ʊ/ with a significantly different vowel height than a subject with a father who speaks Zurich dialect, for instance. With almost complete certainty, this one occurrence of a significant value can again be ascribed to arbitrariness. The ANOVA of the mother's dialect factored by SE means similarly demonstrates only one value below an  $\alpha$  value of  $\leq .05$ , the F1 value of /ɪ/.<sup>149</sup> So far, it is difficult to escape the conclusion that if the sample size, and with it the groups representing particular variables, was enlarged by a factor of two or three, more substantiated claims and results could be presented. Finally, it is left to be answered whether there are possible correlations between the subjects' dialects and their performance in SE.

The majority (11 out of 20) of the students claim to speak Zofingen dialect. Two state that they speak Lucerne dialect, four Bernese dialect, one Basel dialect, one Zurich dialect; one student speaks German.<sup>150</sup> An ANOVA of the subjects' dialects factored by SE means was run.<sup>151</sup> It suggests that the F2 values of /ʌ/ and /ɑ:/ are articulated significantly different, depending on the subject's dialect. Descriptives of the ANOVA specify 947 Hz for the SE F2 value of /ɑ:/ for those who speak Zofingen dialect, Lucerne dialect 1081 Hz, Bernese 1104 Hz, Basel dialect 795 Hz, Zurich dialect 1057, and the one subject who speaks High German 994 Hz. A comparable frequency distribution is found in the SE F2 value of /ʌ/. Furthermore, the following SE vowels are articulated differently between the various dialect groups, with the qualification, however, of having  $p$  values as high as  $.05 < p \leq .2$ : /ɑ:/ F1, /i:/ F1, /ɛ/ F2, /ɛ/ F1, and /ʊ/ F2. Unfortunately, such evidence does not constitute adequate support for the theory that the L1 dialect positively correlates with differences in L2 production, as Barry (1974) has persuasively argued.

<sup>147</sup> Cf. *Appendix F 5.1.1 Dialect Influence Frequencies*.

<sup>148</sup> Cf. *Appendix F 5.1.1 Dialect Influence Frequencies*, *Appendix F 5.1.4 Descriptives to ANOVA of SE Means Factored by the Subject's Father's Dialect*, and *Appendix F 5.1.5 ANOVA of SE Means Factored by the Subject's Father's Dialect*.

<sup>149</sup> Cf. *Appendix F 5.1.1 Dialect Influence Frequencies*, *Appendix F 5.1.6 Descriptives to ANOVA of SE Means Factored by the Subject's Mother's Dialect*, and *Appendix F 5.1.7 ANOVA of SE Means Factored by the Subject's Mother's Dialect*.

<sup>150</sup> Cf. *Appendix F 5.1.1 Dialect Influence Frequencies*.

<sup>151</sup> Cf. *Appendix F 5.1.8 Descriptives to ANOVA of SE Means Factored by the Subject's Dialect* and *Appendix F 5.1.9 ANOVA of SE Means Factored by the Subject's Dialect*.

### 11.1.2 Age

The subchapter 8.1.1.2 *Age* lay emphasis on the fact that age as an external variable is particularly difficult to explore for the impossibility of its isolation and falsification (Moyer 2004: 20). It seems hard to deny that scholars are undecided as to the degree to which age causes phonological variation with second language speakers. Several authors (e.g. Flege and Davidian 1984; Flege 1987a; Flege, Munro, and MacKay 1997; Flege, Schirru, and MacKay 2002; MacKay and Flege 2004; and Birdsong and Molis 2001) call attention to the uncertainty of a critical period and to the lack of evidence for age-related variation. In the attempt to explore whether age occupies a crucial role in the present study, a major problem in running statistical analyses was encountered.

#### 11.1.2.1 Methodological Difficulty in Using Differences Values

Correlations of age and foreign accent in SPSS can best be established with bivariate correlations, which are run in order to investigate the relationship between two variables. The first variable would apparently be age, while the second variable, foreign accent, is more intricate to determine. Technically speaking, the foreign accent in SE vowels is comprised of the perceptible F1, F2, and F3 frequency differences between the spoken vowel and the target vowel. In adherence to this principle, the differences in frequency between SE and RP, and SE and GA were calculated absolutely and relatively for each vowel in its F1 and F2 values.<sup>152</sup> SPSS supplies an output of a correlation matrix for the correlations requested. Table 11.4 presents an excerpt of an SPSS output.

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<sup>152</sup> Cf. *Appendix F 5.3 Absolute Frequency Differences between SE and GA, SE and RP* and *Appendix F 5.4 Relative Frequency Differences between SE and GA, SE and RP*.

|             |                     | alter | hadf1diffam | hadf1diffrp | hadf2diffam | hadf2diffrp |
|-------------|---------------------|-------|-------------|-------------|-------------|-------------|
| alter       | Pearson Correlation | 1     | -.097       | -.097       | .137        | .137        |
|             | Sig. (2-tailed)     | .     | .683        | .683        | .566        | .566        |
|             | N                   | 20    | 20          | 20          | 20          | 20          |
| hadf1diffam | Pearson Correlation | -.097 | 1           | 1.000**     | .152        | .152        |
|             | Sig. (2-tailed)     | .683  | .           | .000        | .521        | .521        |
|             | N                   | 20    | 20          | 20          | 20          | 20          |
| hadf1diffrp | Pearson Correlation | -.097 | 1.000**     | 1           | .152        | .152        |
|             | Sig. (2-tailed)     | .683  | .000        | .           | .521        | .521        |
|             | N                   | 20    | 20          | 20          | 20          | 20          |
| hadf2diffam | Pearson Correlation | .137  | .152        | .152        | 1           | 1.000**     |
|             | Sig. (2-tailed)     | .566  | .521        | .521        | .           | .           |
|             | N                   | 20    | 20          | 20          | 20          | 20          |
| hadf2diffrp | Pearson Correlation | .137  | .152        | .152        | 1.000**     | 1           |
|             | Sig. (2-tailed)     | .566  | .521        | .521        | .           | .           |
|             | N                   | 20    | 20          | 20          | 20          | 20          |

\*\* . Correlation is significant at the 0.01 level (2-tailed).

**Table 11.3:** Correlation matrix of bivariate correlations between age and relative differences between SE and GA, and SE and RP.

Butler (1985) describes the interpretation of correlation coefficients by saying that “a value of +1 is obtained for perfect positive correlation, a value for -1 for perfect negative correlation, and a value of zero for no correlation at all” (Butler 1985: 141).<sup>153</sup> Pearson Correlation, which takes into account “the exact magnitude of each score on each variable” (ibid: 142), of age and *hadf1diffam*, for example,<sup>154</sup> shows that the frequency difference does not significantly correlate with age as shown in its significance value of .683 and its correlation coefficient of -.097. The fundamental dilemma with these correlations is that the correlation coefficients (Pearson Correlation) and significance values (Sig. (2-tailed)) turn out identical for RP and GA, e.g. *hadf1diffam* and *hadf1diffrp* are apparently the same. Although bivariate analyses correctly calculate correlation values, these findings must be justified, as they imply identical formant values for RP and GA.

Authorities in statistics of the University of Berne were able to explain these phenomena.<sup>155</sup> The problem can be summarized as follows: One would like to determine the alteration of formant values in connection to variable X, e.g. age. To ascertain the degree to which SE formant values deviate from GA values and RP values, the differences in formant values between the subjects’

<sup>153</sup> If a correlation coefficient approximates +1 or -1 its significance value approximates zero.

<sup>154</sup> *hadf1diffam* represents the average frequency discrepancy between the Swiss speakers’ F1 values of /æ/ and General American F1 of /æ/.

<sup>155</sup> Several people from the Psychology Department, Physics Department, and General Linguistics Department of the University of Berne were involved in this endeavor.

SE values and the GA values, and between the subjects' SE values and the RP values are calculated. Within these calculations, the formant values are twice involved in a shift by a constant value.<sup>156</sup> If one wants to calculate standard deviations and correlation coefficient values of a sample that is involved in a shift by a constant value, standard deviations and correlation coefficients will remain unchanged. Therefore, significance values and correlation coefficients in Table 11.3 are identical for the RP and GA differences values. The experts in statistics hinted that one should handle these correlations very carefully and suggested, if possible, that one should not make use of them.<sup>157</sup>

The only plausible approach then is to resort to SE mean values instead of applying differences values. Regrettably, by means of this rather generic approach, one will not be able to specifically approach questions regarding an increase or reduction of vowel discrepancies. Even more importantly, it will be difficult to draw conclusions as to whether the presence of an external variable X induces an RP-like or a GA-like pronunciation, because, for such computations, difference values are required. The fundamental drawback in running statistical tests with SE means is that the results are of limited use in the sense that they only offer an insight as to whether groups of SE speakers, factored by an external variable X, demonstrate significant differences between each other.

#### 11.1.2.2 ANOVA of SE Means Factored by Age

The students varied in their age from 17 to 20. Six students were 17, ten students 18, three students 19, and one student 20 years old.<sup>158</sup> Significance values of the conducted ANOVA suggest that out of 20 significance values, only two are below the predefined threshold of  $p \leq .05$ , and six approximating significance, i.e.  $p$  is between .05 and .02.<sup>159</sup> The significant values are found in F1 /ɔ:/,  $p = .009$  and F2 /ʊ/,  $p = .036$ . Those values that approximate significance are the F1s of /ɛ/, /ɪ/, /ɑ:/, /ʊ/, /ʌ/, and the F2 of /u:/. In observation of descriptives of these analyses,<sup>160</sup> one can infer that the F1 mean values of /ɔ:/ for the six 17 year-old subjects is 563 Hz, for the eight 18-year-old

<sup>156</sup> The constant value is the frequency difference between the RP and GA value.

<sup>157</sup> An alternative solution to test foreign accent for age, for example, was proposed by copying and pasting the data from SPSS into Microsoft EXCEL and arranging the difference values into respective age groups. This procedure would allow for a gross comparison between groups, with the apparent drawback, however, of not knowing whether potential differences are statistically significant or not. To keep this study on a scientific level, however, significance values are crucial, which eventually led to the conclusion not to use difference values for further statistical tests.

<sup>158</sup> Cf. *Appendix F 5.2 Age Frequency Table*.

<sup>159</sup> Cf. *Appendix F 5.6 ANOVA of SE Means Factored by Age*.

<sup>160</sup> Cf. *Appendix F 5.5 Descriptives to ANOVA of SE Means Factored by Age*.

subjects 476 Hz, for the two 19 year-olds 485 Hz, and for the 20-year-old 370 Hz. A nearly identical age-related variation occurs in the F2 value of /ʊ/. Five 17-year-old students articulated /ʊ/ with an F2 value of 999 Hz, ten 18-year-old students with 843 Hz, three 19-year-olds with 1021 Hz, and the 20 year-old with 1055 Hz.

If these results are viewed in light of Deterding's (1997) and Hillenbrand et al.'s (1995) suggested benchmark values, with RP/ɔ:/ F1 at 415 Hz and GA /ɔ:/ F1 at 652 Hz, it cannot be concluded that the older or the younger the subjects the more they articulate it native-like because the frequency discrepancy between GA and RP is too great. For the value of /ʊ/ F2, on the other hand, such conclusions are more substantiated as RP /ʊ/ F2 1173 Hz and GA /ʊ/ F2 1122 Hz are much closer. There is a slight trend that the older the student, the more he articulates /ʊ/ native-like in terms of the vowel's backness-frontedness. However, the explanatory power of this finding is weakened due to the small number of speakers per age group.

### 11.1.3 Preliminary Conclusions: External Variables I

This chapter started with an analysis of the amount of differences among the three attempts found in the articulation of SG tokens. It concluded that where the same analyses of SE provided interesting results, insofar as SE front vowels demonstrated more variation than SE back vowels, the analyses of SG turned out differently. By means of Flege's (1995) outlined principles,<sup>161</sup> the chapter also accounted for the phenomenon of sound substitution of the English vowels /i:/, /ɑ:/, and /ʌ/ with the respective SG vowels. It was stressed, however, that such assumptions are to be understood with some provisions due to the Swiss' heterogeneous L1 dialect backgrounds. Consecutively, the area where the subject was raised, his father's dialect, his mother's dialect, and the subject's own dialect were tested for correlation with SE vowel quality. Although the ANOVA provides several significant values, further interpretations are hardly possible due to the small sample and group sizes.

In establishing correlations between age and SE mean values, methodology-related difficulties of applying bivariate correlations were thoroughly discussed. In order to test the Swiss' foreign accent in vowels with a further variable, it would have been beneficial to use difference values. It was closely examined, however, that such calculations are in fact useless as the correlation coefficient as well as the calculated significance values turned out identical for GA and RP differences values. On these grounds, it was decided to only rely on SE mean values in this study. An ANOVA of SE mean values factored by the subjects' age showed that although significant differences in English pronunciation between different age groups are established, they are likely to be attributable to arbitrariness as a result of the small sample size.

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<sup>161</sup> Cf. 8.2.1 *Similarity in Language*.

## 11.2 External Variables II: Language Background

### 11.2.1 L2 Experience

The view put forward in 8.1.2.1 *L2 Experience* explained why the discovery of patterns of L2 experience effects on L2 production is virtually unattainable by means of statistical tests. Moyer has been among the supporters of the view that L2 experience happens primarily in the classroom and in residing in the country where the L2 is spoken (2004: 61).<sup>162</sup> There are, however, a number of other, often random, input factors that equally add to L2 experience. It was noted how the amount of speaking, reading, and listening in the target language may also exert an influence on L2 vowel quality.

Whether there is a positive correlation between the external variables, speaking, reading, listening and SE vowel quality will be addressed first, followed by an in depth investigation as to whether listening to music in the target language, cf. Medina (1993), or watching movies in the target language has any effects on L2 production. Finally, this subchapter ends with an analysis of how L2 classroom experience and LOR affect vowel quality in the present study.

#### 11.2.1.1 Independent-Samples T Test of SE Means Grouped by Frequency of Speaking English

The findings of the questionnaire show that 14 out of 20 subjects had conversed with an English-speaking person for 1-30 minutes over the previous month.<sup>163</sup> The students were asked to tick off whether this type of communication happens at a very regular, rather regular, rather irregular, or very irregular basis. Two students checked rather regular, two rather irregular, and two very irregular.<sup>164</sup> Six out of 20 subjects specify that they had not spoken to any native English speaker over the previous month. According to Flege, Frieda, and Nozawa's (1996) findings, those subjects who frequently speak English ought to produce English sounds more accurately than those who seldom speak English. This hypothesis can be tested by running an independent-samples t test with the SE means as testing values, grouped by the variable of whether or not the subjects frequently converse in English.<sup>165</sup>

The results suggest that the mean values of those who verbally communicate with a native speaker 0-30 minutes a month do not significantly differ from the mean values of those who do not converse with native

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<sup>162</sup> Residing in the country where the L2 is spoken indirectly refers to LOR.

<sup>163</sup> Cf. Appendix F 6.1 *Frequency Table Speaking English*.

<sup>164</sup> Cf. Appendix F 6.2 *Frequency Table Speaking English on What Basis*.

<sup>165</sup> The independent-samples t test is a statistical test performed to determine if there is a statistically relevant difference between the means of two independent samples.

speakers.<sup>166</sup> The F2 values of /ɜ/ and /ʌ/ are articulated significantly different between the two groups, /ɜ/ with a p value of .049, and /ʌ/ of .023. If the descriptives of this independent t test are considered, we observe that the difference of frequency between the two groups is only 73 Hz for the F2 value of /ɜ/ and 86 Hz for the F2 value of /ʌ/.<sup>167</sup> Moreover, the F1 values of /ɛ/, /ɔ:/, and /ɑ:/ as well as the F2 value of /ɑ:/ approximate significance with p values between .05 and .2. Judged by this measure, it might be argued that the two significant values once more emerged due to the small sample size. Flege, Frieda, and Nozawa's (1996) hypothesis, i.e. that speaking in the target language positively affects phonological accurateness, can thus neither be verified nor falsified.

### 11.2.1.2 ANOVA of SE Means Factored by Reading in English

Four out of 20 students assert that they had not read anything in English outside the school context. Ten subjects read 1-30 minutes per week, four 30-60 minutes, one 1-5 hours, and yet one asserts that he spent more than 5 hours reading in the English language per week.<sup>168</sup> Twelve students claim that this reading happens on a rather regular basis, while seven say this happens on a rather irregular basis.<sup>169</sup> Since more than two groups are involved, an ANOVA was run with the SE mean values as dependent variables factored by reading in the English language.

According to the ANOVA, those who read outside of the classroom, regardless of the amount of time spent reading, in nearly all instances do not significantly differ in their pronunciation of English vowels than those who claim not to read in English.<sup>170</sup> The only value in which the groups differ is in /ɔ:/ F2, with a p value of .0001. The values of /ɑ:/ F2 and /ʊ/ F1 show approximating significance p values  $.05 < p \leq .2$ . Descriptives to the ANOVA again show that those who do not read, present an /ɔ:/ F2 value of 831 Hz, those who read 1-30 minutes have 885 Hz, 30-60 minutes 1040 Hz, 1-5 hours 702 Hz, and more than 5 hours 830 Hz.<sup>171</sup> Taking into account that the peak values occur in the group that claims to read 30-60 minutes per week, which is neither most nor least on the continuum, one can neither falsify nor verify the hypothesis that reading correlates with accuracy in L2 vowel articulation.

<sup>166</sup> Cf. Appendix F 6.4 Independent T Test SE Means Grouped by Speaking English.

<sup>167</sup> Cf. Appendix F 6.3 Descriptives to Independent T Test SE Means Grouped by Speaking English.

<sup>168</sup> Cf. Appendix F 6.5 Frequency Table Reading English.

<sup>169</sup> Cf. Appendix F 6.6 Frequency Table Reading English on What Basis.

<sup>170</sup> Cf. Appendix F 6.8 ANOVA of SE Means Factored by Reading English.

<sup>171</sup> Cf. Appendix F 6.7 Descriptives to ANOVA of SE Means Factored by Reading English.

### 11.2.1.3 ANOVA of SE Means and Listening to English

Out of 20 students, three claim to listen to the English language 1-30 minutes per week, one says he listens to English 30-60 minutes a week, five students assert that they listen to English one to five hours a week, and finally more than half of the students (eleven) claim to listen to English more than five hours per week.<sup>172</sup> The majority of the subjects (eleven) state this happens on a very regular basis, five say this happens rather regularly, and 4 say this is a rather irregular practice.<sup>173</sup> Whether or not this type of experience with the English language exerts any significant influences on the production of English vowels was again tested by means of an ANOVA.<sup>174</sup> None of the p values turned out below the threshold value of .05, which entails that subjects who consume large amounts of English language auditorily do not demonstrate significantly different articulation of English vowels than those who listen to English in merely small amounts. The only vowels that demonstrate p values  $.05 < p \leq .2$  are /æ/ F2, /ɜ/ F1, and /ɪ/ F1. Apart from asking the students about their broad-spectrum music consumption, two further questions specifically addressed music and movie consumption in English.

The students were prompted to make an estimate in percent as to the number of American English-speaking artists. 19 out of 20 students answered that at least half if not more of the music they listen to is sung by artists who the group reckons speak American English.<sup>175</sup> One subject claimed to listen only to American English singers, ten noted that 75% of the music they enjoy is sung in American English, and eight asserted that half of the music they listen to is performed in American English. An ANOVA shows that whether subjects listen entirely to American English music or half of their music is in American English does not significantly affect their English vowel articulation.<sup>176</sup> None of the p values are below or equal to the conventional  $\alpha$  value of .05. F1 /ɔ:/ demonstrates the only p value (.171) that is  $.05 < p \leq .2$ . Despite the small sample and group sizes, one could speculate that these findings oppose Medina's (1993) postulate that listening to music positively affects second language learning, as zero significant differences could be established. Comparable results are found in the analysis of potential effects of watching English movies.

Six out of 20 subjects watch movies predominantly in English.<sup>177</sup> Those six students further gave information as to the number of movies they watch in

<sup>172</sup> Cf. Appendix F 6.9 Frequency Table Listening to English.

<sup>173</sup> Cf. Appendix F 6.10 Frequency Table Listening to English on What Basis.

<sup>174</sup> Cf. Appendix F 6.12 ANOVA of SE Means Factored by Listening English and Appendix F 6.11 Descriptives to ANOVA of SE Means Factored by Listening to English

<sup>175</sup> Cf. Appendix F 6.13 Frequency Table Listening to American English Music.

<sup>176</sup> Cf. Appendix F 6.15 ANOVA of SE Means Factored by Listening to American English Music and Appendix F 6.14 Descriptives to ANOVA of SE Means Factored by Listening to American English Music.

<sup>177</sup> Cf. Appendix F 6.16 Frequency Table of Watching Movies in English.

English per month. Three claim to consume one movie and the other three state that they watch 2-4 movies in English.<sup>178</sup> The ANOVA suggests that differences in SE vowel articulation between those who watch one movie are not significant from those who watch 2-4 movies in English.<sup>179</sup> The F2 values of /æ/ and /ɛ/, as well as the F1 values of /ɜ/ and /ɪ/ present p values  $.05 < p \leq .2$ .

According to the statistics run with the variables pertinent to listening experiences, it has been shown that significant differences between groups could not be established. Furthermore, it was illustrated how potential correlations between speaking in the L2 or reading in the L2 with SE means equally do not offer significant values that would yield sufficient evidence for between group differences. Despite the apparent pessimism of these results, one should not overlook that the final correlations between SE means and listening to English addresses speech perception rather than speech production.<sup>180</sup> These findings propose that significantly decisive results could neither be evoked in speech production nor in speech perception, which in turn may be considered as corroborating Strange et al. (2004), Llisterini's (1995), and Jacewicz's (2002) belief in an inextricable link between speech perception and speech production.

#### 11.2.1.4 ANOVA of SE Means Factored by LOR

Frequency counts demonstrate that 14 out of 20 students have not visited an English-speaking country, 5 say they have been to an English-speaking country 1-4 weeks, and 1 person says he spent 1-3 months in an English-speaking country.<sup>181</sup> Unfortunately, it remains undetermined as to where the students have been since only nine out of 20 cared to fill in the corresponding blank in the questionnaire. Out of those who designated their destinations, one person was in Scotland, five in England, two in the U.S., and one in Australia.<sup>182</sup> The ANOVA suggests that significant correlations between length of residence and SE mean values could not be established,<sup>183</sup> except for the F1 value of /ɪ/. The F1 values of /i:/, /ɑ:/, and /ɔ:/ present p values  $.05 < p \leq .2$ . If one considers descriptives of the ANOVA it becomes clear that the significance value of /ɪ/ F1 is subject to arbitrariness:<sup>184</sup> Those who have not been abroad illustrate mean values of 333 Hz (14 subjects), 1-4 weeks 369 Hz (5), and one person who has been abroad for 1-3 months presents 307 Hz. It does not seem

<sup>178</sup> Cf. Appendix F 6.17 Frequency Table of the Amount of Movies in English.

<sup>179</sup> Cf. Appendix F 6.18 Descriptives to ANOVA of SE Means Factored by Amount of Movies in English and Appendix F 6.19 ANOVA of SE Means Factored by Amount of Movies in English.

<sup>180</sup> Cf. 8.1.2.1 L2 Experience.

<sup>181</sup> Cf. Appendix F 6.20 Frequency Table LOR.

<sup>182</sup> Cf. Appendix F 6.21 Frequency Table Abroad Where.

<sup>183</sup> Cf. Appendix F 6.23 ANOVA of SE Means Factored by LOR.

<sup>184</sup> Cf. Appendix F 6.22 Descriptives to ANOVA of SE Means Factored by LOR.

implausible to suggest that the final value is a statistical outlier which consequently explains why /ɪ/ F1 exhibits such a high degree of significance.

### 11.2.1.5 Correlations between Classroom Instruction Factors and Swiss English

Two types of classroom instruction factors need to be distinguished both of which are explored by the answers gained from the questionnaire:<sup>185</sup> Whether the subjects were taught American English or British English and whether they were exposed to direct pronunciation training during classroom instruction. As the frequency counts suggest, it turns out that five subjects believe that they have been taught American English whereas 15 subjects think they have been instructed in British English.<sup>186</sup> An independent-samples t test reveals that subjects who were taught American English do not significantly differ from those who were taught British English, as none of the p values are below .05.<sup>187</sup> The vowel /i:/ demonstrates a p value  $.05 < p \leq .2$  in both its F1 and F2 values, a pattern reflected in the F2 value of /ɪ/. Although the test again only makes available statistically insignificant results, it is important to realize that these findings carry more statistical weight than previous ones because the groups of speakers that have been taught GA or RP are comparatively large.

Subsection 8.1.2.1 *L2 Experience* discussed how Catford and Pisoni (1970) as well as Matthews (1997) found evidence that students who receive systematic training in articulatory phonetics perform much closer to target forms than those who are not given the same instruction. For this reason, the questionnaire asked the students whether their English teachers principally lay focus on lexis, syntax, pronunciation, or interactional competence. Frequency counts indicate that eleven out of 20 students state that their teachers pay particular attention to pronunciation. 13 claim to be predominantly trained in vocabulary, ten in syntax, and seven with an interactional focus.<sup>188</sup> In order to test whether the Swiss who were given pronunciation instruction indeed perform more native-like, as postulated by Catford and Pisoni (1970) and Matthews (1997), an independent-samples t test was run. It turns out that there are two p values below .05, i.e. the F1 values of /ʊ/ and /u:/. The F2 value for /i:/ equally shows differences between the two groups yet only with a p value of .135.<sup>189</sup> If we consider the descriptives of the t test,<sup>190</sup> we learn that

<sup>185</sup> Cf. 8.1 *Questionnaire*.

<sup>186</sup> Cf. Appendix F 6.24 *Frequency Table Taught RP/GA*.

<sup>187</sup> Cf. Appendix F 6.26 *Independent T Test of SE Means Grouped by Taught RP/GA* and Appendix 6.25 *Descriptives to Independent T Test of SE Means Grouped by Taught RP/GA*.

<sup>188</sup> Cf. Appendix F 6.27 *Frequency Table Focus in English Classroom*.

<sup>189</sup> Cf. Appendix F 6.29 *Independent T Test of SE Means Grouped by Pronunciation Focus*.

<sup>190</sup> Cf. Appendix F 6.28 *Descriptives to Independent T Test of SE Means Grouped by Pronunciation Focus*.

those who were instructed in pronunciation show 342 Hz in the F1 for /ʊ/, as opposed to 303 Hz for those who did not have pronunciation training. Deterding's (1997) and Hillenbrand et al.'s (1995) values propose an F1 value for /ʊ/ RP of 379 Hz and an F1 values for /ʊ/ GA of 469 Hz, which proves that those students who were taught pronunciation in effect are closer to the standard varieties than those who were not instructed. This claim finds further support when looking at the F1 values of /u:/ of the two groups: Those subjects who were trained in pronunciation show F1 values of 323 Hz where the others show 294 Hz. Once more, if the values of RP and GA are taken into consideration, 316 Hz RP and 378 Hz GA, it is evident that those who were given training are closer to the standard varieties than those who were not.

In spite of the fact that only two out of 20 p values are significant, these results bear a comparatively high explanatory power as they rely on almost evenly distributed group sizes. It should be added that it is not a coincidence the vowels /ʊ/ and /u:/ emerge as points of distinction between the two groups. In many English classrooms where teachers are concerned about their students' pronunciation, practice in the discrimination between these two vowels is a frequently revisited spot. Accordingly, it may well be the case that those who were given pronunciation training articulate these two vowels differently from those who were not, which are findings in support of Catford and Pisoni's (1970) as well as Matthews' (1997) assumptions.

### 11.2.2 Motivation

Subsection 8.1.2.3 *Cognitive and Aptitude Factors* shed light on how motivation can have an effect on L2 production. Spolsky (1989: 148) notes that a high degree of motivation leads to spending more time in learning an aspect of the L2, which could be pronunciation. The questionnaire asked the subjects to name their degree of motivation on a scale from -1 to +5. The majority of the students (14) assert that they were motivated +4, one person claims to be motivated +2, two took a neutral stance towards motivation, i.e. level 3, and five subjects say they are highly motivated, which corresponds to level of +5.<sup>191</sup> The ANOVA of SE mean values factored by the subjects' motivation level proposes that the subjects' SE vowel quality does not significantly differ within groups of varied motivation levels.<sup>192</sup> The F2 value of /ɛ/ and the F1 value of /ɪ/ present the only p values  $.05 < p \leq .2$ . According to these results, motivation, as part of cognitive affective factors, does not give the impression of assuming a central role in ultimate phonological attainment.

<sup>191</sup> Cf. Appendix F 6.30 Frequency Table Motivation.

<sup>192</sup> Cf. Appendix F 6.32 ANOVA of SE Means Factored by Motivation and Appendix F 6.31 Descriptives to ANOVA of SE Means Factored by Motivation.

### 11.2.3 Preliminary Conclusions: External Variables II

The presentation and discussion of the results in this chapter has frequently alluded to the fact that too small a sample size hampers the drawing of substantial conclusions. Statistical tests to comparing means such as ANOVAs and independent-samples *t* tests have revealed that the SE means do not significantly differ if a subject speaks much or not, reads much or not, or listens to music in the English language or not. When tested for LOR, results again turned out largely insignificant. If significant differences emerged, they were often attributed to arbitrariness and group size-related causes. Catford and Pisoni's (1970) as well as Matthews' (1997) postulation that students who are given pronunciation training perform closer to the target language than those who are not, was in part verified in the present study. Although significance was established for two *p* values only, the implications of the results are particularly interesting as the two values concern the F1s of /u:/ and /ʊ/ - two vowels that English teachers habitually practice with their students. Finally, an ANOVA of the subjects' SE means factored by their motivation levels did not provide significant results.

### 11.3 External Variables III: Language Attitudes

Language attitudes can potentially influence verbal behavior (cf. Lasagabaster et al. 2004, Giles et al. 1987). The final part of the questionnaire attends to exactly this issue, i.e. whether or not the Swiss group's language attitudes towards English affect their articulation of English. The order of the following analyses is analogous to that applied in the questionnaire section on language attitudes.

#### 11.3.1 Subject Preference for RP or GA

The first question in section C of the questionnaire, language attitudes, invites the students to tick off their preference of English, i.e. RP or GA. Frequencies of the subjects' answers show that eight of the 20 subjects prefer GA and twelve favor RP.<sup>193</sup> The independent-samples *t* test proposes an interesting picture:<sup>194</sup> The F1 values of /æ/, /ɔ:/, /i:/, /ɑ:/, and /u:/ between those who state preference for RP and those who like GA better are significantly different. Also, the F1 values of /ɛ/, /ɜ:/, and /ʌ/ as well as the F2 value of /i:/ approximate significance with *p* values .05 < *p* ≤ .2.

Descriptives show that the F1 value of /æ/ is 514 Hz for those who prefer GA and 641 Hz for the students who claim to prefer RP.<sup>195</sup> The F1 values of /æ/ RP and /æ/GA are 690 Hz for RP (Deterding 1997) and 588 for GA

<sup>193</sup> Cf. Appendix F 7.1 Frequency Table Preference for GA or RP.

<sup>194</sup> Cf. Appendix F 7.3 Independent T Test of SE Means Grouped by Preference for GA/RP.

<sup>195</sup> Cf. Appendix F 7.2 Descriptives to Independent T Test of SE Means Grouped by Preference for GA/RP.

(Hillenbrand et al. 1995). With regard to /æ/ F1, one can infer that those who favor RP indeed speak closer to RP, and those who say they favor GA speak more GA-like. With regard to the F1 value of /ɔ:/, descriptives show that speakers who favor American English present 453 Hz whereas those who have a preference for British English demonstrate 536 Hz. The F1 values for /ɔ:/ in the standard varieties are 415 Hz in RP and 652 Hz in GA. These findings, however, contradict the previous principle in that those who say they prefer GA are in fact closer to RP and vice versa. As regards the F1 of /i:/, the group that likes GA better shows 285 Hz as opposed to those favoring RP who show 270 Hz. The values of the Standard Englishes are 342 Hz in GA and 280 Hz in RP. Although the Hz discrepancy in the F1 value of /i:/ between the subjects who prefer GA and the subjects who like RP is only 10 Hz, it could again be argued that the subjects who state preference for RP are closer to RP, and those who find more pleasure in American English speak more GA-like. In the vowel /ɑ:/, the Swiss who prefer GA show 461 Hz and those who favor RP show 543 Hz. The Standard English values of /ɑ:/ F1 is 646 Hz in RP and 768 Hz in GA. These findings again propose a negative correlation, i.e. those who claim to prefer GA are in fact closer to RP and those who like RP better in fact sound more GA-like. Finally, the F1 value of /u:/ presented significant differences as well. The subjects who like GA present 332 Hz and those who prefer British English show 298 Hz. The mean values of the /u:/ F1s for the Standard Englishes are 316 for RP and 378 for GA, which allows for the conclusion that those who prefer GA are closer to GA and analogously, those who prefer RP speak more RP-like.

In the big scheme of things, one could argue that these findings suggest that preference for RP or GA shapes English articulation. Had more subjects been scrutinized such claims could be corroborated more systematically. These results, in turn, contradict Lasagabaster et al.'s (2004: 401) claim that "there seems to be a gap between what people say (their expressed attitudes) and what they do (their actual behaviour)," as the Swiss under scrutiny in fact do seem to form their actual verbal behavior according to their preference for GA or RP.

### 11.3.1.1 Subject Perception of Nice English

When asked to judge the English of celebrities, the subjects probably formed their answers according to whether or not a particular person appealed to them politically, educationally, ethically, economically, ideologically, etc. (Lasagabaster et al. 2004: 403).<sup>196</sup> Frequency counts indicate that nine subjects like the speech of celebrities who speak British English whereas nine subjects like the English of celebrities collectively labeled as speaking American English.<sup>197</sup> Seven subjects think Tony Blair's English sounds nice, six like

<sup>196</sup> Cf. 8.1.3.2 *Researching Language Attitudes with the Questionnaire*.

<sup>197</sup> Cf. Appendix F 7.4 *Frequency Tables Preferred English of Celebrities*.

Robbie Williams' English, five BBC presenters', five Al Pacino's, four CNN presenters', two Queen Elizabeth's, one Madonna's, one Ozzy Osbourne's, one Bridget Jones', one MTV's, and one black Hip Hop celebrities'. Predictably, none of the subjects claims to like George W. Bush's English. It is remarkable to see how, apart from two missing values, these frequencies resemble those of *11.3.1 Subject Preference for RP or GA*, which speaks for a sincere cooperation on the part of the subjects. Given the scope of this paper, statistical tests were not run with these data.

### 11.3.1.2 Subject Preference for RP or GA in the English Text

The evaluation of the text introduced in 6.2 *Text with RP and GA Tokens* intends to reveal potential correlations between the students' articulation of SE vowels and their articulation of tokens that are said to be different in pronunciation between GA and RP. To briefly review the procedure: the text contains tokens that differ in /j u:/ vs. /u:/ (3 tokens), /ɑ:/ vs. /æ/ (3 tokens), /α/ vs. /ɔ/ (2 tokens), /t/ vs. /r/ (3 tokens), possible rhotacisms (3 tokens), as well as specific words that are regularly pronounced differently in GA and RP, such as *secretaries*,<sup>198</sup> *garage*, *centre*, *advertisement*, *schedule*, and *vase* (each token occurring once). The recordings were narrowly transcribed and contrasted to Wells' (2000) RP and GA transcriptions as found in his *Longman Pronunciation Dictionary*.<sup>199</sup> To further statistically analyze the data with SPSS, the transcriptions were grouped according to whether the SE transcription resembled RP or GA.

Before presenting the findings of the analyses, it is important to realize that the articulation of these English tokens is equally subject to language internal factors, such as similarity in language and markedness universals, as is the reading of English word lists.<sup>200</sup> Consider the aspect of similarity in language, for example. If the English words *fast* and *car* are articulated as if they were SG words, i.e. [fɑ:st] and [k<sup>h</sup>ɑ:r], it shows that the vowel in *fast* is pronounced more similar to RP[ɑ:] than to GA [æ]; conversely, the word *car* is articulated more GA-like as the final r in *car* is articulated as in GA *car*, except non-rhotic. To all intents and purposes, this entails that in some of the text tokens the Swiss are inherently prone to use one or the other pronunciation of English.

In terms of markedness, particularly statistical frequencies of words, it is again assumed that marked words, which are statistically infrequent, are more difficult to pronounce than unmarked, statistically frequent, words.<sup>201</sup> The

<sup>198</sup> Different stress assignment between RP and GA.

<sup>199</sup> Cf. *Appendix A 2 Transcription of Text with RP and GA Tokens*.

<sup>200</sup> Cf. *6.1 Word Lists with English Vowels*.

<sup>201</sup> For a detailed disquisition of the concepts of markedness and statistical frequency, see *8.2.2 Markedness Universals*.

Brown Corpus of Standard American English, compiled at Brown University (Providence, RI), offers frequency counts for the 5000 most frequently occurring words in American English from a corpus total of 1,015,9454 words.<sup>202</sup> It shows that the word *new* has 64 counts, whereas the word *garage* presents 4690 counts. The likely corollary of such frequency distributions is that the Swiss are expected to pronounce *new* very English-like because of the markedness, where the word *garage*, in contrast, with almost all certainty yields problems in articulation caused by the word's relative unmarkedness.

The 20 subjects were requested to read the English text containing 22 target tokens once. Three out of 20 subjects were inclined to American pronunciation, twelve to English pronunciation, and five subjects articulated eleven tokens rather RP-like and eleven tokens more GA-like.<sup>203</sup> Table 11.4 illustrates the frequency counts of the Swiss' RP and GA pronunciations. According to these values, the Swiss group slightly prefers RP (12 tokens) pronunciation over GA pronunciation (11 tokens). This result is in support of the assumption noted in 10.2.2.3 *Discussion* that the Swiss scrutinized are minimally closer to RP than to GA in their articulation of English vowels.

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<sup>202</sup> Retrieved from <http://www.edict.com.hk/TextAnalyser/wordlists.htm>, 3.4.2007. A possible drawback of the Brown Corpus is that it omits taking into consideration dialects as well as language external factors such as age and educational levels of the speakers.

<sup>203</sup> Cf. *Appendix F 7.5 Frequency Tables English Text Tokens*.

| Token         | GA | RP |
|---------------|----|----|
| new           | 0  | 20 |
| car           | 17 | 3  |
| on            | 4  | 16 |
| mistreated    | 5  | 15 |
| new           | 2  | 18 |
| fast          | 3  | 17 |
| not           | 6  | 14 |
| work          | 12 | 8  |
| chance        | 14 | 6  |
| temporary     | 13 | 7  |
| jobs          | 6  | 14 |
| tuning        | 0  | 20 |
| city          | 6  | 14 |
| secretaries   | 17 | 2  |
| centre        | 20 | 0  |
| advertisement | 1  | 19 |
| garage        | 9  | 11 |
| schedule      | 6  | 14 |
| vase          | 17 | 3  |
| can't         | 14 | 6  |
| better        | 17 | 3  |
| four          | 20 | 0  |

**Table 11.4:** Frequency counts for RP and GA pronunciation of the text tokens.

The question remains, however, whether or not those Swiss who read the English text tokens with rather RP pronunciation significantly differ in SE vowel quality from those who read the words with American English pronunciation. One way to tackle this question is by means of running an ANOVA of SE means factored by RP articulation of the text tokens, GA articulation of the text tokens, or equally distributed RP and GA articulation of the text tokens. The ANOVA presents only one p value below .05 in /i:/ F2.<sup>204</sup> Descriptives to the ANOVA suggest that the group who read the words with rather GA-like pronunciation demonstrates mean values for /i:/ F2 of 2101 Hz, those who articulated the words more towards RP 2131 Hz, those who pronounced the words with equally many RP and GA features show 2273 Hz.<sup>205</sup> It is clear that the final value, possibly caused by a statistical outlier within the half-GA half-RP pronunciation group, is the reason why difference turned out significant. In addition to this one significant value, there

<sup>204</sup> Cf. *Appendix F 7.7 ANOVA of SE Means Factored by Text Pronunciation.*

<sup>205</sup> Cf. *Appendix F 7.6 Descriptives to ANOVA of SE Means Factored by Text Pronunciation.*

are two p values for the F1s of /ɪ/ and /ʌ/ that approximate significance, while the rest of the significance values all virtually tend towards 1.0. It is difficult to escape the conclusion that significant differences in vowel quality between groups who generally articulate the text tokens with GA, or RP, or half-GA half-RP pronunciation could not be established.

### 11.3.2 Embarrassment in Speaking English

The purpose of the question of whether the subject feels embarrassed when conversing in English with a Swiss, an American, or a Briton is to examine whether such sensations exert influence on the subject's SE vowel quality. In 8.1.3.2 *Researching Language Attitudes with the Questionnaire* it was emphasized that embarrassment is the outcome if a learner fears negative evaluation of his performance in the target language (Horwitz and Gregersen 2002: 562). Two subjects responded by saying that they feel embarrassed when conversing with an American, three in speaking with an Englishman, and one in conversing with a Swiss.<sup>206</sup> Although it would be relevant to the issue at hand to run statistical tests with these values, it would not be appropriate as groups of such small sizes cannot provide adequate evidence to substantiate potential claims or hypotheses.

### 11.3.3 Finding Pleasure in Speaking English

Whether or not a Swiss subject likes to speak English depends on a medley of factors, such as those presented in Schumann's (1978: 10) taxonomy of factors affecting SLA. Unfortunately, a question in the form it was applied in this questionnaire does not address the grounds on which the subjects made their choice to answer this question. Frequency counts of the subjects' answers show that 18 out of 20 subjects like speaking English while two do not.<sup>207</sup> An independent t test of the SE means grouped by whether they enjoy speaking in English or not indicates that no significant differences in SE vowel quality could be established between the two groups; only the F2 values of /i:/ and /u:/ show p values  $.05 < p \leq .2$ .<sup>208</sup> Although these results are straightforward, their explanatory power is weak due to the adversarial circumstances that only two out of 20 speakers do not enjoy speaking in English. Were the distribution more equally balanced between the two groups, the explanatory power of the results would be remarkably higher.

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<sup>206</sup> Cf. Appendix F 7.8 *Frequency Tables L2 Embarrassment*.

<sup>207</sup> Cf. Appendix F 7.9 *Frequency Table Finding Pleasure in Speaking English*.

<sup>208</sup> Cf. Appendix F 7.11 *Independent T Test of SE Means Grouped by Finding Pleasure in Speaking English* and Appendix F 7.10 *Descriptives to Independent T Test of SE Means Grouped by Finding Pleasure in Speaking English*.

### 11.3.4 Attaining the Same Pronunciation as Your English Teacher

In 8.1.3.2 *Researching Language Attitudes with the Questionnaire* it was discussed what effects the question *Do you attempt to attain the same pronunciation as your current English teacher?* might have on the subjects.<sup>209</sup> It was outlined how the students' choice to answer this question is partially contingent on their views of the teacher as an authority (cf. Ray 1971). The subjects were asked to check answers denoting "Yes, I want to attain the same pronunciation as my teacher but only during classroom instruction," "Yes, I want to sound like my teacher in and outside of classroom," "No, I do not want to sound like my teacher neither in nor outside of classroom." Three subjects opted for the first answer, seven for the second answer, and ten claim that they do not want to sound like their English teacher.<sup>210</sup> The ANOVA of the SE mean values factored by whether the students try to attain the teacher's pronunciation does not provide significant results. The F1 values of /ɔ:/ and /ʌ/, and the F2 values of /ʊ/ and /u:/ show approximate significance. According to these results, then, those who do and those who do not want to sound like their English teacher do in reality not differ in their SE vowel quality.<sup>211</sup> Given the decent group sizes, 3, 7, and 10 subjects per group, these results are credible and statistically reliable.

### 11.3.5 Testing Anti-Americanism with L2 Speech Production

In an earlier account it was noted how Bourhis (1983) concluded that changes in the political, social, and cultural climate can crucially affect attitudes towards languages. Therefore, one of the final questions delved into the issue as to whether the tendency of anti-Americanism has had or continues to have an impact on the subjects' pronunciation of English. Out of 20 subjects, three claimed that anti-Americanism has a direct effect on their English pronunciation while 17 students assert that this change in the political and ideological climate does not affect their English pronunciation.<sup>212</sup> An independent-samples t test and its descriptives allow for the conclusions that this change in the political and ideological climate does not lead to significance between group differences in

<sup>209</sup> In chronologically progressing through the questionnaire, the next questions that ought to be examined are *Do you like the pronunciation of the people on the classroom audiotapes?* and *Do you consciously try to attain the same pronunciation as the majority of the people that you hear on classroom audiotapes?* It was decided neither to run statistical tests nor to perform frequency counts with the answers obtained. The rationale behind this decision was the uncertainty as to the ratio between GA and RP speakers on school audio tapes.

<sup>210</sup> Cf. Appendix F 7.12 *Frequency Table Attaining the Teacher's Pronunciation*.

<sup>211</sup> Cf. Appendix F 7.14 *ANOVA of SE Means Factored by Attaining the Teacher's Pronunciation* and Appendix F 7.13 *Descriptives to ANOVA of SE Means and Attaining the Teacher's Pronunciation*.

<sup>212</sup> Cf. Appendix F 7.15 *Frequency Table of Anti-Americanism*.

SE articulation.<sup>213</sup> There are, however, five p values close to the  $\alpha$  value of .05; as the F2 value of /ɜ:/ and the F1 values of /ɔ:/, /i:/, /ɑ:/, and /ʌ/. Such values approximating significance may be taken to suggest that were the sample size larger, particularly of the group that claims anti-Americanism to have an impact on their pronunciation (3 students), more reliable conclusions about the findings of this test could be drawn.

### 11.3.6 Testing if What the Students Say they Speak is Spoken in Reality

The students were given a choice of three answers to the question *Do you try to speak American English, British English?*: “American English,” “British English,” and “Neither of them.” Six students claim to want to speak American English, three British English, and eleven students assert that they want to speak neither of them.<sup>214</sup> An ANOVA of the SE mean values factored by what the subjects claim as their target variety shows few significant values.<sup>215</sup> The only p value below the significance threshold is that of /ɑ:/ F1. There are, however, seven approximating significance p values: The F1 values of /æ/, /ɔ:/, /ɜ:/, /ɪ/, and /ʌ/, as well as the F2 values of /ɛ/ and /ʌ/. Although not statistically verified, it appears as if these findings corroborate the conclusion stated in 11.3.1 *Subject Preference for RP or GA* - there seems to be a correlation between the speakers’ attitudes towards English and their SE verbal performance. These findings can again be considered as contradicting Lasagabaster et al.’s (2004: 401) claim that one’s attitudes often do not overlap with one’s actual behavior.

### 11.3.7 Preliminary Conclusions: External Variables III

Essentially, this chapter was geared at detecting whether the Swiss group’s attitudes towards English affect their English pronunciation. It started with the subchapter about the students’ preference for either GA or RP. Frequency counts of two statistical tests show a preference for British English pronunciation on the part of the subjects. According to the statistics, it turns out that if a subject prefers GA or RP his articulation of a few vowels positively resembles these attitudes: Students who prefer GA speak more GA-like and the subjects who favor RP articulate certain vowels more RP-like. The discussion of the text with English tokens underscores the subjects’ preference for RP English. In sum, twelve tokens were articulated with RP-like pronunciation as opposed to eleven tokens with GA-like articulation. The ANOVA shows, however, that subjects

<sup>213</sup> Cf. Appendix F 7.17 ANOVA of SE Means Factored by Anti-Americanism and Appendix F 7.16 Descriptives to ANOVA of SE Means Factored by Anti-Americanism.

<sup>214</sup> Cf. Appendix F 7.18 Frequency Table of Trying to Speak GA, RP, or Neither.

<sup>215</sup> Cf. Appendix F 7.20 ANOVA of SE Means Factored by Trying To Speak GA, RP, or Neither, and Appendix F 7.19 Descriptives to ANOVA of SE Means Factored by Trying To Speak GA, RP, or Neither.

who, in the context of the text with English tokens, speak more GA-like do not significantly differ in SE vowel quality from those who articulated English tokens more RP-like. The question of whether embarrassment in speaking English and finding pleasure in speaking English affect the articulation of English vowels could not be answered. In both instances, the group sizes are too small to obtain substantial results. Consequently, it was examined if those who want to attain the same pronunciation as their English teachers differ in SE vowel quality from those who do not. Statistically credible results demonstrate that significant differences between the two groups could not be established. In virtue of the present study's statistics, it was further shown that anti-Americanism does not significantly affect SE articulation of vowels. Finally, it was tested whether the variety of English the students claim they speak is spoken in reality. Evidence, although not statistically significant but approximating significance, suggests that in effect there are differences in articulation of SE vowels between those who want to speak RP and those who want to speak GA.

## 12 Final Conclusions and Outlook

This paper scrutinized the English vowel quality of 20 Swiss German EFL speakers. It began with a critical approach towards standardized sociolinguistic procedures, which traced the notions of authenticity in language, language variation, standard language, and methodological difficulties in acoustic phonetic analysis. The discussion of authenticity in researched language, as well as the discussion of language variation and the use of standardized language as a reference point, suggests that the present study's results need to be taken critically. Finally, the critical approaches section attended to problems related to measurement techniques in acoustic phonetics. It was noted how background noise, differing voice qualities, and coarticulatory effects seriously put into question the reliability of results in acoustic phonetic studies.

While the rationale behind section A was to detect topic-related moot points in the context of a general sociolinguistic and acoustic phonetic framework, Section B was primarily concerned to discuss the repercussions of these aspects on the present study. As regards authenticity in language, the low degree of authenticity seems hard to deny, which is largely a corollary of inauthentic contexts and inauthentic speakers. Subsequently, a catalogue of external, internal, and stylistic variables that are believed to cause variation in SLA was outlined. Emphasis was laid on the fact that the scrutinized variables in the present study were subjectively selected and only represent the nexus around which further, and even more specific, variables evolve. Language internal factors for variation in SLA such as similarity in language and markedness universals, as well as stylistic factors were subsequently introduced in the present study. Finally, a few words were devoted to how the present study tackled audio segmentation and acoustic phonetic measurement problems. Sections A and B provided the necessary theoretical foundation for Section C, the sociophonetic analysis and statistical evaluation of 20 Swiss EFL speakers.

The working hypothesis of the present study, i.e. the pronunciation of SG NNSs of English deviates from that of RP and GA, was verified. Virtually all SE vowels are pronounced significantly different from their targets in RP and GA. One-sample t tests showed that in comparison with the F1 values of RP, the Swiss articulated /ɑ:/, /ʊ/, /ɪ/, /æ/, /ɜ:/, /ʌ/, /i:/ /u:/ (the last two not statistically significant) significantly higher than an RP speaker does, whereas the vowels /ɛ/ and /ɔ/ were pronounced significantly lower than in RP. In terms of backness and frontedness, the Swiss articulated /i:/, /ɜ:/, /ʌ/, /ɑ:/, /ʊ/, and /u:/ significantly further back than in RP and /ɪ/, /ɛ/, /æ/, and /ɔ/ significantly more fronted. As far as the comparison with General American English is concerned, the Swiss articulated all English vowels significantly higher than GA,<sup>216</sup> except /æ/ (not statistically

<sup>216</sup> With the exception of the vowel /ʌ/, which demonstrated a p value between .05 and .2.

significant) which was pronounced slightly lower. As for the F2 values, the subjects articulated each vowel significantly further back, except for /ɜ/ (not statistically significant), which was minimally more fronted. Based on these criteria, the chapter concluded that there seems to be a trend of the Swiss group to pronounce English vowels more RP-like than GA-like, specifically in terms of their vowels' F2 values. The second part of Section C was concerned with finding ways to account for these results – an approach heavily reliant on the external, internal, and stylistic factors outlined in Section B.

Statistical tests, primarily independent-samples t tests and One-Way ANOVAs, were run to test SE mean values for correlation with variables gathered from a questionnaire. To begin with, variation among the three attempts of articulating the English word lists was scrutinized. Statistics demonstrate that particularly SE front vowels show more variation than SE back vowels, a claim which could later not be corroborated in the identical analyses with SG vowels. All of this showed that SG interference specifically occurred in the articulation of the English vowels /i:/, /ɑ:/, and /ʌ/. It was further verified that students who were given pronunciation training articulate the F1s of /u:/ and /ʊ/ more native-like than those who were not exposed to articulatory training. Finally, statistically reliable evidence was found that the subjects' preference of RP over GA exerts influence on their SE vowel quality. It was shown how students who favor GA do speak more GA-like with respect to a few vowels, while subjects who favor RP articulate certain vowels more RP-like. Generally, the portrayal of the results gained from the statistical tests was negatively-flavored in that too small a sample size and group size repeatedly impinged on the drawing of explanatory and powerful conclusions.

Finally, I would like to devote a few words to the implications and use of these results. According to the available statistics, this study has, if only in part, corroborated the hypotheses that L2 articulation is affected by attitudes towards the target language and that L2 pronunciation instruction is likely to induce a reduction of foreign accent. The importance of explicitly instructing L2 pronunciation could be elaborated on in a proposal to EFL teachers who are concerned about their students' pronunciation of English. In case a foreign accent is deemed undesirable, most modern research suggests that it could, for example, be alleviated by means of applying electronic visual feedback (EVF) (Lambacher 2001: 1).

Lambacher (*ibid*: 1) deals with foreign accent reduction and is concerned to show that “EVF greatly assists teachers and students in identifying speech errors and progress than by just listening to students' pronunciation in class or from recorded tapes.” Such computer programs provide the language learner with real-time information about the saliency of the acoustic properties of their L2 articulation (Lambacher 1996: 1). Given that students have a basic understanding of the computer program, they would be able to perform

“acoustic analysis of their speech with functions for measuring amplitude, intonation, pitch, duration, and frequency” (ibid: 1), not to forget the acoustic analysis of vowels. Formant patterns are clearly visualized on the screen, which enables the students to observe and to measure the vowels they are producing (Lambacher 2001: 3). Lambacher further maintains that the student’s first three formants of the target vowels can be compared instantaneously with those of the teacher’s (ibid: 3). According to Lambacher, the benefits of such analyses outweigh any potential shortcomings. In several respects, the application of spectrogram analyses in foreign language classrooms certainly has merits. Moreover, the significance of training teachers in how to instruct pronunciation as well as the importance of foreign language pronunciation training on the part of the teachers must not be overlooked. In case teachers would like to assist learners to reduce their Swiss German-flavored accent of English, they need a thorough understanding of articulatory phonetics and didactically useful strategies to communicate this knowledge.

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## Appendices

See: <http://www.isw.unibe.ch/metanav/pers/aleemann.html>



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